

LARGE SCALE PROPERTIES OF THE ELEPHANT RANDOM WALK

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ABSTRACT

This thesis studies the large-scale behaviour of the Elephant Random Walk (ERW), a one-dimensional discrete-time stochastic process with complete memory introduced by Schütz and Trimper in 2004, in which each step repeats a uniformly chosen past increment with probability p and reverses it with probability $1 - p$. After reviewing the Simple Random Walk as a baseline, we derive exact formulas for the expectation and second moment of the ERW and show that the latter undergoes a phase transition at the critical value $p_c = 3/4$, separating a diffusive, a critical, and a superdiffusive regime. Within a martingale framework, we establish a Law of Large Numbers valid across all regimes, together with regime-dependent Central Limit Theorems: Gaussian limits arise for $p \leq 3/4$ with appropriate normalizations, while for $p > 3/4$ the rescaled walk converges almost surely to a non-degenerate, non-Gaussian random variable. We further prove the Quadratic Strong Law and the Law of the Iterated Logarithm, and use them to show that the ERW is recurrent for $p \leq 3/4$ and transient for $p > 3/4$. Finally, we construct a quasi-maximum likelihood estimator of p from a single observed path, prove its strong consistency, and conclude with a Monte Carlo study of confidence intervals for the memory parameter p .

NOTATION

$\mathbb{Z}^d$ integer lattice in d dimensions.

$\mathcal{F}_n$ filtration, $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$.

$\mathbb{E}[\cdot \mid \mathcal{F}_n]$ conditional expectation.

ΔM_k martingale increment, $M_k - M_{k-1}$.

$\langle M \rangle_n$ predictable quadratic variation.

$n \rightarrow \infty$ asymptotics.

$\sim$ asymptotic equivalence.

$o(\cdot)$ little-o.

Γ Gamma function.

H_n n -th harmonic number.

$(X_n)_{n \geq 0}$ Simple Random Walk, $X_n = \sum_{i=1}^n \eta_i$.

$\xrightarrow{a.s.}$ almost sure convergence.

$\xrightarrow{\mathcal{D}}$ convergence in distribution.

$\mathcal{N}(\mu, \sigma^2)$ Gaussian distribution with mean μ and variance σ^2 .

$\mathcal{L}_\infty(x)$ total occupation time at x .

$(\xi_n)_{n \geq 1}$ increments of the Elephant Random Walk.

$(S_n)_{n \geq 0}$ Elephant Random Walk, $S_n = \sum_{i=1}^n \xi_i$.

p memory parameter.

q initial bias.

α $\alpha := 2p - 1$.

β $\beta := 2q - 1$.

p_c critical value, $p_c = \frac{3}{4}$.

$\hat{p}_n$ quasi-maximum likelihood estimator of p .

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INTRODUCTION

The **Simple Random Walk** is among the most thoroughly understood processes in probability. It is a stochastic process in which each step is a fresh draw, independent of the entire history that precedes it, and this independence is the key tool running through the classical proofs. The Simple Random Walk lies at the origin of **Brownian motion**, a process applied across fields ranging from mathematical finance to biology [Shr04, CPB08]. But suppose the walker remembers every step it has taken, so that its next step depends on the entire path before it, with the strength of that memory tuned through a parameter p . What does memory do to a random walk?

The **Elephant Random Walk** (ERW) is precisely such a process. The walk was introduced in 2004 by the physicists Schütz and Trimper [ST04]. Its name alludes to the legendary memory of elephants: at every step, the walker recalls its full history and chooses accordingly. The appeal of the ERW lies in the simplicity of its rule and the richness of the mathematics it produces. Already in their original paper, Schütz and Trimper found that the behaviour of the walk changes fundamentally at the critical memory parameter $p_c = \frac{3}{4}$. Beyond its mathematical interest, the ERW serves as a simple model for real world processes in which the past shapes the future, with applications in markets where customers choose between competing products [SPR26], while closely related random walks are used to model how animals move using memory of where they have been [FMK23]. Since its introduction, the ERW has continued to attract mathematical attention, and a number of open questions about the walk are still being researched today.

In this thesis, we study a collection of these results, with a focus on the large-scale behaviour of the walk. The aim is to build up a clear picture of how the ERW behaves at large scales: its asymptotic growth, its limiting distributions, its recurrence behaviour, and the way these features depend on the parameters. By the end, the reader will not only have become acquainted with this stochastic process, but will also have seen how the **martingale**, a process whose expected next value equals its current one, provides a framework on which most of the proofs rest.

Before turning to the ERW itself, **Chapter 1** collects the analytical tools that appear repeatedly throughout the thesis: filtrations, martingales, asymptotic notation, and the Gamma function. **Chapter 2** then introduces the **Simple Random Walk**, a special case of the ERW,

which we treat first to set the scale against which the complexity of the ERW and its results can later be measured. [Chapter 3](#) introduces the main object of study itself: we give the formal definition, compute its expectation and second moment, and reveal the [phase transition](#) at $p_c = \frac{3}{4}$. This is also where the three regimes (diffusive, critical, and superdiffusive), separated by p_c , are defined.

Building on this, [Chapter 4](#) establishes a [Law of Large Numbers](#) and regime-dependent [Central Limit Theorems](#) for the ERW. We will prove that in the diffusive and critical regimes the rescaled walk converges to a Gaussian limit, while in the superdiffusive regime it converges instead to a non-degenerate, non-Gaussian random variable. [Chapter 5](#) moves beyond distributional results and establishes two almost-sure limit theorems: the Quadratic Strong Law and the Law of the Iterated Logarithm. These are then used to show that the ERW is [recurrent](#) in the diffusive and critical regimes and [transient](#) in the superdiffusive regime. [Chapter 6](#) reverses the question: rather than studying the walk for a given p , we ask how p can be recovered from a single observed path. We introduce a [quasi-maximum likelihood](#) estimator and prove that it is consistent. We then use this estimator to construct confidence intervals for p , and conclude the chapter with a Monte Carlo study that investigates how these intervals perform for finite sample sizes.

A word on my own contribution. The thesis is for the most part expository, and credit for the results presented here is given throughout. Still, a few things are my own contribution. Throughout the thesis, I have aimed to complement the results with simulations and illustrations that bring out their underlying intuition. All figures in the thesis, whether numerical simulations or schematic illustrations, were produced by me. Reconstructing the cited proofs proved to be one of the most demanding parts of the project. The original papers often leave out intermediate steps, and working these out in full took a considerable amount of time. Several proofs that appear without a citation are my own (e.g. the proof of [Theorem 5.4](#)). The example in [Section 6.3](#) was originally intended as an original contribution, but in the course of working it out I discovered that Bercu and Laulin [[BL24](#)] already treat the same question in greater depth. The Monte Carlo study presented there, which assesses how the confidence intervals behave for finite n , is nonetheless my own. I designed the simulation setup and carried out the resulting analysis of their finite-sample coverage.

PRELIMINARIES

This chapter collects the main concepts and notation used throughout the thesis, so that later chapters can proceed without repeated detours into background material. We introduce filtrations, martingales, asymptotic notation, and the Gamma function, all of which are essential to the analysis of the Elephant Random Walk.

1.1 FILTRATIONS

Throughout this thesis, we work on a fixed **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$. A **discrete-time filtration** is a sequence $(\mathcal{F}_n)_{n \geq 0}$ of sub- σ -algebras of $\mathcal{F}$ such that

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

The σ -algebra $\mathcal{F}_n$ is interpreted as the collection of events that can be determined from the information available up to time n . A stochastic process $(Y_n)_{n \geq 0}$ is called **adapted** to $(\mathcal{F}_n)_{n \geq 0}$ if Y_n is $\mathcal{F}_n$ -measurable for every $n \geq 0$. In other words, the value of Y_n is determined by the information available at time n . Given a process $(Y_n)_{n \geq 0}$, a natural choice of filtration is its **natural filtration**,

$$\mathcal{F}_n^Y := \sigma(Y_0, \dots, Y_n), \quad n \geq 0,$$

which is the smallest filtration with respect to which $(Y_n)_{n \geq 0}$ is adapted.

Let Y be an integrable random variable and $(\mathcal{F}_n)_{n \geq 0}$ a filtration. The **conditional expectation** $\mathbb{E}[Y \mid \mathcal{F}_n]$ may be interpreted as the expectation of Y given the information available up to time n , and we will repeatedly use a few of its standard properties.

First, conditional expectation is **linear**. That is, for integrable Y, W and constants $a, b \in \mathbb{R}$, one has

$$\mathbb{E}[aY + bW \mid \mathcal{F}_n] = a \mathbb{E}[Y \mid \mathcal{F}_n] + b \mathbb{E}[W \mid \mathcal{F}_n]. \quad (1.1.1)$$

Moreover, if Y is $\mathcal{F}_n$ -measurable then

$$\mathbb{E}[Y \mid \mathcal{F}_n] = Y.$$

More generally, any quantity that is already determined by the information in $\mathcal{F}_n$ can be pulled outside the conditional expectation: if H is $\mathcal{F}_n$ -measurable and bounded, then

$$\mathbb{E}[HY \mid \mathcal{F}_n] = H \mathbb{E}[Y \mid \mathcal{F}_n]. \quad (1.1.2)$$

Conditional expectations also satisfy an iterated conditioning rule, known as the **tower property**: whenever $\mathcal{F}_n \subseteq \mathcal{F}_m$, conditioning first on $\mathcal{F}_m$ and then on $\mathcal{F}_n$ is the same as conditioning directly on $\mathcal{F}_n$, that is,

$$\mathbb{E}[\mathbb{E}[Y \mid \mathcal{F}_m] \mid \mathcal{F}_n] = \mathbb{E}[Y \mid \mathcal{F}_n]. \quad (1.1.3)$$

As a special case, taking expectations removes the conditioning. This is known as the **law of total expectation**:

$$\mathbb{E}[\mathbb{E}[Y \mid \mathcal{F}_n]] = \mathbb{E}[Y]. \quad (1.1.4)$$

1.2 MARTINGALES

Building on the notion of filtrations, the next key object is a **martingale**, which plays an essential role in almost all the proofs in this thesis. Informally, a martingale is a stochastic process whose best prediction for its next value, given all information available up to the present time, is its current value. More concretely:

Definition 1.1 (Martingale). Let $(M_n)_{n \geq 0}$ be a stochastic process adapted to a filtration $(\mathcal{F}_n)_{n \geq 0}$. We call $(M_n)_{n \geq 0}$ a martingale with respect to $(\mathcal{F}_n)_{n \geq 0}$ if each M_n is integrable, that is,

$$\mathbb{E}[|M_n|] < \infty,$$

and if, for every $n \geq 0$, the **martingale property**

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$$

holds.

This identity means that, once the information up to time n is fixed, the conditional expectation of the next value equals the current one. In particular, taking expectations on both sides and using [Property 1.1.4](#), we obtain

$$\mathbb{E}[M_{n+1}] = \mathbb{E}[\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]] = \mathbb{E}[M_n],$$

so a martingale has constant expectation over time.

Two more objects appear throughout: the increments and the predictable quadratic variation. For a martingale $(M_n)_{n \geq 0}$, let

$$\Delta M_k := M_k - M_{k-1}, \quad k \geq 1,$$

denote its k -th **increment**. Then

$$\mathbb{E}[\Delta M_k \mid \mathcal{F}_{k-1}] = 0,$$

and therefore also

$$\mathbb{E}[\Delta M_k] = 0.$$

If, in addition, $(M_n)_{n \geq 0}$ is square-integrable, meaning that $\mathbb{E}[M_n^2] < \infty$ for all $n \geq 0$, then the process

$$\langle M \rangle_n := \sum_{k=1}^n \mathbb{E}[(\Delta M_k)^2 \mid \mathcal{F}_{k-1}], \quad n \geq 1, \quad (1.2.1)$$

is called the **predictable quadratic variation** of $(M_n)_{n \geq 0}$. The quantity $\langle M \rangle_n$ represents the accumulated conditional second moments of the martingale increments up to time n . The word predictable refers to the fact that each term is $\mathcal{F}_{k-1}$ -measurable, and is therefore determined by the information available before the k -th step.

1.3 ASYMPTOTIC NOTATION

When analyzing the long term behaviour of sequences, we will frequently need to compare their growth rates as $n \rightarrow \infty$. This section introduces the **asymptotic** notation used throughout the thesis and collects the properties used in the subsequent sections.

Let $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ be sequences of real numbers with $b_n \neq 0$ for all sufficiently large n .

We write $a_n = o(b_n)$ if

$$\frac{a_n}{b_n} \rightarrow 0, \quad \text{as } n \rightarrow \infty,$$

and say that a_n is eventually **negligible** compared to b_n .

We write $a_n \sim b_n$ if

$$\frac{a_n}{b_n} \rightarrow 1, \quad \text{as } n \rightarrow \infty,$$

and in this case say that a_n and b_n are **asymptotically equivalent**. Equivalently, $a_n \sim b_n$ if and only if $a_n = b_n(1 + o(1))$.

Asymptotic equivalence is preserved under products: if $a_n \sim b_n$ and $c_n \sim d_n$, then $a_n c_n \sim b_n d_n$. It is also preserved under powers: if $a_n \sim b_n$, then $a_n^\gamma \sim b_n^\gamma$ for any fixed $\gamma \in \mathbb{R}$, provided $a_n > 0$ and $b_n > 0$ for all sufficiently large n . Moreover, if $a_n \sim b_n$ and $b_n \neq 0$ for all sufficiently large n , then

$$\frac{1}{a_n} \sim \frac{1}{b_n}.$$

In general, if $a_n \sim b_n$ and one of the two sequences converges to a nonzero limit L , then so does the other, and to the same value as well. Also, if either one of the sequences diverges to infinity, then so does the other.

A specific asymptotic equivalence that will be used repeatedly throughout this thesis concerns sums of powers:

Lemma 1.2. For every $r > -1$, one has

$$\sum_{j=1}^n j^r \sim \frac{n^{r+1}}{r+1}, \quad n \rightarrow \infty.$$

Proof. Set

$$a_n := \sum_{j=1}^n j^r, \quad b_n := \frac{n^{r+1}}{r+1}.$$

Since $r > -1$, we have $b_n \rightarrow \infty$. Hence, by the **Stolz–Cesàro theorem** (Lemma A.1),

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^r}{\frac{(n+1)^{r+1} - n^{r+1}}{r+1}}.$$

By the **Mean Value Theorem**, for each n there exists $\phi_n \in (n, n+1)$ such that

$$(n+1)^{r+1} - n^{r+1} = (r+1)\phi_n^r.$$

Therefore,

$$\frac{(n+1)^r}{\frac{(n+1)^{r+1} - n^{r+1}}{r+1}} = \left(\frac{n+1}{\phi_n} \right)^r \rightarrow 1,$$

since $\phi_n/(n+1) \rightarrow 1$. We conclude that

$$\sum_{j=1}^n j^r \sim \frac{n^{r+1}}{r+1}, \quad n \rightarrow \infty.$$

□

Another important asymptotic relation is that of the **Harmonic series**, defined by

$$H_n := \sum_{k=1}^n \frac{1}{k}.$$

A classical result states that

$$H_n \sim \log n, \quad (1.3.1)$$

as $n \rightarrow \infty$. In particular, the Harmonic series diverges, since $\log n$ diverges.

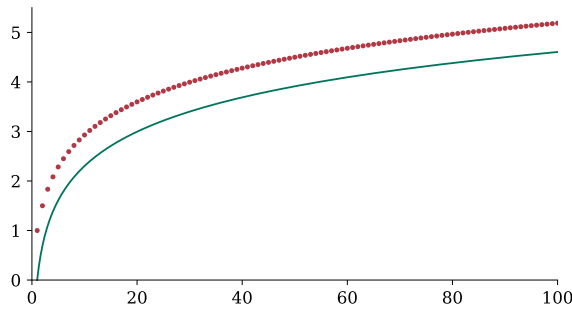


Figure 1.1: The harmonic series H_n (dots) and $\log n$ (solid line).

The asymptotic rules collected above will be used repeatedly in combination with Gamma function identities in the next section.

1.4 THE GAMMA FUNCTION

The **Gamma function** originates in Euler's work of the late 1720s, motivated by the question of interpolating the factorial $n \mapsto n!$ to a smooth function of a variable. The symbol Γ and the modern convention, in which $\Gamma(n) = (n-1)!$, are due to Legendre. For $\Re(z) > 0$, $\Gamma(z)$ is defined as

$$\Gamma(z) := \int_0^\infty t^{z-1} e^{-t} dt, \quad (1.4.1)$$

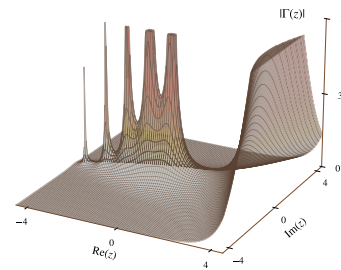
and admits a **meromorphic continuation** to $\mathbb{C} \setminus \{-n : n \in \mathbb{Z}_{\geq 0}\}$. The function will be used frequently throughout this thesis. A fundamental property is the functional relation

$$\Gamma(x+1) = x\Gamma(x), \quad (1.4.2)$$

established by a single integration by parts in (1.4.1). A consequence of this is that for all $n \in \mathbb{N}$, one has

$$(n-1)! = \Gamma(n).$$

Another consequence that will be used throughout is the conversion of products into Gamma ratios.



Lemma 1.3. Let $a \in \mathbb{R}_{>-1}$ and $n \geq 2$. One has the identity

$$\prod_{k=1}^{n-1} \frac{k+a}{k} = \frac{\Gamma(n+a)}{\Gamma(1+a)\Gamma(n)}.$$

Proof. We may write

$$\prod_{k=1}^{n-1} \frac{k+a}{k} = \frac{\prod_{k=1}^{n-1} (k+a)}{\prod_{k=1}^{n-1} k} = \frac{\prod_{k=1}^{n-1} (k+a)}{\Gamma(n)}.$$

To evaluate the numerator, we apply the functional equation of the Gamma function repeatedly to obtain an expression for the product in terms of Gamma functions

$$\begin{aligned} \Gamma(n+a) &= (n+a-1)\Gamma(n+a-1) \\ &= \left(\prod_{k=1}^{n-1} (k+a) \right) \Gamma(1+a). \end{aligned}$$

Therefore

$$\prod_{k=1}^{n-1} (k+a) = \frac{\Gamma(n+a)}{\Gamma(1+a)}.$$

Substituting this expression into the denominator yields the final result

$$\prod_{k=1}^{n-1} \frac{k+a}{k} = \frac{\Gamma(n+a)}{\Gamma(1+a)\Gamma(n)}.$$

□

To approximate the growth rate as $n \rightarrow \infty$, we use **Stirling's approximation** (Lemma A.2) to obtain the following estimate.

Lemma 1.4. *Let $a, b \in \mathbb{R}$ be fixed. Then, as $n \rightarrow \infty$*

$$\frac{\Gamma(n+a)}{\Gamma(n+b)} \sim n^{a-b}.$$

Proof. By Stirling's approximation (Lemma A.2),

$$\Gamma(x) \sim \sqrt{2\pi} x^{x-\frac{1}{2}} e^{-x}, \quad x \rightarrow \infty.$$

Applying this with $x = n+a$ and $x = n+b$, we obtain

$$\frac{\Gamma(n+a)}{\Gamma(n+b)} \sim \frac{(n+a)^{n+a-\frac{1}{2}} e^{-(n+a)}}{(n+b)^{n+b-\frac{1}{2}} e^{-(n+b)}}.$$

Factoring out n from both powers yields

$$\frac{\Gamma(n+a)}{\Gamma(n+b)} \sim n^{a-b} e^{b-a} \frac{\left(1 + \frac{a}{n}\right)^{n+a-\frac{1}{2}}}{\left(1 + \frac{b}{n}\right)^{n+b-\frac{1}{2}}}.$$

It is well-known that

$$\left(1 + \frac{a}{n}\right)^{n+a-\frac{1}{2}} \rightarrow e^a, \quad \left(1 + \frac{b}{n}\right)^{n+b-\frac{1}{2}} \rightarrow e^b,$$

as $n \rightarrow \infty$. Therefore, the remaining factor converges to

$$e^{b-a} \frac{e^a}{e^b} = 1.$$

We conclude that

$$\frac{\Gamma(n+a)}{\Gamma(n+b)} \sim n^{a-b},$$

as claimed. □

Moreover, the identity

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2} \quad (1.4.3)$$

will be used repeatedly.

Lemma 1.5. *Let $\Gamma(\cdot)$ be defined as in (1.4.1), then we have the identity*

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

Proof. We first consider the classical identity

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}. \quad (1.4.4)$$

This identity follows by the following argument. Substituting $t = u^2$ in the definition of the Gamma function (1.4.1) yields

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-1/2} e^{-t} dt = 2 \int_0^\infty e^{-u^2} du.$$

Squaring and passing to polar coordinates, we obtain

$$\Gamma\left(\frac{1}{2}\right)^2 = 4 \int_0^\infty \int_0^\infty e^{-(u^2+v^2)} du dv = 4 \int_0^{\pi/2} \int_0^\infty r e^{-r^2} dr d\theta = \pi.$$

Since $\Gamma(1/2) > 0$, we conclude that $\Gamma(1/2) = \sqrt{\pi}$. Together with [Property 1.4.2](#), this gives

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2},$$

the desired result. □

THE SIMPLE RANDOM WALK

Before introducing the main object of study, it is both natural and useful to first look at a special case: the **Simple Random Walk** (SRW). Because its structure is much simpler than that of the Elephant Random Walk, it provides a useful reference point for the more complex behaviour studied later.

It is worth noting that, as we will see in [Chapter 3](#), the Simple Symmetric Random Walk is in fact a special case of the **Elephant Random Walk**, obtained by setting $p = \frac{1}{2}$ and $q = \frac{1}{2}$, and we will verify this at the appropriate moments.

The chapter proceeds as follows. We give the definition, expectation and second moment of the Simple Random Walk, then state and prove the Law of Large Numbers and the Central Limit Theorem, before closing with a discussion of recurrence and transience. As the focus of this thesis lies elsewhere, most proofs are presented as proof sketches rather than in full rigorous detail.

2.1 DEFINITION

We begin by making the notion of the Simple Random Walk precise, working throughout in arbitrary dimension $d \in \mathbb{N}$.

Definition 2.1 (The Simple Random Walk). Fix a dimension $d \in \mathbb{N}$, and let $e_1, \dots, e_d$ denote the standard basis vectors of $\mathbb{Z}^d$. Let $(\eta_n)_{n \geq 1}$ be independent and identically distributed random variables with values in

$$\{\pm e_1, \dots, \pm e_d\},$$

such that

$$\mathbb{P}(\eta_n = e_i) = \mathbb{P}(\eta_n = -e_i) = \frac{1}{2d}, \quad 1 \leq i \leq d,$$

for every $n \geq 1$. Define

$$X_0 = 0, \quad X_n = \sum_{i=1}^n \eta_i \quad (n \geq 1).$$

Then $(X_n)_{n \geq 0}$ is called the **d -dimensional Simple Random Walk** on $\mathbb{Z}^d$.

To build intuition for the definition above, [Figure 2.1](#) below displays sample paths of the Simple Random Walk in dimensions $d = 2$ and $d = 3$.



Figure 2.1: Sample paths of the SRW on $\mathbb{Z}^d$, $n = 6,000$. Left: $d = 2$. Right: $d = 3$.

The paths in [Figure 2.1](#) are obtained by repeatedly adding one random increment to the current position. Since

$$X_n - X_{n-1} = \eta_n,$$

the increments of the Simple Random Walk are precisely the random variables $(\eta_n)_{n \geq 1}$. For any $m \geq 1$ and any $v_1, \dots, v_m \in \{\pm e_1, \dots, \pm e_d\}$, [Definition 2.1](#) gives

$$\mathbb{P}(\eta_1 = v_1, \dots, \eta_m = v_m) = \prod_{k=1}^m \mathbb{P}(\eta_k = v_k).$$

Moreover, each η_k has the same distribution on $\{\pm e_1, \dots, \pm e_d\}$, namely

$$\mathbb{P}(\eta_k = e_j) = \mathbb{P}(\eta_k = -e_j) = \frac{1}{2d}, \quad 1 \leq j \leq d.$$

Therefore, the increments $(X_n - X_{n-1})_{n \geq 1}$ are independent and identically distributed, or simply **i.i.d.** This property will be of great use in the proofs of this chapter.

2.2 EXPECTATION & VARIANCE

We now turn to its basic probabilistic properties. We begin by computing the **expected displacement** and **variance** of $(X_n)_{n \geq 0}$.

Theorem 2.2. *Let $(X_n)_{n \geq 0}$ be the d -dimensional Simple Random Walk on $\mathbb{Z}^d$. Then for $n \geq 0$,*

$$\mathbb{E}[X_n] = 0.$$

Proof. By definition, $X_n = \sum_{i=1}^n \eta_i$ for every $n \geq 1$, while $X_0 = 0$. First consider $n = 0$. Then clearly

$$\mathbb{E}[X_0] = \mathbb{E}[0] = 0.$$

Now let $n \geq 1$. Using linearity of expectation, we obtain

$$\mathbb{E}[X_n] = \mathbb{E}\left[\sum_{i=1}^n \eta_i\right] = \sum_{i=1}^n \mathbb{E}[\eta_i].$$

Since $(\eta_i)_{i \geq 1}$ are i.i.d., this becomes

$$\mathbb{E}[X_n] = n \mathbb{E}[\eta_1]. \quad (2.2.1)$$

It therefore remains to compute $\mathbb{E}[\eta_1]$. By construction,

$$\mathbb{P}(\eta_1 = e_j) = \mathbb{P}(\eta_1 = -e_j) = \frac{1}{2d}, \quad 1 \leq j \leq d.$$

Hence

$$\mathbb{E}[\eta_1] = \sum_{j=1}^d \left(\frac{1}{2d} e_j + \frac{1}{2d} (-e_j) \right) = \sum_{j=1}^d 0 = 0.$$

Substituting this into (2.2.1) yields

$$\mathbb{E}[X_n] = n \cdot 0 = 0.$$

This completes the proof. $\square$

We now turn to its **variance**. Recall that the variance of a stochastic process $(Y_n)_{n \geq 0}$ in dimension $d \in \mathbb{N}$ is given by

$$\text{Var}(Y_n) := \mathbb{E}\left[\|Y_n - \mathbb{E}[Y_n]\|^2\right]$$

In other words, the variance measures the expected squared Euclidean distance of the walker from its mean position at time n . Since we have already shown that $\mathbb{E}[X_n] = 0$, in the present setting it simply quantifies how far the walk typically lies from the origin, and thus how widely the sample paths are spread out. The proof of the following theorem is inspired by the argument presented in [Lawnd].

Theorem 2.3. *Let $(X_n)_{n \geq 0}$ be the d -dimensional Simple Random Walk on $\mathbb{Z}^d$. Then, for $n \geq 0$,*

$$\text{Var}(X_n) := \mathbb{E}\left[\|X_n - \mathbb{E}[X_n]\|^2\right] = n.$$

Proof Sketch. First consider the case $n = 0$. Since $X_0 = 0$, we have

$$\text{Var}(X_0) = \mathbb{E}[\|X_0 - \mathbb{E}[X_0]\|^2] = \mathbb{E}[\|0\|^2] = 0.$$

Now let $n \geq 1$. By [Theorem 2.2](#), we have

$$\text{Var}(X_n) = \mathbb{E}[\|X_n\|^2].$$

Since $X_n = \sum_{i=1}^n \eta_i$, we obtain

$$\|X_n\|^2 = \left\langle \sum_{i=1}^n \eta_i, \sum_{j=1}^n \eta_j \right\rangle = \sum_{i=1}^n \|\eta_i\|^2 + 2 \sum_{1 \leq i < j \leq n} \langle \eta_i, \eta_j \rangle.$$

Taking expectations gives

$$\mathbb{E}[\|X_n\|^2] = \sum_{i=1}^n \mathbb{E}[\|\eta_i\|^2] + 2 \sum_{1 \leq i < j \leq n} \mathbb{E}[\langle \eta_i, \eta_j \rangle]. \quad (2.2.2)$$

We now treat the two terms separately. Since each increment η_i takes values in $\{\pm e_1, \dots, \pm e_d\}$, we have $\|\eta_i\| = 1$, and hence

$$\mathbb{E}[\|\eta_i\|^2] = 1 \quad \forall i \geq 1.$$

Moreover, for $i < j$, independence of η_i and η_j yields

$$\mathbb{E}[\langle \eta_i, \eta_j \rangle] = \langle \mathbb{E}[\eta_i], \mathbb{E}[\eta_j] \rangle.$$

Each increment has mean zero, so

$$\mathbb{E}[\eta_i] = 0 \quad \forall i \geq 1.$$

Therefore,

$$\mathbb{E}[\langle \eta_i, \eta_j \rangle] = 0 \quad \text{when } i < j.$$

Substituting these identities into (2.2.2), we find

$$\mathbb{E}[\|X_n\|^2] = \sum_{i=1}^n 1 + 2 \sum_{1 \leq i < j \leq n} 0 = n.$$

Consequently,

$$\text{Var}(X_n) = n$$

for every $n \geq 1$. Together with the case $n = 0$, this completes the proof. $\square$

2.3 LAW OF LARGE NUMBERS

The **Law of Large Numbers** is one of the most fundamental results in probability theory. Informally, it states that the average of a large number of random variables converges to their common expected value as the number of observations grows. For the Simple Random Walk, this means that the position X_n , when divided by n , converges to zero almost surely.

The proof of the theorem presented below is based on the proofs of [Nor, Section 10.1] and [Rom, Section 9.2].

Theorem 2.4. *Let $(X_n)_{n \geq 0}$ be the d -dimensional Simple Random Walk on $\mathbb{Z}^d$. Then*

$$\frac{X_n}{n} \xrightarrow{a.s.} 0, \quad (n \rightarrow \infty).$$

Proof Sketch. We show that

$$\left\| \frac{X_n}{n} \right\| \xrightarrow{a.s.} 0,$$

as this is equivalent to the desired vector convergence. Fix $\varepsilon > 0$. By **Markov's inequality**,

$$\mathbb{P} \left(\left\| \frac{X_n}{n} \right\| > \varepsilon \right) = \mathbb{P}(\|X_n\|^4 > \varepsilon^4 n^4) \leq \frac{\mathbb{E}[\|X_n\|^4]}{\varepsilon^4 n^4}. \quad (2.3.1)$$

We proceed by writing

$$X_n = (X_n^{(1)}, \dots, X_n^{(d)}),$$

where

$$X_n^{(r)} := \sum_{k=1}^n \eta_k^{(r)}, \quad 1 \leq r \leq d,$$

and $\eta_k^{(r)}$ denotes the r -th coordinate of η_k . Then

$$\|X_n\|^2 = \sum_{r=1}^d (X_n^{(r)})^2,$$

so that by the **Cauchy-Schwarz inequality**,

$$\|X_n\|^4 = \left(\sum_{r=1}^d (X_n^{(r)})^2 \right)^2 \leq d \sum_{r=1}^d (X_n^{(r)})^4.$$

Taking expectations yields

$$\mathbb{E}[\|X_n\|^4] \leq d \sum_{r=1}^d \mathbb{E}[(X_n^{(r)})^4]. \quad (2.3.2)$$

We now estimate the fourth moment of a fixed coordinate. For each $r \in \{1, \dots, d\}$, the random variables $(\eta_k^{(r)})_{k \geq 1}$ are independent and identically distributed, centered, and satisfy

$$\mathbb{P}(\eta_k^{(r)} = 1) = \mathbb{P}(\eta_k^{(r)} = -1) = \frac{1}{2d}, \quad \mathbb{P}(\eta_k^{(r)} = 0) = 1 - \frac{1}{d}.$$

Hence

$$\mathbb{E}[\eta_k^{(r)}] = 0, \quad \mathbb{E}[(\eta_k^{(r)})^2] = \frac{1}{d}, \quad \mathbb{E}[(\eta_k^{(r)})^4] = \frac{1}{d}. \quad (2.3.3)$$

Expanding the fourth power and using independence together with the fact that all mixed terms containing an odd power have expectation zero, we obtain

$$\mathbb{E}[(X_n^{(r)})^4] = n \mathbb{E}[(\eta_1^{(r)})^4] + 6 \sum_{1 \leq i < j \leq n} \mathbb{E}[(\eta_i^{(r)})^2] \mathbb{E}[(\eta_j^{(r)})^2].$$

Substituting the moments computed in (2.3.3) gives

$$\mathbb{E}[(X_n^{(r)})^4] = \frac{n}{d} + 6 \binom{n}{2} \frac{1}{d^2} = \frac{n}{d} + \frac{3n(n-1)}{d^2} \leq Cn^2$$

for some constant $C > 0$ independent of n . Consequently, together with (2.3.2),

$$\mathbb{E}[\|X_n\|^4] \leq C'n^2$$

for some constant $C' > 0$. Returning to the estimate in (2.3.1), we find

$$\mathbb{P}\left(\left\|\frac{X_n}{n}\right\| > \varepsilon\right) \leq \frac{C'}{\varepsilon^4 n^2}.$$

Since

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty,$$

it follows that

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\left\|\frac{X_n}{n}\right\| > \varepsilon\right) < \infty.$$

By the **Borel-Cantelli Lemma** (Lemma A.3), for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left\|\frac{X_n}{n}\right\| > \varepsilon \text{ infinitely often}\right) = 0.$$

Thus, for every $\varepsilon > 0$, one has

$$\left\|\frac{X_n}{n}\right\| \leq \varepsilon$$

for all sufficiently large n , almost surely. Hence

$$\left\|\frac{X_n}{n}\right\| \xrightarrow{a.s.} 0.$$

Therefore,

$$\frac{X_n}{n} \xrightarrow{a.s.} 0,$$

as $n \rightarrow \infty$. □

Figure 2.2 illustrates this convergence. Despite early fluctuations, all paths are visibly drawn toward zero as n grows.

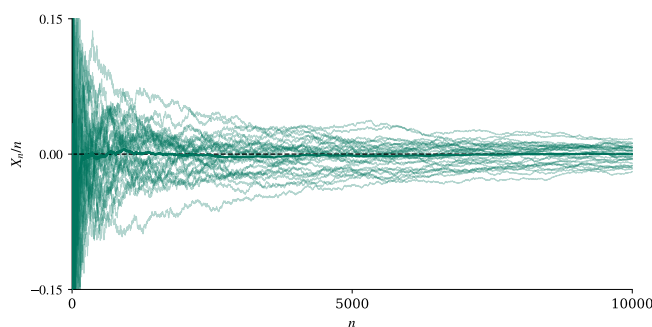


Figure 2.2: Paths of X_n/n with $n = 10,000$ and $d = 1$. Solid line: empirical mean (0.002 at $n = 10,000$); dashed line: zero.

2.4 CENTRAL LIMIT THEOREM

The next natural question is: what is the limiting distribution of X_n ? The **Central Limit Theorem** shows that $X_n/\sqrt{n}$ converges in distribution to a centered Gaussian law on $\mathbb{R}^d$.

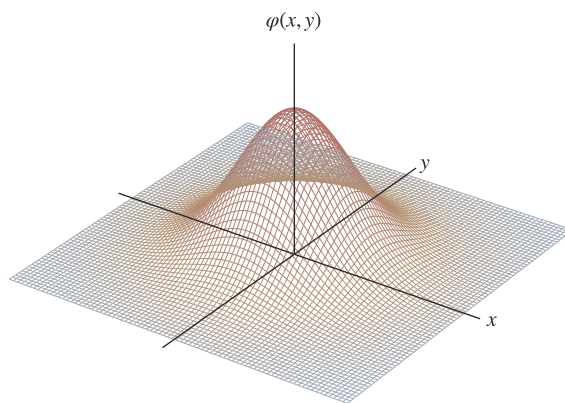


Figure 2.3: Graph of the standard Gaussian density $\varphi(x, y) = \frac{1}{2\pi} \exp(-\frac{1}{2}(x^2 + y^2))$ in $\mathbb{R}^2$.

The proof uses **characteristic functions**, which are particularly useful because, by **Lévy's Uniqueness Theorem** (Lemma A.4), they completely determine the distribution of a random variable. The proof of Theorem 2.5 is inspired by the work of [Gad20].

Theorem 2.5. Let $(X_n)_{n \geq 0}$ be the d -dimensional Simple Random Walk. Then

$$\frac{X_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}_d\left(0, \frac{1}{d} I_d\right),$$

as $n \rightarrow \infty$, where $\mathcal{N}_d(\mu, \Sigma)$ denotes the d -dimensional **Gaussian distribution** with **mean vector** $\mu \in \mathbb{R}^d$ and **covariance matrix** $\Sigma \in \mathbb{R}^{d \times d}$.

Proof Sketch. We establish the convergence using **characteristic functions**. For $t \in \mathbb{R}^d$, consider

$$\varphi_n(t) := \mathbb{E} \left[\exp \left(i \left\langle t, \frac{X_n}{\sqrt{n}} \right\rangle \right) \right].$$

Since the increments are i.i.d., we have

$$\varphi_n(t) = \prod_{k=1}^n \mathbb{E} \left[\exp \left(i \left\langle t, \frac{\eta_k}{\sqrt{n}} \right\rangle \right) \right] = \left(\mathbb{E} \left[\exp \left(i \left\langle t, \frac{\eta_1}{\sqrt{n}} \right\rangle \right) \right] \right)^n.$$

It therefore suffices to compute the characteristic function of one increment. Since $\eta_1 = \pm e_r$ with probability $\frac{1}{2d}$ for each $1 \leq r \leq d$, we obtain

$$\mathbb{E} \left[\exp \left(i \left\langle t, \frac{\eta_1}{\sqrt{n}} \right\rangle \right) \right] = \frac{1}{2d} \sum_{r=1}^d \exp \left(i \frac{t_r}{\sqrt{n}} \right) + \frac{1}{2d} \sum_{r=1}^d \exp \left(-i \frac{t_r}{\sqrt{n}} \right).$$

Using

$$\frac{e^{ix} + e^{-ix}}{2} = \cos x,$$

this simplifies to

$$\mathbb{E} \left[\exp \left(i \left\langle t, \frac{\eta_1}{\sqrt{n}} \right\rangle \right) \right] = \frac{1}{d} \sum_{r=1}^d \cos \left(\frac{t_r}{\sqrt{n}} \right).$$

Hence

$$\varphi_n(t) = \left(\frac{1}{d} \sum_{r=1}^d \cos \left(\frac{t_r}{\sqrt{n}} \right) \right)^n.$$

We now use the **Taylor expansion**,

$$\cos x = 1 - \frac{x^2}{2} + o(x^2), \quad x \rightarrow 0.$$

Since $t_r/\sqrt{n} \rightarrow 0$ for each r , it follows that

$$\cos \left(\frac{t_r}{\sqrt{n}} \right) = 1 - \frac{t_r^2}{2n} + o \left(\frac{1}{n} \right).$$

Summing over $r = 1, \dots, d$, we find

$$\frac{1}{d} \sum_{r=1}^d \cos \left(\frac{t_r}{\sqrt{n}} \right) = 1 - \frac{1}{2dn} \sum_{r=1}^d t_r^2 + o \left(\frac{1}{n} \right).$$

Since $\sum_{r=1}^d t_r^2 = \|t\|^2$, this becomes

$$\frac{1}{d} \sum_{r=1}^d \cos \left(\frac{t_r}{\sqrt{n}} \right) = 1 - \frac{\|t\|^2}{2dn} + o \left(\frac{1}{n} \right).$$

Therefore,

$$\varphi_n(t) = \left(1 - \frac{\|t\|^2}{2dn} + o\left(\frac{1}{n}\right)\right)^n.$$

Using the standard limit $(1 + \frac{a_n}{n})^n \rightarrow e^a$, when $a_n \rightarrow a$ we conclude that

$$\varphi_n(t) \rightarrow \exp\left(-\frac{\|t\|^2}{2d}\right), \quad n \rightarrow \infty.$$

But $\exp\left(-\frac{\|t\|^2}{2d}\right)$ is precisely the characteristic function of the **centered Gaussian law** $\mathcal{N}_d\left(0, \frac{1}{d}I_d\right)$. Hence, by **Lévy's Uniqueness Theorem** (Lemma A.4),

$$\frac{X_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}_d\left(0, \frac{1}{d}I_d\right),$$

which completes the proof. $\square$

As we will see later, the Elephant Random Walk has a dependent increment structure. For that reason, this approach is no longer available, and the proofs of Central Limit Theorems obtained in Chapter 4 require different methods.

2.5 RECURRENCE & TRANSIENCE

We close this chapter with a fundamental question about the Simple Random Walk: does it keep returning to where it started, or does it eventually wander off never to come back? These two behaviours are known as recurrence and transience, respectively.

To make this precise, for each site $x \in \mathbb{Z}^d$, define the total number of visits to x over the entire lifetime of the walk by

$$\mathcal{L}_\infty(x) := \lim_{n \rightarrow \infty} \sum_{k=0}^n \mathbb{1}_{\{X_k=x\}}.$$

The walk is said to be **recurrent** if $\mathbb{P}(\mathcal{L}_\infty(0) = \infty) = 1$, meaning it returns to the origin infinitely often almost surely, and **transient** if $\mathbb{P}(\mathcal{L}_\infty(0) < \infty) = 1$, meaning it visits the origin only finitely many times almost surely. Every random walk is either one or the other. The two cases are mutually exclusive.

In the proof below, we work with an equivalent criterion: the walk is recurrent if and only if $\sum_{n=0}^\infty p_n(0) = \infty$, where $p_n(0) := \mathbb{P}(X_n = 0)$, and transient if and only if this series converges. That this is equivalent to the occupation time definition above is established in [Nor97].

As Theorem 2.6 shows, the answer turns out to depend entirely on the dimension d . Inspiration for the proof of this theorem is taken from [Nor97].

Theorem 2.6 (Pólya's Theorem). *Let $(X_n)_{n \geq 0}$ be the d -dimensional Simple Random Walk on $\mathbb{Z}^d$. Then the walk is **recurrent** for $d = 1, 2$, and **transient** for $d \geq 3$.*

Proof Sketch. For $x \in \mathbb{Z}^d$, we make an estimation for $p_n(0)$.

Case $d = 1$. Since the walk moves in steps of size one, it can only return to the origin at even times. More precisely,

$$p_{2n}(0) = \binom{2n}{n} 2^{-2n}, \quad p_{2n+1}(0) = 0.$$

By **Stirling's formula** (Lemma A.2),

$$\binom{2n}{n} 2^{-2n} \sim \frac{1}{\sqrt{\pi n}}, \quad n \rightarrow \infty.$$

Hence

$$\sum_{n=0}^{\infty} p_n(0) = \sum_{n=0}^{\infty} p_{2n}(0) \sim \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \infty.$$

Therefore the one-dimensional SRW is **recurrent**.

Case $d = 2$. Write

$$X_n = (Z_n, Y_n),$$

and define

$$U_n := Z_n + Y_n, \quad V_n := Z_n - Y_n.$$

At each step, the pair (U_n, V_n) changes by one of

$$(1, 1), (1, -1), (-1, 1), (-1, -1),$$

each with probability $\frac{1}{4}$. Thus $(U_n)_{n \geq 0}$ and $(V_n)_{n \geq 0}$ are two independent one-dimensional Simple Random Walks. Since

$$Z_n = \frac{U_n + V_n}{2}, \quad Y_n = \frac{U_n - V_n}{2},$$

we have

$$X_n = 0 \iff U_n = 0 \text{ and } V_n = 0.$$

Therefore, for even times,

$$p_{2n}(0) = \mathbb{P}(U_{2n} = 0) \mathbb{P}(V_{2n} = 0) = \left(\binom{2n}{n} 2^{-2n} \right)^2,$$

while $p_{2n+1}(0) = 0$. Again by Stirling's formula (Lemma A.2),

$$p_{2n}(0) \sim \frac{1}{\pi n}, \quad n \rightarrow \infty.$$

Hence

$$\sum_{n=0}^{\infty} p_n(0) = \sum_{n=0}^{\infty} p_{2n}(0) \sim \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Thus the two-dimensional SRW is **recurrent**.

Case $d \geq 3$. Again, $p_{2n+1}(0) = 0$. To return to the origin after $2n$ steps, the walk must take a_i steps in the direction $+e_i$ and a_i steps in the direction $-e_i$, for $i = 1, \dots, d$, where

$$a_1 + \dots + a_d = n.$$

Hence

$$p_{2n}(0) = \frac{(2n)!}{(2d)^{2n}} \sum_{a_1 + \dots + a_d = n} \frac{1}{a_1!^2 \dots a_d!^2}.$$

Rewrite this as

$$p_{2n}(0) = \frac{(2n)!}{n!^2 (2d)^{2n}} \sum_{a_1 + \dots + a_d = n} \left(\frac{n!}{a_1! \dots a_d!} \right)^2.$$

Set

$$m(a_1, \dots, a_d) := \frac{n!}{a_1! \dots a_d!}.$$

Then

$$\sum_{a_1 + \dots + a_d = n} m(a_1, \dots, a_d) = d^n$$

by the **Multinomial Theorem**. Moreover, by **Stirling's formula** (Lemma A.2), the maximal multinomial coefficient satisfies

$$\max_{a_1 + \dots + a_d = n} m(a_1, \dots, a_d) \leq C \frac{d^n}{n^{(d-1)/2}}$$

for some constant $C > 0$ independent of n . Consequently,

$$\begin{aligned} \sum_{a_1 + \dots + a_d = n} m(a_1, \dots, a_d)^2 &\leq \left(\max_{a_1 + \dots + a_d = n} m(a_1, \dots, a_d) \right) \sum_{a_1 + \dots + a_d = n} m(a_1, \dots, a_d) \\ &\leq C \frac{d^{2n}}{n^{(d-1)/2}}. \end{aligned}$$

Using also

$$\frac{(2n)!}{n!^2 (2d)^{2n}} = \frac{\binom{2n}{n}}{(2d)^{2n}} \leq C' \frac{4^n}{\sqrt{n} (2d)^{2n}} = C' \frac{1}{d^{2n} \sqrt{n}},$$

we obtain

$$p_{2n}(0) \leq \frac{C''}{n^{d/2}}$$

for some constant $C'' > 0$. Since $d \geq 3$,

$$\sum_{n=1}^{\infty} p_{2n}(0) \leq C'' \sum_{n=1}^{\infty} \frac{1}{n^{d/2}} < \infty.$$

Thus

$$\sum_{n=0}^{\infty} p_n(0) < \infty,$$

and the walk is **transient**. □

Theorem 2.6 reveals that the **dimension** d alone completely determines whether the walk returns to its starting point infinitely often or escapes to infinity forever.

This section is perhaps best captured by a remark credited to Kakutani (U.C.L.A. colloquium talk): “*A drunk man will eventually find his way home, but a drunk bird may get lost forever.*”

THE ELEPHANT RANDOM WALK

In this chapter we introduce the main object of study: the **Elephant Random Walk**, a discrete-time stochastic process with complete memory. At each time step, the walker selects one of its previous increments uniformly at random and then either repeats it with probability p , or reverses its sign with probability $1 - p$. The name alludes to the well-known saying that elephants never forget. The Elephant Random Walk was introduced in 2004 by the physicists Schütz and Trimper [ST04], who discovered that complete memory gives rise to a phase transition in the large-scale behaviour of the walk, governed by the memory parameter p . We will refer to it as the **ERW** throughout this thesis.

Beyond its mathematical interest, the ERW serves as a simple model for real world processes in which the past shapes the future, with applications in markets where customers choose between competing products [SPR26], while closely related random walks are used to model how animals move using memory of where they have been [FMK23]. This chapter develops the fundamental probabilistic properties of the ERW. After giving the formal definition and building intuition for the role of the parameters p and q , we derive exact formulas for the expected displacement and the second moment, and analyze their asymptotic behaviour. We close the chapter by examining the **correlation** structure of the increments. These results form the analytical foundation on which the limit theorems of [Chapter 4](#) and the almost sure results of [Chapter 5](#) are built.

3.1 DEFINITION

We make this notion precise by providing a formal definition. Although the ERW can be defined more generally on $\mathbb{Z}^d$ for arbitrary **dimension** d , this thesis restricts attention to the one-dimensional case $d = 1$. Moreover, without loss of generality, we take the **origin** as the starting point of the walk.

The formal definition of the **Elephant Random Walk** is as follows:

Definition 3.1 (The Elephant Random Walk). Fix parameters $q \in [0, 1]$ and $p \in (0, 1)$. Let $(\xi_n)_{n \geq 1} \in \{-1, +1\}^{\mathbb{N}}$ be the **increments** and define

$$S_0 = 0, \quad S_n = \sum_{i=1}^n \xi_i \quad (n \geq 1).$$

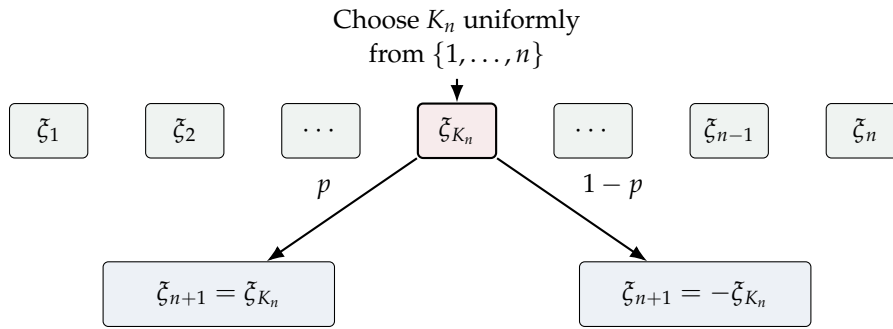
The first increment ξ_1 is given by:

$$\mathbb{P}(\xi_1 = +1) = q, \quad \mathbb{P}(\xi_1 = -1) = 1 - q.$$

For each $n \geq 1$, conditional on $\xi_1, \dots, \xi_n$, pick an index K_n uniformly from $\{1, \dots, n\}$ and set

$$\xi_{n+1} = \begin{cases} \xi_{K_n}, & \text{with probability } p, \\ -\xi_{K_n}, & \text{with probability } 1 - p. \end{cases}$$

The construction of the n -th increment, for $n \geq 1$, is illustrated schematically in the figure below:



Before entering into the analysis of the ERW, we develop some intuition for how p and q shape the behaviour of $(S_n)_{n \geq 0}$. To illustrate the role of p , the two simulations in [Figure 3.1](#) below contrast typical sample paths for two representative values.

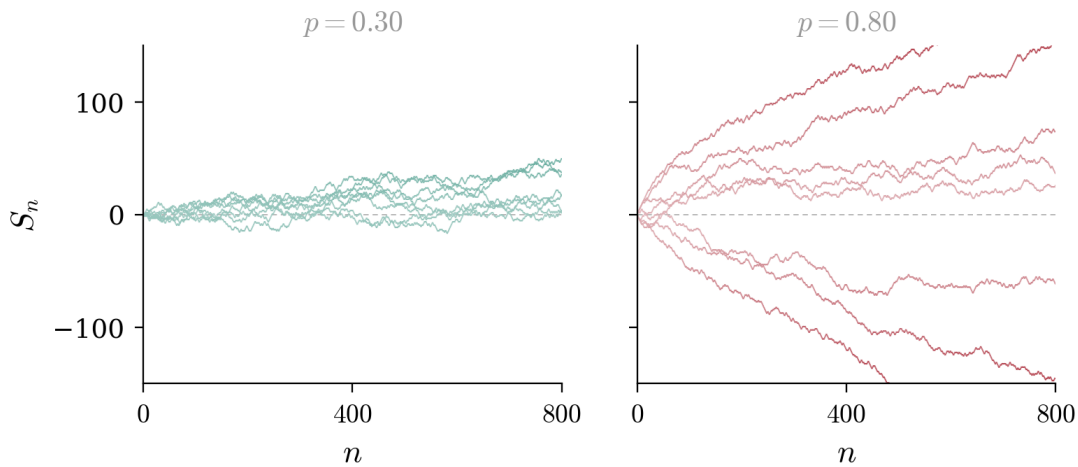


Figure 3.1: Sample paths of the ERW with $q = \frac{1}{2}$, $n = 800$ and $p \in \{0.30, 0.80\}$.

The influence of the initial bias parameter $q \in [0, 1]$ and the memory parameter $p \in (0, 1)$ are fairly intuitive.

On the one hand, the **initial bias parameter** q directly affects only the distribution of the first step. Nevertheless, since the future evolution of the walk depends on its entire history of increments, the process $(S_n)_{n \geq 0}$ carries a dependence on q for all $n \geq 1$.

On the other hand, the **memory parameter** p controls whether $(S_n)_{n \geq 0}$ tends to repeat or reverse its past increments. At time $n \geq 1$, the ERW samples one of its previous increments uniformly at random and either repeats it (with probability p) or flips its sign (with probability $1 - p$). When $p > 1/2$, repetitions of past movements are more likely than reversals, meaning the walk tends to persist in the same direction for longer periods. When $p < 1/2$, sign flips dominate, producing a walk that tends to move in the opposite direction of its past increments.

Let us consider some edge cases for the parameter p :

- When $p = \frac{1}{2}$, the memory mechanism becomes neutral: at each time $n \geq 1$, the increment ξ_{n+1} is equally likely to be $+1$ or -1 , regardless of the past. Hence the increments $(\xi_n)_{n \geq 2}$ are **i.i.d. Rademacher random variables**, and after the first step, the ERW evolves as a **simple symmetric random walk** started from the random position $S_1 = \xi_1$. In particular, if additionally $q = \frac{1}{2}$, then the entire process $(S_n)_{n \geq 0}$ is exactly a simple symmetric random walk started at the origin. This case is discussed extensively in [Chapter 2](#).
- The cases $p \in \{0, 1\}$ could in principle be included in the definition. When $p = 1$, the walk is entirely determined by its first increment ξ_1 : every subsequent step repeats the same sign, rendering the walk essentially **deterministic**. When $p = 0$, the walk is maximally **anti-persistent**: at each step it selects a past increment uniformly at random and moves in the opposite direction. We exclude these boundary values from the definition, as they give rise to a number of tedious edge cases in many of the proofs that follow.

In summary, q determines the initial bias of the walk, whereas p dictates whether the walk tends to follow its history ($p > 1/2$), balance it ($p = 1/2$), or reverse it ($p < 1/2$).

Throughout the rest of the thesis, we use the natural filtration

$$\mathcal{F}_0 := \{\emptyset, \Omega\}, \quad \mathcal{F}_n := \sigma(\xi_1, \dots, \xi_n), \quad n \geq 1.$$

Thus, $\mathcal{F}_n$ represents all information contained in the path up to time n .

3.2 EXPECTED DISPLACEMENT

Now that we have established the formal definition of the ERW and developed an understanding of how the initial bias and memory parameters influence its behaviour, we proceed to analyze its probabilistic properties. The first property we examine is the **conditional expectation** of the processes $(\xi_n)_{n \geq 1}$ and $(S_n)_{n \geq 0}$.

We first analyze the expected value of ξ_{n+1} for some $n \geq 1$, conditioned on its natural filtration $\mathcal{F}_n$. Recall that we fixed the **filtration** $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ throughout the thesis. However, the uniformly chosen index K remains random. Conditioning additionally on $K_n = k$, we obtain

$$\begin{aligned} \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n, K_n = k] &= p\xi_k + (1-p)(-\xi_k) \\ &= (2p-1)\xi_k. \end{aligned}$$

Applying the **Law of Total Expectation** (1.1.4), we find

$$\begin{aligned} \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] &= \sum_{k=1}^n \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n, K_n = k] \mathbb{P}(K_n = k \mid \mathcal{F}_n) \\ &= \frac{(2p-1)}{n} \sum_{k=1}^n \xi_k \\ &= \frac{2p-1}{n} S_n \end{aligned}$$

Thus we have established that, conditioned on the full history of past increments, the expected value of ξ_{n+1} depends on the displacement of $(S_n)_{n \geq 0}$ at time n and the parameter p , and is given by the formula

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = \frac{(2p-1)}{n} S_n. \quad (3.2.1)$$

Recall that by definition $S_{n+1} = S_n + \xi_{n+1}$, so by the linearity **Property 1.1.1** of the conditional expectation and **Property 1.1.2** one has, for $n \geq 1$,

$$\begin{aligned} \mathbb{E}[S_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[S_n \mid \mathcal{F}_n] + \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] \\ &= S_n + \frac{2p-1}{n} S_n \\ &= \left(1 + \frac{2p-1}{n}\right) S_n. \end{aligned} \quad (3.2.2)$$

Here we used the fact that S_n is completely determined when conditioned on the filtration $\mathcal{F}_n$. Note that this shows $(S_n)_{n \geq 0}$ is not a **martingale**, since $1 + \frac{2p-1}{n} \neq 1$ except when $p = \frac{1}{2}$, in which case the ERW reduces to a SRW.

Following Schütz and Trimper [ST04], we now derive a general formula for the unconditional expected value of S_n .

Theorem 3.2 (Expected Value). *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with $(p, q) \in (0, 1) \times [0, 1]$. Define $\alpha := 2p - 1$ and $\beta := 2q - 1$. Then, for all $n \geq 1$,*

$$\mathbb{E}[S_n] = \beta \frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1)\Gamma(n)}.$$

Proof. Taking expectations on both sides of (3.2.2) and using the law of total expectation (1.1.4) yields

$$\mathbb{E}[\mathbb{E}[S_{n+1} \mid \mathcal{F}_n]] = \mathbb{E}\left[\left(1 + \frac{2p-1}{n}\right) S_n\right].$$

Since $1 + \frac{2p-1}{n}$ is **deterministic**, this gives the recursion

$$\mathbb{E}[S_{n+1}] = \left(1 + \frac{2p-1}{n}\right) \mathbb{E}[S_n].$$

Iterating this recursion from step 1 to step $n - 1$ gives

$$\begin{aligned} \mathbb{E}[S_n] &= \prod_{k=1}^{n-1} \left(1 + \frac{2p-1}{k}\right) \mathbb{E}[S_1] \\ &= \prod_{k=1}^{n-1} \left(\frac{k+\alpha}{k}\right) \mathbb{E}[S_1]. \end{aligned}$$

Expressing the numerator in terms of the **Gamma function** (Property 1.4.2) and substituting $\mathbb{E}[S_1] = \beta$, we obtain

$$\mathbb{E}[S_n] = \beta \frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1)\Gamma(n)},$$

which is what we set out to show. $\square$

Throughout the remainder of this thesis, we adopt the shorthand $\alpha := 2p - 1$ and $\beta := 2q - 1$, as these quantities will appear repeatedly. Note that $\alpha > 0$ if and only if $p > 1/2$, and $\beta > 0$ if and only if $q > 1/2$, a correspondence that will prove useful in later calculations.

In view of Theorem 3.2, we can make the following observation.

Corollary 3.3. *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk, with the corresponding increment process $(\xi_n)_{n \geq 1}$. For $n \geq 1$,*

$$\mathbb{E}[\xi_{n+1}] = \frac{\alpha \beta}{n} \frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1)\Gamma(n)}. \quad (3.2.3)$$

Proof. Taking expectations on both sides of (3.2.1) and using the fact that $\frac{\alpha}{n}$ is deterministic, we obtain

$$\mathbb{E}[\xi_{n+1}] = \frac{\alpha}{n} \mathbb{E}[S_n].$$

Substituting the expression for $\mathbb{E}[S_n]$ from Theorem 3.2 yields

$$\mathbb{E}[\xi_{n+1}] = \frac{\alpha \beta}{n} \frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1) \Gamma(n)},$$

as claimed. $\square$

An interesting detail not captured by Corollary 3.3 is $\mathbb{E}[\xi_1] = \beta$.

Applying the asymptotic estimate (Lemma 1.4) to Theorem 3.2, we derive the **asymptotic equivalence**

$$\mathbb{E}[S_n] \sim \beta \frac{n^\alpha}{\Gamma(\alpha + 1)}. \quad (3.2.4)$$

Theorem 3.2 allows us to analyze the asymptotic behaviour of the expected displacement as a function of the parameter p . We distinguish three cases:

First, consider the case $p < 1/2$. In this regime $\alpha < 0$, so $n^\alpha \rightarrow 0$ as $n \rightarrow \infty$. Since $\frac{\beta}{\Gamma(\alpha+1)}$ is a constant, it follows that $\mathbb{E}[S_n] \rightarrow 0$. This reflects the anti-persistent nature of the walk in this regime: reversals of past steps are more likely than repetitions, causing the walk to remain close to its starting point over time. A simulation illustrating this behaviour is shown in Figure 3.2:

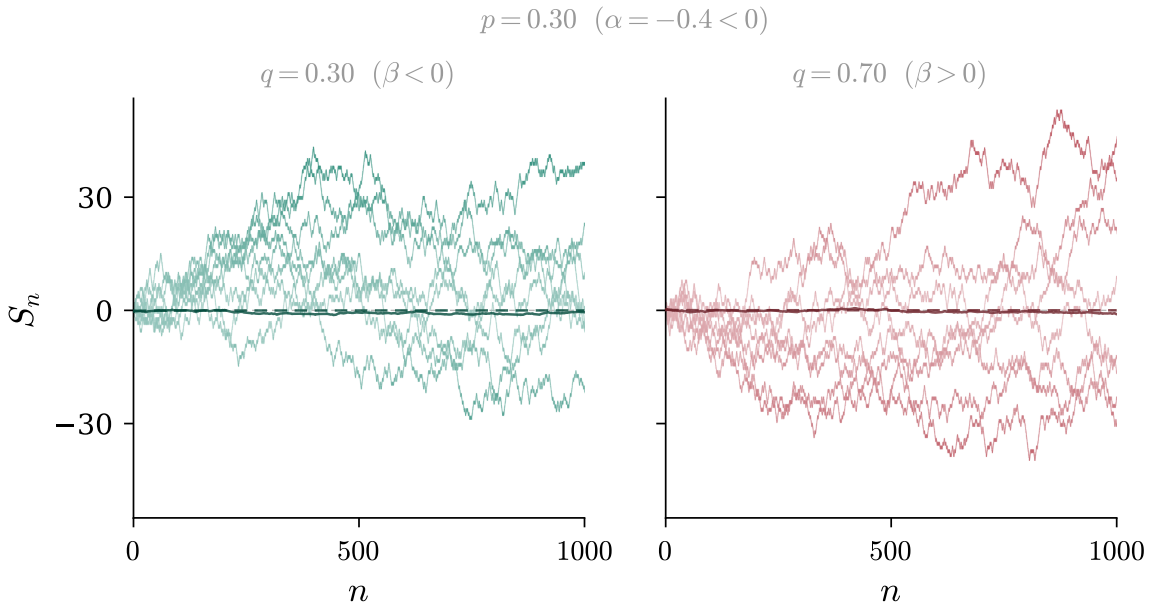


Figure 3.2: Sample paths of the ERW with $p = 0.30$, $n = 1000$, for $q \in \{0.30, 0.70\}$. Solid and dashed lines show the empirical mean and asymptotic $\beta n^\alpha / \Gamma(\alpha + 1)$; both converge to zero as $\alpha < 0$.

Second, when $p = \frac{1}{2}$, we have $\mathbb{E}[S_n] = \beta$ for all n , which is consistent with the results of [Chapter 2](#).

Finally, consider the case $p > 1/2$. Here $0 < \alpha < 1$, so $n^\alpha \rightarrow \infty$. The asymptotic behaviour of the expected displacement then depends on q :

- When $q < 1/2$, we have $\beta < 0$, and so

$$\mathbb{E}[S_n] \sim \beta \frac{n^\alpha}{\Gamma(\alpha + 1)} \rightarrow -\infty, \quad \text{as } n \rightarrow \infty.$$

- When $q = 1/2$, we have $\beta = 0$, and thus

$$\mathbb{E}[S_n] \sim \beta \frac{n^\alpha}{\Gamma(\alpha + 1)} \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

- When $q > 1/2$, we have $\beta > 0$, and therefore

$$\mathbb{E}[S_n] \sim \beta \frac{n^\alpha}{\Gamma(\alpha + 1)} \rightarrow \infty, \quad \text{as } n \rightarrow \infty.$$

This behaviour is illustrated in [Figure 3.3](#).

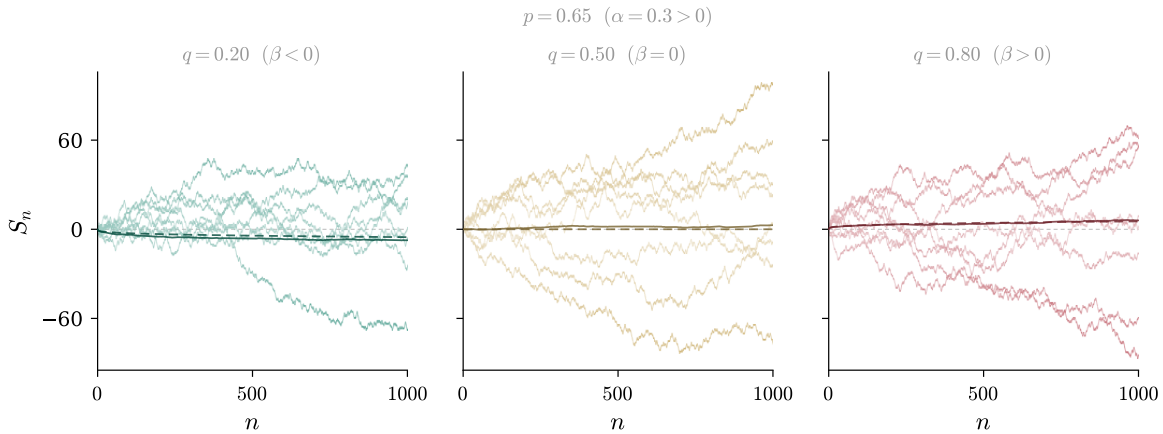


Figure 3.3: Simulated ERW paths with $p = 0.65$ for $q \in \{0.20, 0.50, 0.80\}$. The solid and dashed lines show the **empirical mean** and $\beta n^\alpha / \Gamma(\alpha + 1)$, respectively.

[Figure 3.3](#) highlights a key feature of the ERW in the persistent regime $p > \frac{1}{2}$: the influence of the initial bias does not disappear over time. Instead, because the walk repeatedly samples from its own history, the sign of $\beta = 2q - 1$ is reinforced, and the expected displacement grows on the polynomial scale n^α . This already highlights an important difference from the SRW, where the effect of the initial bias does not continue to grow over time.

3.3 SECOND MOMENT

We have now derived the expected displacement of the ERW and described its asymptotic behaviour. We now proceed to analyze the **second moment** of $(S_n)_{n \geq 0}$.

We begin by computing the **conditional expectation** of the second moment of $(S_n)_{n \geq 0}$. Recall that by definition we have

$$(S_{n+1})^2 = (S_n + \xi_{n+1})^2 = (S_n)^2 + 2S_n\xi_{n+1} + (\xi_{n+1})^2. \quad (3.3.1)$$

Since $\xi_n \in \{-1, 1\}$, we have $\xi_{n+1}^2 = 1$. Taking the conditional expectation and applying linearity and [Property 1.1.2](#) yields

$$\begin{aligned} \mathbb{E}[(S_{n+1})^2 \mid \mathcal{F}_n] &= \mathbb{E}[(S_n)^2 \mid \mathcal{F}_n] + \mathbb{E}[2S_n\xi_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[(\xi_{n+1})^2 \mid \mathcal{F}_n] \\ &= S_n^2 + 2S_n\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] + 1 \\ &= S_n^2 + 2\left(\frac{\alpha}{n}\right)S_n^2 + 1 \\ &= S_n^2\left(1 + \frac{2\alpha}{n}\right) + 1. \end{aligned}$$

As a result,

$$\mathbb{E}[(S_{n+1})^2 \mid \mathcal{F}_n] = S_n^2\left(1 + \frac{2\alpha}{n}\right) + 1.$$

Taking unconditional expectations on both sides of (3.3.1) and applying linearity yields

$$\mathbb{E}[S_{n+1}^2] = \mathbb{E}[S_n^2] + 2\mathbb{E}[S_n\xi_{n+1}] + 1.$$

By the **Law of Total Expectation** and [Property 1.1.2](#) we obtain

$$\begin{aligned} \mathbb{E}[S_n\xi_{n+1}] &= \mathbb{E}[\mathbb{E}[S_n\xi_{n+1} \mid \mathcal{F}_n]] \\ &= \mathbb{E}[S_n\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n]] \\ &= \mathbb{E}\left[S_n \cdot \frac{\alpha}{n}S_n\right] \\ &= \frac{\alpha}{n}\mathbb{E}[S_n^2]. \end{aligned}$$

Substituting this, we obtain the recursive formula

$$\mathbb{E}[S_{n+1}^2] = 1 + \left(1 + \frac{2\alpha}{n}\right)\mathbb{E}[S_n^2]. \quad (3.3.2)$$

Throughout, we use the fact that the **reciprocal Gamma function** $1/\Gamma(\cdot)$ extends to an **entire function** with zeros at $0, -1, -2, \dots$ [[Pro19](#)]. In particular,

$$\frac{1}{\Gamma(-1)} = 0, \quad \frac{1}{\Gamma(0)} = 0.$$

Following Schütz and Trimper [ST04], we are now ready to state the theorem concerning the second moment of the Elephant Random Walk.

Theorem 3.4. *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with $(p, q) \in (0, 1) \times [0, 1]$. Then $\mathbb{E}[S_1^2] = 1$. Moreover, for $n \geq 2$ and $p \neq \frac{3}{4}$,*

$$\mathbb{E}[S_n^2] = \frac{n}{2\alpha - 1} \left(\frac{\Gamma(n + 2\alpha)}{\Gamma(n + 1)\Gamma(2\alpha)} - 1 \right).$$

If $p = 3/4$, then

$$\mathbb{E}[S_n^2] = n \sum_{k=1}^n \frac{1}{k} = n H_n,$$

*where H_n denotes the **Harmonic number**.*

Proof. We treat the two cases separately. First assume $p \neq \frac{3}{4}$. For brevity, set $m_n := \mathbb{E}[S_n^2]$ and define

$$\phi_n := \frac{\Gamma(n + 2\alpha)}{\Gamma(n + 1)\Gamma(2\alpha)}.$$

Using $\Gamma(1 + 2\alpha) = 2\alpha \Gamma(2\alpha)$, one computes directly that $\phi_1 = 2\alpha$. We claim that

$$m_n = \frac{n}{2\alpha - 1} (\phi_n - 1),$$

and verify this by induction. At $n = 1$, the right-hand side equals $\frac{1}{2\alpha - 1} (2\alpha - 1) = 1 = \mathbb{E}[S_1^2]$, so the base case holds. Now suppose the formula holds for some $n \geq 1$. The recurrence (3.3.2) yields

$$m_{n+1} = 1 + \left(1 + \frac{2\alpha}{n} \right) m_n.$$

Substituting the inductive hypothesis and applying the Gamma recursion $\phi_{n+1} = \phi_n \cdot \frac{n+2\alpha}{n+1}$, or equivalently $(n+1)\phi_{n+1} = (n+2\alpha)\phi_n$, we obtain

$$m_{n+1} = \frac{n+2\alpha}{2\alpha - 1} (\phi_n - 1) + 1 = \frac{n+2\alpha}{2\alpha - 1} \phi_n - \frac{n+1}{2\alpha - 1}.$$

Replacing $(n+2\alpha)\phi_n = (n+1)\phi_{n+1}$ in the first term yields

$$m_{n+1} = \frac{n+1}{2\alpha - 1} \phi_{n+1} - \frac{n+1}{2\alpha - 1} = \frac{n+1}{2\alpha - 1} (\phi_{n+1} - 1),$$

completing the induction. Assume $p = \frac{3}{4}$. Here $\alpha = \frac{1}{2}$, and the recurrence (3.3.2) reduces to

$$m_{n+1} = \frac{n+1}{n} m_n + 1.$$

Setting $u_n := m_n/n$ and dividing both sides by $n+1$ yields

$$u_{n+1} = u_n + \frac{1}{n+1}.$$

Since $u_1 = 1$, iterating this recursion yields $u_n = \sum_{k=1}^n \frac{1}{k}$, and the result $m_n = n u_n = n H_n$ follows immediately. $\square$

Having established an exact expression for the second moment in [Theorem 3.4](#), we now turn to its asymptotic behaviour via [Lemma 1.4](#). As we shall see, the behaviour depends critically on the memory parameter p , and we distinguish three cases.

When $p < 3/4$, one has $2\alpha - 1 < 0$, so that $n^{2\alpha-1} \rightarrow 0$ as $n \rightarrow \infty$. Applying [Lemma 1.4](#) to [Theorem 3.4](#) therefore gives

$$\mathbb{E}[S_n^2] \sim \frac{n}{3-4p}, \quad (p < 3/4).$$

A simulation consistent with this finding is shown in [Figure 3.4](#):

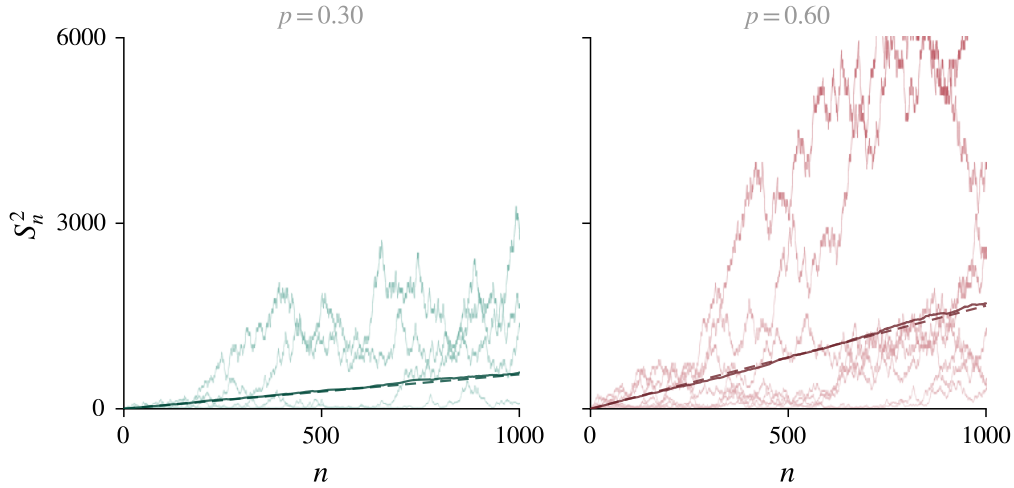


Figure 3.4: Sample paths of S_n^2 with $q = \frac{1}{2}$, $n = 1000$. Solid and dashed lines show the empirical mean $\hat{\mathbb{E}}[S_n^2]$ and $n/(3-4p)$.

At the boundary value $p = 3/4$, one has $\alpha = 1/2$, and [Theorem 3.4](#) combined with [\(1.3.1\)](#) directly yields

$$\mathbb{E}[S_n^2] = n H_n \sim n \log n, \quad (p = 3/4),$$

where H_n denotes the harmonic number. Moreover, [Figure 3.5](#) provides a visual confirmation of this result:

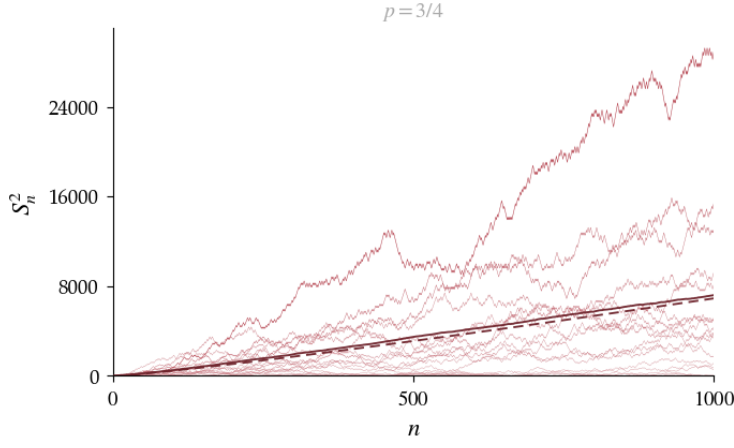


Figure 3.5: Sample paths of S_n^2 for the critical ERW with $q = \frac{1}{2}$, $n = 1000$. Solid and dashed lines show the empirical mean $\widehat{\mathbb{E}}[S_n^2]$ and $n \log n$.

Finally, when $p > 3/4$, one has $2\alpha - 1 > 0$, and applying [Lemma 1.4](#) gives

$$\mathbb{E}[S_n^2] \sim \frac{n^{4p-2}}{(4p-3)\Gamma(4p-2)}, \quad (p > 3/4).$$

The above is corroborated by the simulation in [Figure 3.6](#):

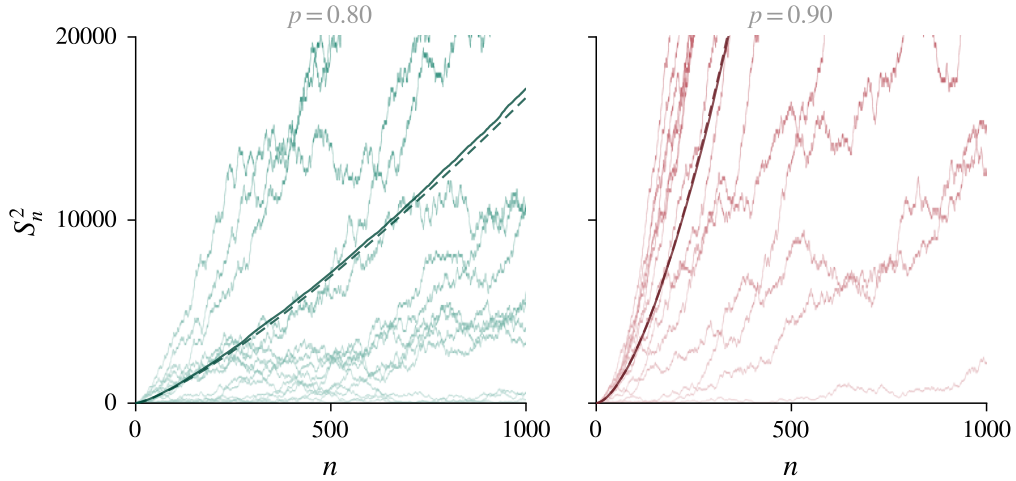


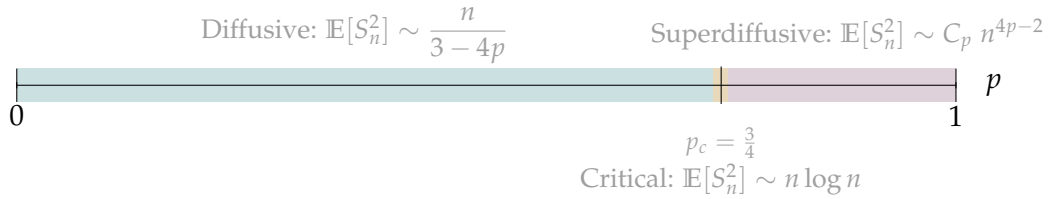
Figure 3.6: Sample paths of S_n^2 for the superdiffusive ERW with $q = \frac{1}{2}$, $n = 1000$. Solid and dashed lines show the empirical mean $\widehat{\mathbb{E}}[S_n^2]$ and $\frac{n^{4p-2}}{(4p-3)\Gamma(4p-2)}$.

These three cases reveal that the asymptotic growth of $\mathbb{E}[S_n^2]$ undergoes a qualitative change at the value $p_c := 3/4$, which we call the **critical point**. This change is not merely quantitative: below p_c the second moment grows linearly in n , at p_c a logarithmic correction appears, and above p_c the growth is polynomial of degree $4p - 2 > 1$.

This phase transition at p_c divides the parameter space into three distinct regimes, each characterized by a different type of asymptotic behaviour.

- The interval $p \in (0, \frac{3}{4})$ is called the **diffusive regime**. In this regime the second moment grows linearly, $\mathbb{E}[S_n^2] \sim \frac{n}{3-4p}$, which is the same order of growth as for the Simple Random Walk. The walk is therefore said to spread diffusively.
- The value $p = \frac{3}{4}$ is called the **critical regime**, as it marks the boundary between diffusive and superdiffusive behaviour. The second moment $\mathbb{E}[S_n^2] \sim n \log n$ grows slightly faster than linear, with the logarithmic factor signaling that the walk sits precisely at the transition between the two regimes.
- The interval $p \in (\frac{3}{4}, 1)$ is called the **superdiffusive regime**. A process is called superdiffusive when its variance grows faster than that of a Simple Random Walk, which scales as n . Here $\mathbb{E}[S_n^2] \sim n^{4p-2}$, and since $4p-2 > 1$ throughout this regime, the second moment indeed grows faster than linearly, justifying the name.

To summarize these three cases, we present the following schematic overview. Here we denote $C_p := \frac{1}{(4p-3)\Gamma(4p-2)}$.



Throughout the remainder of this thesis, the regime of p will repeatedly prove to be the crucial factor governing the behaviour of the Elephant Random Walk.

3.4 CORRELATION STRUCTURE OF THE INCREMENTS

With the first and second moments in hand, we now turn to a more refined property of the ERW: how are the increments $(\xi_n)_{n \geq 1}$ correlated with one another? Recall that the covariance of two random variables X and Y is

$$\text{Cov}(X, Y) := \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

For clarity, we restrict to $q = \frac{1}{2}$ in this section. It forces $\beta = 0$, so that by [Corollary 3.3](#), $\mathbb{E}[\xi_n] = 0$ for all $n \geq 1$, and the covariance reduces to

$$\text{Cov}(\xi_n, \xi_{n+k}) = \mathbb{E}[\xi_n \xi_{n+k}], \quad n \geq 1, k \geq 1. \quad (3.4.1)$$

This keeps the calculations clean without losing any essential structure. Following Coletti, Gava, and Schütz [CGS17a], the main result of this section gives an explicit formula for $\mathbb{E}[\xi_n \xi_{n+k}]$ in terms of Gamma functions.

We begin by introducing the auxiliary quantity, defined for $n \geq 2$, $k \geq 1$ as

$$\rho_k := (n + k - 1) \mathbb{E}[\xi_{n+k} S_n]. \quad (3.4.2)$$

Although this expression may appear random at first, it will prove to be a useful construction as the argument proceeds. Since $n \leq n + k$, the random variable S_n is $\mathcal{F}_{n+k}$ -measurable. Applying the tower property of conditional expectation yields

$$\mathbb{E}[\xi_{n+k+1} S_n] = \mathbb{E}[\mathbb{E}[\xi_{n+k+1} S_n \mid \mathcal{F}_{n+k}]] = \mathbb{E}[S_n \mathbb{E}[\xi_{n+k+1} \mid \mathcal{F}_{n+k}]].$$

Combining this with (3.2.1) yields

$$\rho_{k+1} = (n + k) \frac{\alpha}{n + k} \mathbb{E}[S_{n+k} S_n] = \alpha \mathbb{E}[S_{n+k} S_n].$$

Expanding $S_{n+k} = S_{n+k-1} + \xi_{n+k}$ then gives

$$\rho_{k+1} = \alpha \mathbb{E}[S_{n+k-1} S_n] + \alpha \mathbb{E}[\xi_{n+k} S_n]. \quad (3.4.3)$$

Applying the same reasoning to the definition of ρ_k gives

$$\rho_k = (n + k - 1) \mathbb{E}[\xi_{n+k} S_n] = \alpha \mathbb{E}[S_{n+k-1} S_n].$$

Substituting this into the expression (3.4.3), we obtain

$$\rho_{k+1} = \rho_k + \alpha \mathbb{E}[\xi_{n+k} S_n] = \rho_k + \frac{\alpha}{n + k - 1} \rho_k,$$

which simplifies to the recurrence

$$\rho_{k+1} = \left(1 + \frac{\alpha}{n + k - 1}\right) \rho_k.$$

It remains to determine the initial value ρ_1 . By definition,

$$\rho_1 = n \mathbb{E}[\xi_{n+1} S_n], \quad (3.4.4)$$

and conditioning on $\mathcal{F}_n$ yields $\rho_1 = \alpha \mathbb{E}[S_n^2]$. Combining these results, we arrive at the recurrence relation

$$\rho_1 = \alpha \mathbb{E}[S_n^2], \quad \rho_{k+1} = \left(1 + \frac{\alpha}{n + k - 1}\right) \rho_k. \quad (3.4.5)$$

The general solution to this recurrence is given by

$$\rho_k = \rho_1 \prod_{j=1}^{k-1} \left(1 + \frac{\alpha}{n + j - 1}\right) = \rho_1 \prod_{j=1}^{k-1} \left(\frac{n + j + 2p - 2}{n + j - 1}\right).$$

The two products can be evaluated in closed form using the Gamma function:

$$\prod_{j=1}^{k-1} (n+j+2p-2) = \frac{\Gamma(n+k+2p-2)}{\Gamma(n+2p-1)}, \quad \prod_{j=1}^{k-1} (n+j-1) = \frac{(n+k-2)!}{(n-1)!}.$$

Substituting these expressions yields

$$\rho_k = \frac{\Gamma(n+k+2p-2) (n-1)!}{\Gamma(n+2p-1) (n+k-2)!} \rho_1.$$

Recalling that $\rho_k = (n+k-1) \mathbb{E}[\xi_{n+k} S_n]$ by definition, we obtain

$$\mathbb{E}[\xi_{n+k} S_n] = \frac{1}{n+k-1} \cdot \frac{\Gamma(n+k+2p-2) (n-1)!}{\Gamma(n+2p-1) (n+k-2)!} \rho_1.$$

Using the identity $(n+k-1)(n+k-2)! = (n+k-1)!$, this simplifies to

$$\mathbb{E}[\xi_{n+k} S_n] = \frac{\Gamma(n+k+2p-2) (n-1)!}{\Gamma(n+2p-1) (n+k-1)!} \rho_1.$$

Substituting (3.4.5), we find

$$\mathbb{E}[\xi_{n+k} S_n] = \alpha \frac{\Gamma(n+k+2p-2) (n-1)!}{\Gamma(n+2p-1) (n+k-1)!} \mathbb{E}[S_n^2]. \quad (3.4.6)$$

To compute the increment correlation $\mathbb{E}[\xi_{n+k} \xi_n]$, we use the decomposition $\xi_n = S_n - S_{n-1}$, which gives

$$\mathbb{E}[\xi_{n+k} \xi_n] = \mathbb{E}[\xi_{n+k} S_n] - \mathbb{E}[\xi_{n+k} S_{n-1}].$$

The first term is given by (3.4.6). The second term follows by applying (3.4.6) with k replaced by $k+1$ and n replaced by $n-1$, yielding

$$\mathbb{E}[\xi_{n+k} S_{n-1}] = \alpha \frac{\Gamma(n+k+2p-2) (n-2)!}{\Gamma(n+2p-2) (n+k-1)!} \mathbb{E}[S_{n-1}^2].$$

Observe that

$$\frac{(n-2)!}{\Gamma(2p+n-2)} = \frac{(n-1)!}{\Gamma(2p+n-1)} \cdot \frac{n+2p-2}{n-1} = \frac{(n-1)!}{\Gamma(2p+n-1)} \left(1 + \frac{2p-1}{n-1}\right),$$

so that $\mathbb{E}[\xi_{n+k} S_{n-1}]$ can be rewritten as

$$\mathbb{E}[\xi_{n+k} S_{n-1}] = \alpha \frac{\Gamma(n+k+2p-2) (n-1)!}{\Gamma(n+2p-1) (n+k-1)!} \left(1 + \frac{2p-1}{n-1}\right) \mathbb{E}[S_{n-1}^2].$$

Combining both terms, we arrive at

$$\mathbb{E}[\xi_{n+k} \xi_n] = \alpha \frac{\Gamma(n+k+2p-2) (n-1)!}{\Gamma(n+2p-1) (n+k-1)!} \left(\mathbb{E}[S_n^2] - \left(1 + \frac{2p-1}{n-1}\right) \mathbb{E}[S_{n-1}^2] \right).$$

Writing $\alpha(n-1)! = \frac{\alpha}{n} n!$, the expression becomes

$$\mathbb{E}[\xi_{n+k}\xi_n] = \frac{\Gamma(n+k+2p-2)n!}{\Gamma(n+2p-1)(n+k-1)!} F(n),$$

where we define

$$F(n) = \frac{\alpha}{n} \left(\mathbb{E}[S_n^2] - \left(1 + \frac{2p-1}{n-1}\right) \mathbb{E}[S_{n-1}^2] \right).$$

Recalling the explicit formula for $\mathbb{E}[S_n^2]$ from [Theorem 3.4](#), we can express $F(n)$ purely in terms of p and n . The critical case $p = \frac{3}{4}$ will be treated separately. For $p \neq \frac{3}{4}$, substituting the second moment formula yields

$$F(n) = \frac{2p-1}{3-4p} \left(\frac{2(1-p)}{n} - \frac{(2p-1)\Gamma(n+4p-3)}{\Gamma(4p-2)n!} \right). \quad (3.4.7)$$

In the critical case $p = \frac{3}{4}$, an analogous substitution using [Theorem 3.4](#) gives

$$F_{p_c}(n) = \frac{1}{2n} \left(1 + \frac{1}{2} H_{n-1} \right), \quad (3.4.8)$$

where H_{n-1} denotes the $(n-1)$ -th **Harmonic number**.

We have thus established an explicit formula for the covariance of the increments. To summarize: Let $n \geq 2$ and $k \geq 1$. For $q = \frac{1}{2}$ and $p \neq \frac{3}{4}$,

$$\text{Cov}(\xi_n, \xi_{n+k}) = \mathbb{E}[\xi_{n+k}\xi_n] = \frac{\Gamma(n+k+2p-2)n!}{\Gamma(n+2p-1)(n+k-1)!} F(n),$$

where $F(n)$ is as defined in (3.4.7). In the critical case $p = \frac{3}{4}$, the same formula holds with $F(n)$ replaced by $F_{p_c}(n)$, as defined in (3.4.8).

Coletti, Gava, and Schütz [[CGS17a](#)] further describe, using Stirling's formula ([Lemma A.2](#)), the asymptotic behaviour of the function $F(n)$. They obtain

$$F(n) \sim \begin{cases} \frac{2(2p-1)(1-p)}{3-4p} n^{-1}, & p < \frac{3}{4}, \\ \frac{\log n}{4n}, & p = \frac{3}{4}, \\ \frac{(2p-1)^2}{(4p-3)\Gamma(4p-2)} n^{-4(1-p)}, & p > \frac{3}{4}. \end{cases}$$

This shows that the decay of the increment correlations changes at the critical value $p_c = \frac{3}{4}$. In the diffusive regime, $F(n)$ decays of order n^{-1} . At the critical point, a logarithmic correction appears. In the superdiffusive regime, the decay is slower, since $4(1-p) < 1$. Hence the correlations between increments persist more strongly when $p > \frac{3}{4}$.

LIMIT THEOREMS FOR THE ERW

A natural question is how the ERW behaves on large scales, that is, in the limit $n \rightarrow \infty$. This chapter investigates the distribution of the ERW and how it depends on the memory parameter p .

The main results of this chapter are a **Law of Large Numbers** (LLN) and **Central Limit Theorems** (CLT) for the ERW. The LLN tells us that $(S_n - \mathbb{E}[S_n])/n \rightarrow 0$ almost surely for any $q \in [0, 1]$ and $p \in (0, 1)$, confirming that the walk grows slower than linearly regardless of the memory parameter. The CLT, however, reveals a phase transition at the critical value $p_c = 3/4$. For $p \leq 3/4$, appropriately centered and rescaled, S_n converges to a standard normal distribution, though the correct normalization differs between the diffusive regime $p < 3/4$ and the critical regime $p = 3/4$. For $p > 3/4$, Gaussian behaviour vanishes and the rescaled walk converges almost surely to a non-degenerate, non-Gaussian random variable.

Taken together, these results show that the choice of p has profound consequences for the long term behaviour of the ERW.

4.1 LAW OF LARGE NUMBERS

The **Law of Large Numbers** (LLN) is one of the most fundamental results in probability theory, describing the long run average behaviour of a stochastic process. For the Elephant Random Walk, it tells us that despite the memory dependent structure, the walk still satisfies a LLN. Notably, and in contrast to the Central Limit Theorem, this result exhibits no phase transition at the critical value $p = \frac{3}{4}$. The same qualitative law holds uniformly across the entire parameter space

$$(p, q) \in (0, 1) \times [0, 1].$$

The results established in this section, [Theorem 4.1](#) and [Corollary 4.5](#) in particular, will be used repeatedly in the proofs of the limit theorems.

We now state the Law of Large Numbers for the ERW, the proof of which is due to Coletti, Gava, and Schütz [CGS17a].

Theorem 4.1 (The Law of Large Numbers). *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk, with $(p, q) \in (0, 1) \times [0, 1]$. Then*

$$\frac{S_n - \mathbb{E}[S_n]}{n} \xrightarrow{a.s.} 0, \quad \text{as } n \rightarrow \infty.$$

Prior to proving this theorem, we first establish a framework and derive several auxiliary lemmas that will play a key role in the proof. This framework will not only serve as the backbone of the proof of Theorem 4.1, but will also prove useful in the proof of the Central Limit Theorems for the ERW.

We start by defining the **deterministic** sequence $(\Lambda_n)_{n \geq 1}$ by

$$\Lambda_1 = 1, \quad \Lambda_n = \prod_{k=1}^{n-1} \left(1 + \frac{\alpha}{k}\right), \quad (n \geq 2). \quad (4.1.1)$$

Note that this sequence satisfies the recursive relation

$$\Lambda_{n+1} = \Lambda_n \left(1 + \frac{\alpha}{n}\right).$$

Moreover, since $p > 0$, by Lemma 1.3

$$\begin{aligned} \Lambda_n &= \prod_{k=1}^{n-1} \left(1 + \frac{\alpha}{k}\right) \\ &= \prod_{k=1}^{n-1} \left(\frac{k + \alpha}{k}\right) \\ &= \frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1)\Gamma(n)}. \end{aligned}$$

Furthermore, we define the stochastic process $(M_n)_{n \geq 1}$ by

$$M_n := \frac{S_n - \mathbb{E}[S_n]}{\Lambda_n}, \quad (n \geq 1). \quad (4.1.2)$$

We will show that $(M_n)_{n \geq 1}$ is a **martingale** with respect to the filtration $(\mathcal{F}_n)_{n \geq 0}$, as this will provide us with several useful properties. This proof follows the approach of Coletti, Gava, and Schütz [CGS17a].

Lemma 4.2. *Let $(M_n)_{n \geq 1}$ be the process defined in (4.1.2). Then $(M_n)_{n \geq 1}$ is a **martingale** with respect to the filtration $(\mathcal{F}_n)_{n \geq 0}$, that is,*

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n.$$

Proof. We begin by checking that M_n is integrable. As ξ_i takes values in $\{-1, 1\}$, both S_n and $\mathbb{E}[S_n]$ are bounded in absolute value by n , so $|M_n| \leq 2n/\Lambda_n$. In particular, $\mathbb{E}[|M_n|] < \infty$ for all $n \geq 1$. Recall that $S_{n+1} = S_n + \xi_{n+1}$ by definition. We rewrite M_{n+1} as

$$M_{n+1} = \frac{S_{n+1} - \mathbb{E}[S_{n+1}]}{\Lambda_{n+1}} = \frac{S_n - \mathbb{E}[S_n]}{\Lambda_{n+1}} + \frac{\xi_{n+1} - \mathbb{E}[\xi_{n+1}]}{\Lambda_{n+1}}.$$

As established in (3.2.1),

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = \frac{\alpha}{n} S_n.$$

By linearity of conditional expectation, we obtain

$$\begin{aligned} \mathbb{E}[M_{n+1} \mid \mathcal{F}_n] &= \mathbb{E} \left[\frac{S_n - \mathbb{E}[S_n]}{\Lambda_{n+1}} + \frac{\xi_{n+1} - \mathbb{E}[\xi_{n+1}]}{\Lambda_{n+1}} \mid \mathcal{F}_n \right] \\ &= \frac{1}{\Lambda_{n+1}} (\mathbb{E}[S_n - \mathbb{E}[S_n] \mid \mathcal{F}_n] + \mathbb{E}[\xi_{n+1} - \mathbb{E}[\xi_{n+1}] \mid \mathcal{F}_n]) \\ &= \frac{1}{\Lambda_{n+1}} (S_n - \mathbb{E}[\mathbb{E}[S_n \mid \mathcal{F}_n]] + \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] - \mathbb{E}[\mathbb{E}[\xi_{n+1}] \mid \mathcal{F}_n]) \\ &= \frac{1}{\Lambda_{n+1}} (S_n - \mathbb{E}[S_n]) + \frac{1}{\Lambda_{n+1}} \left(\frac{\alpha}{n} S_n - \frac{\alpha}{n} \mathbb{E}[S_n] \right) \\ &= (S_n - \mathbb{E}[S_n]) \left(\frac{1 + \frac{\alpha}{n}}{\Lambda_{n+1}} \right) \\ &= (S_n - \mathbb{E}[S_n]) \frac{1}{\Lambda_n} \\ &= M_n. \end{aligned}$$

Since $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ for all $n \geq 1$, $(M_n)_{n \geq 1}$ is a martingale, as desired. $\square$

With $(M_n)_{n \geq 1}$ established as a martingale, we define the **increments** of $(M_n)_{n \geq 1}$ as

$$\Delta M_1 := M_1, \quad \Delta M_n := M_n - M_{n-1}, \quad (n \geq 2).$$

By definition of $(S_n)_{n \geq 0}$, we may write

$$\begin{aligned} \Delta M_n &= \left(\frac{S_n - \mathbb{E}[S_n]}{\Lambda_n} \right) - \left(\frac{S_{n-1} - \mathbb{E}[S_{n-1}]}{\Lambda_{n-1}} \right) \\ &= \left(\frac{(S_{n-1} + \xi_n) - \mathbb{E}[S_{n-1} + \xi_n]}{\Lambda_n} \right) - \left(\frac{S_{n-1} - \mathbb{E}[S_{n-1}]}{\Lambda_{n-1}} \right) \\ &= \frac{\xi_n - \mathbb{E}[\xi_n]}{\Lambda_n} + \frac{S_{n-1} - \mathbb{E}[S_{n-1}]}{\Lambda_n} - \left(1 + \frac{\alpha}{n-1} \right) \frac{S_{n-1} - \mathbb{E}[S_{n-1}]}{\Lambda_n} \\ &= \frac{\xi_n - \mathbb{E}[\xi_n]}{\Lambda_n} - \frac{S_{n-1} - \mathbb{E}[S_{n-1}]}{\Lambda_n} \frac{\alpha}{n-1}. \end{aligned}$$

This gives the identity

$$\Delta M_n = \frac{\xi_n - \mathbb{E}[\xi_n]}{\Lambda_n} - \frac{S_{n-1} - \mathbb{E}[S_{n-1}]}{\Lambda_n} \frac{\alpha}{n-1}. \quad (4.1.3)$$

We now derive an absolute bound for ΔM_n . Since $\xi_n \in \{-1, 1\}$, we have $|\xi_n - \mathbb{E}[\xi_n]| \leq |\xi_n| + |\mathbb{E}[\xi_n]| \leq 2$. Moreover, since $S_{n-1} = \sum_{i=1}^{n-1} \xi_i$, it follows that $|S_{n-1}| \leq n-1$, and similarly $|\mathbb{E}[S_{n-1}]| \leq n-1$, so the triangle inequality yields

$$|S_{n-1} - \mathbb{E}[S_{n-1}]| \leq 2(n-1).$$

Since $p \in (0, 1)$, we have $|2p - 1| \leq 1$, that is $|\alpha| \leq 1$. Combining these estimates yields the bound

$$\begin{aligned} |\Delta M_n| &\leq \left| \frac{\xi_n - \mathbb{E}[\xi_n]}{\Lambda_n} \right| + \left| \frac{S_{n-1} - \mathbb{E}[S_{n-1}]}{\Lambda_n} \frac{\alpha}{n-1} \right| \\ &\leq \frac{2}{\Lambda_n} + \frac{2(n-1)}{\Lambda_n} \frac{1}{n-1} \\ &\leq \frac{4}{\Lambda_n}. \end{aligned}$$

It follows that

$$|\Delta M_n| \leq \frac{4}{\Lambda_n}. \quad (4.1.4)$$

Also, because $(M_n)_{n \geq 1}$ is a martingale, the increments have expectation zero, that is

$$\mathbb{E}[\Delta M_n \mid \mathcal{F}_{n-1}] = \mathbb{E}[M_n \mid \mathcal{F}_{n-1}] - \mathbb{E}[M_{n-1} \mid \mathcal{F}_{n-1}] = 0.$$

We now turn our attention to a theorem stated in *Hall, P.; Heyde (1980)* [HH80, Theorem 2.17].

Theorem 4.3. Let $\{T_n = \sum_{i=1}^n X_i, \mathcal{F}_n, n \geq 1\}$ be a *martingale* and let $1 \leq r \leq 2$. Then T_n converges almost surely on the set $\{\sum_{i=1}^{\infty} \mathbb{E}(|X_i|^r \mid \mathcal{F}_{i-1}) < \infty\}$.

With this theorem in mind, we define $N_j := \frac{\Lambda_j}{j} \Delta M_j$ for $j \geq 1$. Since $\frac{\Lambda_j}{j}$ is deterministic, $(N_j)_{j \geq 1}$ is a *martingale difference sequence*. Note that by (4.1.4) we have that

$$\begin{aligned} \sum_{j=1}^{\infty} \mathbb{E}[N_j^2 \mid \mathcal{F}_{j-1}] &= \sum_{j=1}^{\infty} \mathbb{E}\left[\frac{\Lambda_j^2}{j^2} \Delta M_j^2 \mid \mathcal{F}_{j-1}\right] \\ &\leq \sum_{j=1}^{\infty} \mathbb{E}\left[\frac{\Lambda_j^2}{j^2} \left(\frac{4}{\Lambda_j}\right)^2 \mid \mathcal{F}_{j-1}\right] \\ &= \sum_{j=1}^{\infty} \frac{16}{j^2} \\ &= \frac{8\pi^2}{3}. \end{aligned}$$

Here we used the classical fact that $\sum_{j=1}^{\infty} \frac{1}{j^2} = \frac{\pi^2}{6}$. Hence $\sum_{j=1}^{\infty} \mathbb{E}[N_j^2 | \mathcal{F}_{j-1}]$ converges almost surely. Applying [Theorem 4.3](#), we conclude that $\sum_{j=1}^{\infty} N_j$ converges almost surely. And therefore,

$$\sum_{k=1}^{\infty} \frac{\Lambda_k}{k} \Delta M_k < \infty. \quad (4.1.5)$$

To proceed, we state **Kronecker's lemma**:

Lemma 4.4 (Kronecker's Lemma). *Let $(y_n)_{n \geq 1}$ be a sequence of real numbers such that the series $\sum_{n=1}^{\infty} y_n$ converges. Let $(b_n)_{n \geq 1}$ be a sequence of positive real numbers that is nondecreasing and satisfies $b_n \rightarrow \infty$ as $n \rightarrow \infty$. Then*

$$\frac{1}{b_n} \sum_{k=1}^n b_k y_k \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Note that [Lemma 4.4](#) is equivalent to saying that, if all the conditions are met, and we define $z_k := b_k y_k$, so $\sum_{k=1}^{\infty} \frac{z_k}{b_k}$ converges, then

$$\frac{1}{b_n} \sum_{k=1}^n z_k \rightarrow 0, \quad \text{as } n \rightarrow \infty. \quad (4.1.6)$$

We now have all the necessary ingredients to prove [Theorem 4.1](#). We conclude this section with the proof of the Law of Large Numbers for the ERW.

Proof of Theorem 4.1. To apply the reformulation of [Lemma 4.4](#), set

$$b_k := \frac{k}{\Lambda_k}, \quad z_k := \Delta M_k.$$

Then $\frac{z_k}{b_k} = \frac{\Lambda_k}{k} \Delta M_k$. Because of [Lemma 4.4](#), it remains to prove that $\frac{n}{\Lambda_n}$ is non-decreasing and diverges and that $\sum_{j=1}^{\infty} \Delta M_j \frac{\Lambda_j}{j}$ converges. By [\(4.1.1\)](#) one has

$$\begin{aligned} \frac{\Lambda_n}{n} &= \frac{1}{n} \prod_{j=1}^{n-1} \left(1 + \frac{\alpha}{j}\right) \\ &= \frac{1}{n} \prod_{j=1}^{n-1} \left(\frac{j+\alpha}{j}\right) \\ &= \prod_{j=1}^{n-1} \left(\frac{j}{j+1}\right) \prod_{j=1}^{n-1} \left(\frac{j+\alpha}{j}\right) \\ &= \prod_{j=1}^{n-1} \frac{j+\alpha}{j+1} \\ &= \frac{\Gamma(n+\alpha)}{\Gamma(\alpha+1)\Gamma(n+1)} \end{aligned}$$

Proceeding, [Lemma 1.4](#) yields

$$\frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1)\Gamma(n + 1)} \sim \frac{n^{\alpha-1}}{\Gamma(\alpha + 1)}.$$

Since $p \in (0, 1)$ one has $\alpha - 1 < 0$, and therefore

$$\frac{\Lambda_n}{n} \sim \frac{n^{\alpha-1}}{\Gamma(\alpha + 1)} \rightarrow 0, \quad \text{as } n \rightarrow \infty. \quad (4.1.7)$$

and therefore one has $\frac{n}{\Lambda_n} \rightarrow \infty$. Also,

$$\frac{(n + 1)/\Lambda_{n+1}}{n/\Lambda_n} = \frac{n + 1}{n} \cdot \frac{\Lambda_n}{\Lambda_{n+1}} = \frac{n + 1}{n + \alpha} > 1.$$

As a result, $(\frac{n}{\Lambda_n})_{n \geq 1}$ is a non-decreasing sequence. Since $(n/\Lambda_n)_{n \geq 1} \nearrow \infty$, and from (4.1.5) we know that $\sum_{j=1}^{\infty} N_j = \sum_{j=1}^{\infty} \frac{\Lambda_j}{j} \Delta M_j < \infty$, we can use [Lemma 4.4](#) which gives

$$\frac{\Lambda_n}{n} \sum_{k=1}^n \Delta M_k = \frac{\Lambda_n}{n} M_n \rightarrow 0, \quad (n \rightarrow \infty).$$

Finally, note that $\frac{\Lambda_n}{n} M_n = \frac{S_n - \mathbb{E}[S_n]}{n}$, hence we can conclude that

$$\frac{S_n - \mathbb{E}[S_n]}{n} \xrightarrow{a.s.} 0, \quad \text{as } n \rightarrow \infty,$$

for all $(p, q) \in (0, 1) \times [0, 1]$, which completes the proof. $\square$

The almost sure convergence established above is illustrated in [Figure 4.1](#), where all sample paths are visibly drawn toward zero across all three regimes. Note that the scalings of the figures differ across the regimes.

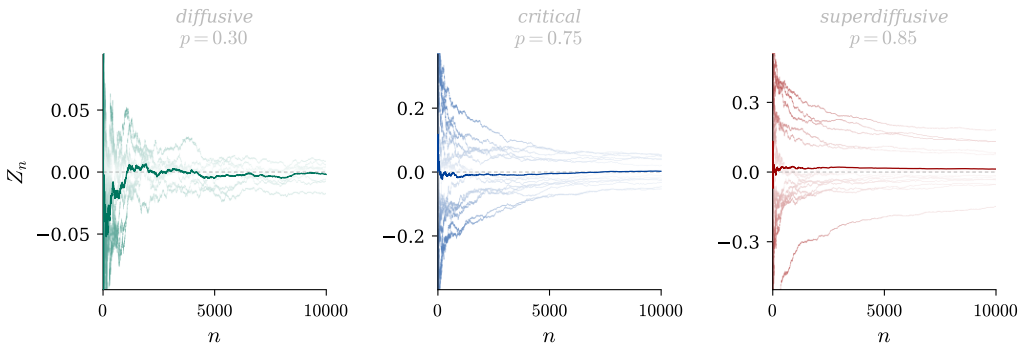


Figure 4.1: Paths of $Z_n := (S_n - \mathbb{E}[S_n])/n$ with $q = 0.5$, $n = 10,000$, for three regimes. Empirical means at $n = 10,000$: $-0.002, 0.003, 0.013$.

A related, yet more straightforward result is given in the following corollary, the proof of which is also due to Coletti, Gava, and Schütz [[CGS17a](#)].

Corollary 4.5. *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk. Then for all $(p, q) \in (0, 1) \times [0, 1]$,*

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} 0, \quad \text{as } n \rightarrow \infty.$$

Proof. Decompose

$$\frac{S_n}{n} = \frac{S_n - \mathbb{E}[S_n]}{n} + \frac{\mathbb{E}[S_n]}{n}.$$

By [Theorem 4.1](#), the first term of the right hand side converges almost surely to 0. For the second term, [\(3.2.4\)](#) gives $\mathbb{E}[S_n] \sim \frac{\beta}{\Gamma(\alpha+1)} n^\alpha$, and since $\alpha < 1$, it follows that

$$\frac{\mathbb{E}[S_n]}{n} \sim \frac{\beta}{\Gamma(\alpha+1)} n^{\alpha-1} \rightarrow 0.$$

We therefore conclude that $S_n/n \xrightarrow{\text{a.s.}} 0$, as $n \rightarrow \infty$. $\square$

The almost sure convergence to zero established in [Corollary 4.5](#) is confirmed by the simulation in [Figure 4.2](#), which, similarly to [Figure 4.1](#), displays sample paths of S_n/n across all three regimes.

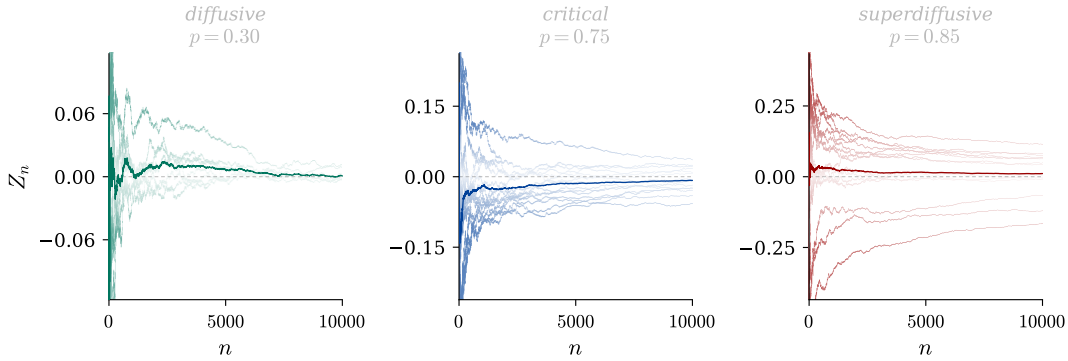


Figure 4.2: Paths of $Z_n := S_n/n$ with $q = \frac{1}{2}$, $n = 10,000$, for three regimes. The solid line is the empirical mean; the dashed line marks zero. Empirical means at $n = 10,000$: 0.001 (diffusive), -0.008 (critical), 0.011 (superdiffusive).

[Corollary 4.5](#) confirms that S_n grows strictly **sublinearly**, meaning the walk grows slower than the linear term n . In hindsight, this is perhaps not so surprising. Since each increment satisfies $\xi_n \in \{-1, +1\}$, the walk moves at most one step at a time, so $|S_n| \leq n$ holds for all trajectories $\omega \in \Omega$ and maximal linear growth would require every single step to point in the same direction, an event of probability zero. This also explains why the convergence appears slower in the superdiffusive regime. When $p > 3/4$, the strong memory causes the walk to spread faster, so the ratio S_n/n takes longer to become small, which is clearly visible in [Figure 4.2](#).

4.2 CENTRAL LIMIT THEOREM

Remarkably, the long-term behaviour of the ERW is not governed by a single **universal law**, but depends critically on the memory parameter p . Accordingly, the parameter space $(0, 1) \times [0, 1]$ splits into three regimes governed by distinct limit laws. For $p < 3/4$, the walk is **diffusive**, meaning that the fluctuations of S_n grow like $\sqrt{n}$, just as in the SRW. At the critical value $p_c = 3/4$, the walk is on the boundary between diffusive and superdiffusive behaviour, and a **logarithmic** correction to the normalization appears, signaling that the walk spreads slightly faster than in the diffusive regime. For $p > 3/4$, the walk is **superdiffusive**, meaning that the fluctuations grow faster than $\sqrt{n}$, Gaussian behaviour breaks down, and the rescaled walk converges to a non-Gaussian random variable. We treat each of these three regimes separately, starting with the diffusive regime.

4.2.1 The Diffusive Regime

We begin with the **diffusive regime**, where $p < 3/4$. Recall from [Section 3.3](#) that in this regime the second moment grows linearly, $\mathbb{E}[S_n^2] \sim \frac{n}{3-4p}$, which is the same order of growth as for the Simple Random Walk and already suggests that the fluctuations of S_n are of order $\sqrt{n}$. The following theorem confirms that, despite the dependence structure introduced by the memory mechanism, a Gaussian limit still holds.

We follow [\[CGS17a\]](#) closely, reproducing both the statement and proof of the following theorem from Coletti, Gava, and Schütz.

Theorem 4.6 (Central Limit Theorem in the Diffusive Regime). *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with $q \in [0, 1]$ and let $p \in (0, \frac{3}{4})$. Then we have the convergence in distribution*

$$\frac{S_n - \frac{\beta}{\Gamma(\alpha+1)} n^\alpha}{\sqrt{\frac{n}{3-4p}}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1),$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the **normal distribution** with mean μ and variance σ^2 .

Remark. In the unbiased case $q = \frac{1}{2}$, we have $\beta = 0$ and the centering term vanishes, so [Theorem 4.6](#) takes the cleaner form: for $p \in (0, \frac{3}{4})$,

$$\frac{S_n}{\sqrt{\frac{n}{3-4p}}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1),$$

or equivalently,

$$\frac{S_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{3-4p}\right).$$

Note that this agrees with [Theorem 2.5](#) in the $p = \frac{1}{2}$ case.

Proof Theorem 4.6. Fix $q \in [0, 1]$ and $p \in (0, \frac{3}{4})$. Recall from [\(4.1.1\)](#) the sequence $(\Lambda_n)_{n \geq 1}$, which was given by

$$\Lambda_n = \frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1) \Gamma(n)}, \quad n \geq 1.$$

Moreover, recall from [Lemma 1.4](#) the asymptotic equivalence

$$\Lambda_n \sim \frac{n^\alpha}{\Gamma(\alpha + 1)}. \quad (4.2.1)$$

In terms of $(\Lambda_n)_{n \geq 1}$, define the sequence $(s_n)_{n \geq 1}$ by

$$s_1^2 := 4q(1 - q), \quad s_n^2 := 4q(1 - q) + \sum_{j=2}^n \frac{1}{\Lambda_j^2}, \quad n \geq 2.$$

The asymptotic behaviour of $(s_n)_{n \geq 1}$ is described by the following lemma and proof presented by [\[CGS17a\]](#).

Lemma 4.7. *Let $(\Lambda_n)_{n \geq 1}$ be defined as in [\(4.1.1\)](#). The series $\sum_{n=1}^{\infty} \frac{1}{\Lambda_n^2}$ converges if and only if $p > \frac{3}{4}$.*

Proof. Let $b_n := \frac{1}{\Lambda_n^2}$ and suppose $p \in (0, \frac{3}{4})$. Since $b_n > 0$, we have $\sum_{j=1}^n b_j > 0$. Recall that

$$\Lambda_{n+1} = \Lambda_n \left(1 + \frac{\alpha}{n}\right).$$

From this it follows that

$$\frac{b_{n+1}}{b_n} = \frac{\Lambda_n^2}{\Lambda_{n+1}^2} = \frac{1}{\left(1 + \frac{\alpha}{n}\right)^2}.$$

A direct computation yields

$$1 - \frac{b_{n+1}}{b_n} = 1 - \frac{1}{1 + \frac{2\alpha}{n} + \frac{\alpha^2}{n^2}} = \frac{\frac{2\alpha}{n} + \frac{\alpha^2}{n^2}}{1 + \frac{2\alpha}{n} + \frac{\alpha^2}{n^2}}.$$

Multiplying both sides by n we obtain

$$n \left(1 - \frac{b_{n+1}}{b_n}\right) = \frac{2\alpha + \frac{\alpha^2}{n}}{1 + \frac{2\alpha}{n} + \frac{\alpha^2}{n^2}}.$$

Passing to the limit $n \rightarrow \infty$ gives

$$R := n \left(1 - \frac{b_{n+1}}{b_n} \right) \rightarrow 2\alpha$$

By **Raabe's criterion** (Lemma A.6), $\sum_{j=1}^{\infty} b_j$ converges if $R > 1$ and diverges if $R < 1$. Hence $\sum_{j=1}^{\infty} b_j$ converges if and only if $p > 3/4$, and diverges if and only if $p < 3/4$.

The case $p = \frac{3}{4}$ is treated separately. For $p = \frac{3}{4}$, one has the asymptotic equivalence

$$\Lambda_n \sim \frac{n^{\frac{1}{2}}}{\Gamma(3/2)} = \frac{2\sqrt{n}}{\sqrt{\pi}},$$

where we used Identity 1.4.3. Therefore

$$\frac{1}{\Lambda_n^2} \sim \frac{\pi}{4n},$$

and consequently

$$\sum_{n=1}^{\infty} \frac{1}{\Lambda_n^2} \sim \frac{\pi}{4} \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Hence $\sum_{n=1}^{\infty} \frac{1}{\Lambda_n^2}$ diverges at $p = \frac{3}{4}$ as well. $\square$

We proceed by establishing some further asymptotic estimates that will be needed in what follows. From (4.2.1) it follows that

$$\frac{1}{\Lambda_j^2} \sim \Gamma(\alpha + 1)^2 j^{-2\alpha}.$$

Since $\sum_{j=1}^n j^{-2\alpha} \sim \frac{n^{1-2\alpha}}{1-2\alpha}$ (Lemma 1.2), the series $\sum_{j=1}^n j^{-2\alpha}$ diverges for $\alpha < \frac{1}{2}$. This determines the asymptotic behaviour of s_n^2 :

$$s_n^2 = 4q(1-q) + \sum_{j=2}^n \frac{1}{\Lambda_j^2} \sim 4q(1-q) + \Gamma(\alpha + 1)^2 \frac{n^{1-2\alpha}}{1-2\alpha}.$$

Since the latter term diverges, the constant $4q(1-q)$ is negligible in comparison, and therefore

$$s_n \sim \frac{\Gamma(\alpha + 1)}{\sqrt{1-2\alpha}} n^{\frac{1-2\alpha}{2}}. \quad (4.2.2)$$

Combining (4.2.1) and (4.2.2) yields

$$\Lambda_n s_n \sim \frac{\sqrt{n}}{\sqrt{1-2\alpha}} = \frac{\sqrt{n}}{\sqrt{3-4p}}. \quad (4.2.3)$$

In particular, it follows that

$$\Lambda_n s_n \rightarrow \infty, \quad (n \rightarrow \infty). \quad (4.2.4)$$

With these estimates in hand, we define

$$D_{n,j} := \frac{\Delta M_j}{s_n}, \quad 1 \leq j \leq n.$$

Observe that

$$\sum_{j=1}^n D_{n,j} = \frac{1}{s_n} \sum \Delta M_j = \frac{M_n}{s_n} = \frac{S_n - \mathbb{E}[S_n]}{\Lambda_n s_n}. \quad (4.2.5)$$

The remainder of the proof rests on the following result, which is [HH80, Corollary 3.1].

Theorem 4.8 (Hall and Heyde, 1980). *Let $\{D_{n,j}, \mathcal{F}_j, 1 \leq j \leq n, n \geq 1\}$ be a zero-mean, square integrable martingale array with differences $D_{n,j} = \Delta M_j / s_n$. Suppose that the following two conditions hold. First, **Lindeberg's condition**: for all $\epsilon > 0$,*

$$\sum_{j=1}^n \mathbb{E} \left[D_{n,j}^2 \mathbf{1}_{\{|D_{n,j}| > \epsilon\}} \mid \mathcal{F}_{j-1} \right] \xrightarrow{\mathcal{P}} 0. \quad (4.2.6)$$

*Second, the **conditional variance condition**:*

$$\sum_{j=1}^n \mathbb{E} \left[D_{n,j}^2 \mid \mathcal{F}_{j-1} \right] \xrightarrow{\mathcal{P}} 1. \quad (4.2.7)$$

Then

$$\sum_{j=1}^n D_{n,j} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

We first verify **Lindeberg's condition**. Fix $\epsilon > 0$. By (4.1.4) one has

$$|D_{n,j}| = \frac{\Delta M_j}{s_n} \leq \frac{4}{\Lambda_j s_n}.$$

We wish to show that

$$\sum_{j=1}^n \mathbb{E} [D_{n,j}^2 \mathbf{1}_{\{|D_{n,j}| > \epsilon\}} \mid \mathcal{F}_{j-1}] \xrightarrow{a.s.} 0.$$

We distinguish two cases. If $p \in [\frac{1}{2}, \frac{3}{4}]$, then $\alpha \geq 0$, so $\Lambda_j \geq 1$ for all $j \geq 1$. Since $s_n \rightarrow \infty$ by Lemma 4.7, it follows that

$$|D_{n,j}| \leq \frac{4}{\Lambda_j s_n} \leq \frac{4}{s_n} \rightarrow 0, \quad (n \rightarrow \infty).$$

Since

$$\{|D_{n,j}| > \epsilon\} \subset \left\{ \frac{4}{s_n} > \epsilon \right\} = \emptyset$$

for all sufficiently large n , we conclude that

$$\mathbb{1}_{\{|D_{n,j}| > \epsilon\}} \stackrel{a.s.}{=} 0,$$

as $n \rightarrow \infty$. Now consider $p \in (0, \frac{1}{2})$, so $\alpha < 0$. In this case $\Lambda_j \geq \Lambda_n$ for all $j \leq n$, and therefore

$$\frac{4}{\Lambda_j s_n} \leq \frac{4}{\Lambda_n s_n} \rightarrow 0,$$

as $n \rightarrow \infty$, by (4.2.4). Hence $|D_{n,j}| \rightarrow 0$ as $n \rightarrow \infty$. Therefore

$$\mathbb{1}_{\{|D_{n,j}| > \epsilon\}} \stackrel{a.s.}{=} 0,$$

as well. In both cases Lindeberg's condition (4.2.6) is satisfied.

It remains to verify the second condition of Theorem 4.8, namely the **conditional variance condition** (4.2.7). We begin by writing

$$\sum_{j=1}^n \mathbb{E}[D_{n,j}^2 \mid \mathcal{F}_{j-1}] = \frac{1}{s_n^2} \sum_{j=1}^n \mathbb{E}[\Delta M_j^2 \mid \mathcal{F}_{j-1}].$$

Expanding $(\Delta M_j)^2$ using (4.1.3), we obtain

$$\begin{aligned} (\Delta M_j)^2 &= \left(\frac{\xi_j - \mathbb{E}[\xi_j]}{\Lambda_j} - \frac{S_{j-1} - \mathbb{E}[S_{j-1}]}{\Lambda_j} \frac{\alpha}{j-1} \right)^2 \\ &= \frac{1}{\Lambda_j^2} \left[(\xi_j - \mathbb{E}[\xi_j])^2 - 2 (\xi_j - \mathbb{E}[\xi_j]) \left(\frac{\alpha (S_{j-1} - \mathbb{E}[S_{j-1}])}{j-1} \right) + \left(\frac{\alpha (S_{j-1} - \mathbb{E}[S_{j-1}])}{j-1} \right)^2 \right]. \end{aligned}$$

This expansion yields three terms. By Theorem 4.1 one has

$$\frac{1}{\Lambda_j^2} \left(\frac{\alpha (S_{j-1} - \mathbb{E}[S_{j-1}])}{j-1} \right)^2 \stackrel{a.s.}{=} o\left(\frac{1}{\Lambda_j^2}\right). \quad (4.2.8)$$

Similarly,

$$\frac{2}{\Lambda_j^2} (\xi_j - \mathbb{E}[\xi_j]) \left(\frac{\alpha (S_{j-1} - \mathbb{E}[S_{j-1}])}{j-1} \right) \stackrel{a.s.}{=} o\left(\frac{1}{\Lambda_j^2}\right). \quad (4.2.9)$$

It remains to evaluate the term $\frac{1}{\Lambda_j^2} (\xi_j - \mathbb{E}[\xi_j])^2$. Since $\xi_j^2 = 1$ for all $j \geq 1$, we obtain

$$\begin{aligned} \frac{1}{\Lambda_j^2} (\xi_j - \mathbb{E}[\xi_j])^2 &= \frac{1}{\Lambda_j^2} (1 - 2\xi_j \mathbb{E}[\xi_j] + \mathbb{E}[\xi_j]^2) \\ &= \frac{1}{\Lambda_j^2} - \frac{2\xi_j}{\Lambda_j^2} \mathbb{E}[\xi_j] + \frac{1}{\Lambda_j^2} \mathbb{E}[\xi_j]^2. \end{aligned} \quad (4.2.10)$$

Conditioning on $\mathcal{F}_{j-1}$ yields

$$\begin{aligned} & \mathbb{E}\left[\frac{1}{\Lambda_j^2} - \frac{2\tilde{\xi}_j}{\Lambda_j^2} \mathbb{E}[\tilde{\xi}_j] + \frac{1}{\Lambda_j^2} \mathbb{E}[\tilde{\xi}_j]^2 \mid \mathcal{F}_{j-1}\right] \\ &= \frac{1}{\Lambda_j^2} - \frac{2}{\Lambda_j^2} \mathbb{E}[\tilde{\xi}_j \mathbb{E}[\tilde{\xi}_j] \mid \mathcal{F}_{j-1}] + \frac{1}{\Lambda_j^2} \mathbb{E}[\mathbb{E}[\tilde{\xi}_j]^2 \mid \mathcal{F}_{j-1}] \\ &= \frac{1}{\Lambda_j^2} - \frac{2}{\Lambda_j^2} \mathbb{E}[\tilde{\xi}_j] \mathbb{E}[\tilde{\xi}_j \mid \mathcal{F}_{j-1}] + \frac{1}{\Lambda_j^2} \mathbb{E}[\tilde{\xi}_j]^2. \end{aligned}$$

By [Corollary 3.3](#) and [Lemma 1.4](#),

$$\mathbb{E}[\tilde{\xi}_j] = \frac{\alpha\beta}{j} \frac{\Gamma(j+\alpha)}{\Gamma(\alpha+1)\Gamma(j)} \sim \frac{\alpha\beta}{\Gamma(\alpha+1)} j^{\alpha-1}.$$

Since $\alpha < 1$ throughout this regime, it follows that

$$\mathbb{E}[\tilde{\xi}_j] \rightarrow 0, \quad \mathbb{E}[\tilde{\xi}_j]^2 \rightarrow 0.$$

Moreover, we have

$$|\mathbb{E}[\tilde{\xi}_j \mid \mathcal{F}_{j-1}]| \leq 1.$$

It follows that,

$$1 - 2\mathbb{E}[\tilde{\xi}_j] \mathbb{E}[\tilde{\xi}_j \mid \mathcal{F}_{j-1}] + \mathbb{E}[\tilde{\xi}_j]^2 = 1 + o(1).$$

Multiplication by $\frac{1}{\Lambda_j^2}$ yields

$$\begin{aligned} \frac{1}{\Lambda_j^2} \left(1 - 2\mathbb{E}[\tilde{\xi}_j] + \mathbb{E}[\tilde{\xi}_j]^2\right) &= \frac{1}{\Lambda_j^2} + \frac{1}{\Lambda_j^2} o(1) \\ &= \frac{1}{\Lambda_j^2} + o\left(\frac{1}{\Lambda_j^2}\right) \end{aligned}$$

It follows that for $j \geq 2$,

$$\mathbb{E}[\Delta M_j^2 \mid \mathcal{F}_{j-1}] = \frac{1}{\Lambda_j^2} + o\left(\frac{1}{\Lambda_j^2}\right) \quad (4.2.11)$$

For $j = 1$, a direct computation gives

$$\mathbb{E}[\Delta M_1^2 \mid \mathcal{F}_0] = 4q(1-q). \quad (4.2.12)$$

Combining [Lemma 4.7](#) with (4.2.11) and (4.2.12) yields

$$\begin{aligned} \frac{1}{s_n^2} \sum_{j=1}^n \mathbb{E}[\Delta M_j^2 \mid \mathcal{F}_{j-1}] &= \frac{4q(1-q) + \sum_{j=1}^n \left(\frac{1}{\Lambda_j^2} + o\left(\frac{1}{\Lambda_j^2}\right)\right)}{4q(1-q) + \sum_{k=2}^n \frac{1}{\Lambda_k^2}} \\ &= 1 + \frac{\sum_{j=1}^n o\left(\frac{1}{\Lambda_j^2}\right)}{4q(1-q) + \sum_{k=2}^n \frac{1}{\Lambda_k^2}} \xrightarrow{a.s.} 1. \end{aligned}$$

Both conditions of [Theorem 4.8](#) are thus satisfied, and we conclude that

$$\sum_{j=1}^n D_{n,j} = \frac{S_n - \mathbb{E}[S_n]}{\Lambda_n s_n} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1). \quad (4.2.13)$$

We have already seen in [\(4.2.3\)](#) that

$$\Lambda_n s_n \sim \sqrt{\frac{n}{3-4p}}.$$

Hence, by definition one has

$$\frac{\Lambda_n s_n}{\sqrt{\frac{n}{3-4p}}} \rightarrow 1,$$

as $n \rightarrow \infty$. In view of [Slutsky's Theorem](#) ([Theorem A.7](#)) one has

$$\frac{S_n - \mathbb{E}[S_n]}{\Lambda_n s_n} \cdot \frac{\Lambda_n s_n}{\sqrt{\frac{n}{3-4p}}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1)$$

Combining this with [\(3.2.4\)](#) and applying Slutsky's Theorem again, yields the desired result

$$\frac{S_n - \frac{\beta n^\alpha}{\Gamma(\alpha+1)}}{\sqrt{\frac{n}{3-4p}}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

□

The conclusion of [Theorem 4.6](#) is supported by the numerical simulation in the following figure. [Figure 4.3](#) displays the empirical distribution of the rescaled walk over a large number of sample paths, and the agreement with the Gaussian density function is clearly visible.

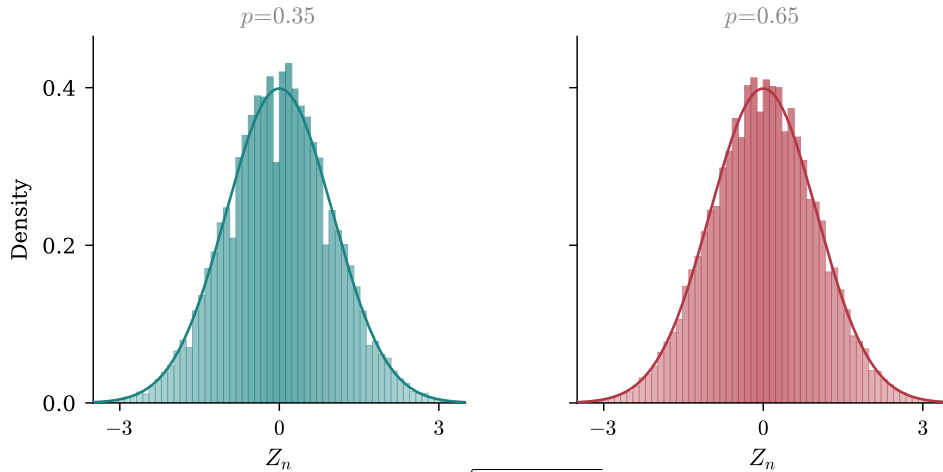


Figure 4.3: Histogram of $Z_n := S_n / \sqrt{n/(3-4p)}$ over 8,000 paths at $n = 7,000$, $q = \frac{1}{2}$, with the $\mathcal{N}(0, 1)$ density overlaid.

In conclusion, the diffusive regime retains classical central limit behaviour: the effect of memory is visible in the centering and scaling, but not in the limiting shape, which remains Gaussian.

4.2.2 The Critical Regime

We now turn to the **critical regime**, where $p = \frac{3}{4}$. Recall from [Section 3.3](#) that at this boundary value the second moment grows as $\mathbb{E}[S_n^2] \sim n \log n$, picking up a logarithmic correction relative to the diffusive case. This correction signals that the walk spreads slightly faster than order $\sqrt{n}$, and accordingly the normalization must be adjusted. The following theorem shows that a Gaussian limit still holds, but with $\sqrt{n \log n}$ replacing $\sqrt{n}$ as the correct scaling.

The following theorem and proof, due to [\[Ber18\]](#), establishes the Central Limit Theorem in the critical regime.

Theorem 4.9 (Central Limit Theorem in the Critical Regime).

Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with $q \in [0, 1]$ and let $p = \frac{3}{4}$. Then we have the convergence in distribution

$$\frac{S_n - \frac{2\beta}{\sqrt{\pi}} \sqrt{n}}{\sqrt{n \log n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1),$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the **normal distribution** with mean μ and variance σ^2 .

Remark. As before, setting $q = \frac{1}{2}$ simplifies the statement considerably, giving

$$\frac{S_n}{\sqrt{n \log n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

The proof of [Theorem 4.9](#) follows the same approach as in the **diffusive** case. The only difference lies in the asymptotic behaviour of $(\Lambda_n)_{n \geq 1}$, which takes a different form in the critical regime.

Proof Theorem 4.9. We fix $p = \frac{3}{4}$, and hence $\alpha = \frac{1}{2}$. Combining this with [\(4.2.1\)](#) yields

$$\Lambda_n \sim \frac{n^{1/2}}{\Gamma(3/2)} = \frac{2\sqrt{n}}{\sqrt{\pi}}, \quad (4.2.14)$$

where we used [Identity 1.4.3](#). It follows that

$$\frac{1}{\Lambda_n^2} \sim \frac{\pi}{4n},$$

and from this, we obtain the following asymptotic behaviour of $(s_n)_{n \geq 1}$:

$$s_n^2 = 4q(1 - q) + \sum_{j=2}^n \frac{1}{\Lambda_j^2} \sim 4q(1 - q) + \frac{\pi}{4} \sum_{j=2}^n \frac{1}{j} \sim \frac{\pi}{4} \log n. \quad (4.2.15)$$

Combining [\(4.2.14\)](#) and [\(4.2.15\)](#) one has

$$\Lambda_n s_n \sim \sqrt{n \log n}.$$

This confirms that $\Lambda_n s_n \rightarrow \infty$ as $n \rightarrow \infty$. Recall that the proof of [Theorem 4.6](#) relied on the auxiliary [Theorem 4.8](#). In the critical regime, this result applies in exactly the same way, and we again obtain

$$\frac{S_n - \mathbb{E}[S_n]}{\Lambda_n s_n} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

In view of [\(4.2.14\)](#), we have

$$\mathbb{E}[S_n] = \beta \Lambda_n \sim \frac{2\beta}{\sqrt{\pi}} \sqrt{n}.$$

By [Slutsky's Theorem](#) we have as a result

$$\frac{S_n - \frac{2\beta}{\sqrt{\pi}} \sqrt{n}}{\sqrt{n \log n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

This completes the proof. $\square$

As in the diffusive regime, [Theorem 4.9](#) can be verified numerically. The corresponding simulation is shown in [Figure 4.4](#):

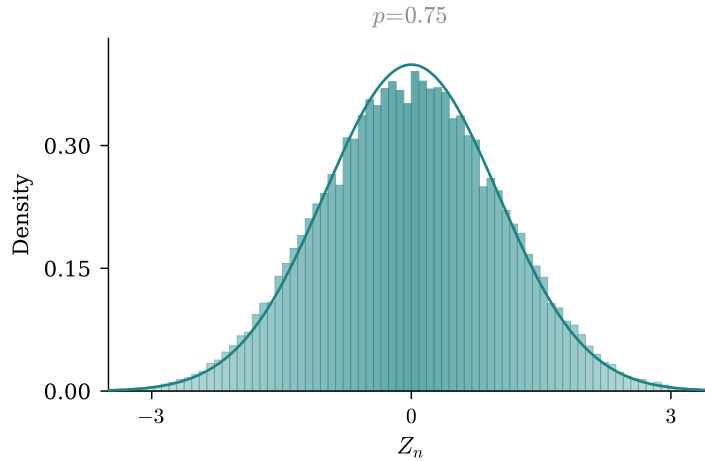


Figure 4.4: Histogram of $Z_n := S_n / \sqrt{n \log n}$ over 15,000 paths at $n = 7,000$, $q = \frac{1}{2}$, with the $\mathcal{N}(0, 1)$ density overlaid.

Taken together, [Theorem 4.6](#) and [Theorem 4.9](#) present a unified picture: for all $p \leq 3/4$, the walk eventually behaves like a Gaussian in the limit, with only the normalization needing an adjustment. It may therefore be tempting to think that this is simply an inherent feature of the ERW. In the [superdiffusive regime](#), however, the behaviour is different. When $p > 3/4$, the memory is strong enough to fundamentally change the limiting distribution, and the central limit theorem breaks down. The next section makes this precise.

4.2.3 The Superdiffusive Regime

We now turn to the [superdiffusive regime](#), where $p > 3/4$. Recall from [Section 3.3](#) that in this regime the second moment grows as

$\mathbb{E}[S_n^2] \sim C_p n^{4p-2}$, with exponent $4p - 2 > 1$, meaning the walk spreads strictly faster than $\sqrt{n}$. This faster growth signals that the **Gaussian** behaviour might break down. Instead, as the following theorem shows, it converges almost surely to a finite, non-degenerate, zero-mean random variable.

Recall that a random variable X is called **non-degenerate** if $\text{Var}(X) > 0$, or equivalently, if it does not concentrate all of its probability mass on a single value.

That the limiting distribution is non-Gaussian can be seen from simulation. This is illustrated in [Figure 4.5](#), which displays the empirical distribution of the unrescaled walk across a large number of sample paths with p in the superdiffusive regime.

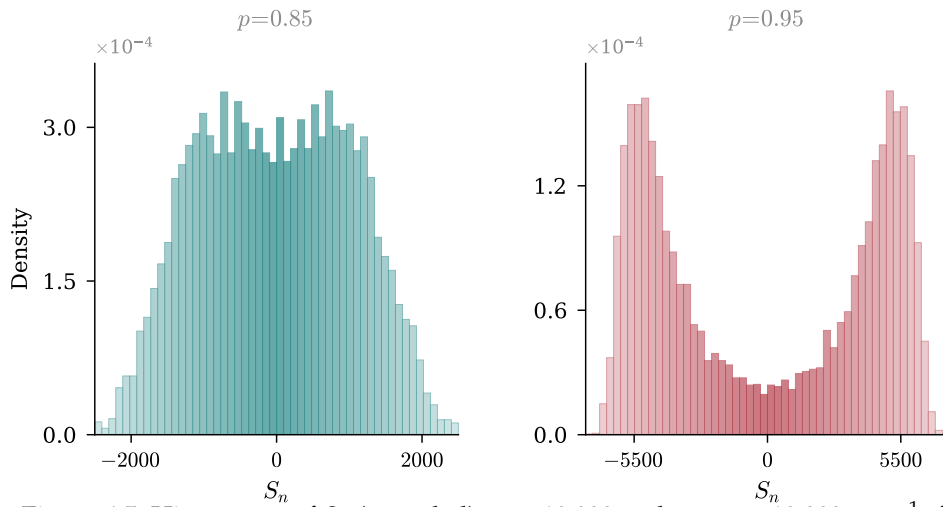


Figure 4.5: Histograms of S_n (unscaled) over 10,000 paths at $n = 10,000$, $q = \frac{1}{2}$, for two values of p in the superdiffusive regime.

The following theorem, due to [\[Ber18, Theorem 3\]](#), makes this precise.

Theorem 4.10 (Almost Sure Limit in the Superdiffusive Regime). *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk, with $p \in (\frac{3}{4}, 1)$ and $q \in [0, 1]$. Then*

$$\frac{S_n}{n^\alpha \Gamma(\alpha + 1)^{-1}} - \beta \xrightarrow{\text{a.s.}} M,$$

where M is a non-Gaussian, non-degenerate, zero-mean random variable.

Remark. Note that we have **almost sure convergence** here, which is stronger than merely convergence in distribution. Convergence in distribution only guarantees that the law of the rescaled walk approaches that of M . It says nothing about individual paths. Almost sure convergence means that for almost every realization ω , the

quantity $S_n(\omega)/(n^\alpha \Gamma(\alpha + 1)^{-1}) - \beta$ converges to the same random variable $M(\omega)$.

Before giving the proof, we recall the definition of $\mathcal{L}^p$ spaces, as the argument relies on the martingale $(M_n)_{n \geq 1}$ being bounded in $\mathcal{L}^2$.

Definition ($\mathcal{L}^p$ spaces). A random variable Z is in $\mathcal{L}^p$ with $1 \leq p < \infty$, if and only if

$$\|Z\|_p = \mathbb{E}[|Z|^p]^{\frac{1}{p}} < \infty.$$

Proof Theorem 4.10. In (4.1.2) we defined the martingale $M_n := \frac{S_n - \mathbb{E}[S_n]}{\Lambda_n}$, where $(\Lambda_n)_{n \geq 1}$ is defined as in (4.1.1). One easily checks

$$\begin{aligned} \|M_n\|_2 &= \left(\mathbb{E} \left[\left| \frac{S_n - \mathbb{E}[S_n]}{\Lambda_n} \right|^2 \right] \right)^{1/2} \\ &\leq \left(\mathbb{E} \left[\frac{4n^2}{|\Lambda_n|^2} \right] \right)^{1/2} \\ &< \infty. \end{aligned}$$

Hence $M_n \in \mathcal{L}^2$. In this regime we have $\alpha > \frac{1}{2}$, and therefore, by Lemma 4.7 one has

$$\sum_{n=1}^{\infty} \frac{1}{\Lambda_n^2} < \infty. \quad (4.2.16)$$

From (4.1.4) we deduce that $(\Delta M_n)^2 \leq \frac{16}{\Lambda_n^2}$. This, together with (4.2.16) yields

$$\sum_{n=1}^{\infty} \mathbb{E}[(\Delta M_n)^2] \leq \sum_{n=1}^{\infty} \frac{16}{\Lambda_n^2} < \infty.$$

From [Wil91, Chapter 12.1] we have the following theorem:

Theorem 4.11. Let $(M_n)_{n \geq 1}$ be a martingale for which $M_n \in \mathcal{L}^2$, $\forall n$. Then $(M_n)_{n \geq 1}$ is bounded in $\mathcal{L}^2$ if and only if

$$\sum_{n=1}^{\infty} \mathbb{E}[(\Delta M_n)^2] < \infty;$$

and when this obtains,

$$M_n \rightarrow M \quad \text{almost surely and in } \mathcal{L}^2.$$

Therefore, Theorem 4.11 ensures that

$$M_n \xrightarrow{a.s.} M,$$

where M is some random variable in $\mathcal{L}^2$. Since $\mathbb{E}[S_n] = \beta\Lambda_n$ and $\mathbb{E}[S_n] \sim \beta \frac{n^\alpha}{\Gamma(\alpha+1)}$, it follows

$$\frac{\Lambda_n}{n^\alpha \Gamma(\alpha+1)^{-1}} \rightarrow 1. \quad (4.2.17)$$

From this we can conclude that $\Lambda_n \sim n^\alpha \Gamma(\alpha+1)^{-1}$. Moreover, since $\mathbb{E}[S_n] = \beta\Lambda_n$, we obtain

$$\frac{S_n}{n^\alpha \Gamma(\alpha+1)^{-1}} - \beta = \frac{\Lambda_n}{n^\alpha \Gamma(\alpha+1)^{-1}} \left(\frac{S_n - \mathbb{E}[S_n]}{\Lambda_n} \right) + \beta \left(\frac{\Lambda_n}{n^\alpha \Gamma(\alpha+1)^{-1}} - 1 \right).$$

Hence, using $M_n = (S_n - \mathbb{E}[S_n]) / \Lambda_n$,

$$\frac{S_n}{n^\alpha \Gamma(\alpha+1)^{-1}} - \beta = \frac{\Lambda_n}{n^\alpha \Gamma(\alpha+1)^{-1}} M_n + \beta \left(\frac{\Lambda_n}{n^\alpha \Gamma(\alpha+1)^{-1}} - 1 \right).$$

Since $M_n \rightarrow M$ almost surely and using (4.2.17) it follows that

$$\frac{S_n}{n^\alpha \Gamma(\alpha+1)^{-1}} - \beta \xrightarrow{\text{a.s.}} M.$$

To show M is a **non-degenerate** random variable, we must show that $\text{Var}(M) > 0$. Recall that $\mathbb{E}[M_n] = 0$ for every n . Since $M_n \rightarrow M$ in $\mathcal{L}^2$, the **Cauchy-Schwarz inequality** gives

$$|\mathbb{E}[M]| = |\mathbb{E}[M - M_n]| \leq \mathbb{E}[|M - M_n|] \leq \left(\mathbb{E}[|M - M_n|^2] \right)^{1/2} \rightarrow 0,$$

as $n \rightarrow \infty$. Hence

$$\mathbb{E}[M] = 0.$$

From $\mathcal{L}^2$ -convergence it follows that

$$\text{Var}(M) = \lim_{n \rightarrow \infty} \text{Var}(M_n) = \sum_{j=1}^{\infty} \mathbb{E}[(\Delta M_j)^2] > 0.$$

Hence, M is a non-degenerate and mean-zero random variable. To show the non-Gaussian nature of M , we observe the closed form formula for the **skewness** and **kurtosis** of M . These explicit computations are made in [Ber18, Theorem 3.8], and we borrow these results for our proof. We define

$$L := \lim_{n \rightarrow \infty} \frac{S_n}{n^\alpha}. \quad (4.2.18)$$

Then we see that

$$M = \Gamma(\alpha+1)L - \beta.$$

Since skewness and kurtosis are **invariant** under shifts and positive rescaling, M and L have the same skewness and kurtosis. In [Ber18] they computed the first four moments for the $\alpha > \frac{1}{2}$ regime

$$\begin{aligned} \mathbb{E}[L] &= \frac{\beta}{\Gamma(\alpha+1)}, & \mathbb{E}[L^2] &= \frac{1}{(2\alpha-1)\Gamma(2\alpha)}, \\ \mathbb{E}[L^3] &= \frac{(\alpha+1)\beta}{\alpha(2\alpha-1)\Gamma(3\alpha)}, & \mathbb{E}[L^4] &= \frac{6(2\alpha^2+2\alpha-1)}{(4\alpha-1)(2\alpha-1)^2\Gamma(4\alpha)}. \end{aligned}$$

Recall the formulas for **skewness** and **kurtosis**, respectively,

$$\gamma(L) = \frac{\mathbb{E}[(L - \mathbb{E}[L])^3]}{\text{Var}(L)^{\frac{3}{2}}}, \quad \kappa(L) = \frac{\mathbb{E}[(L - \mathbb{E}[L])^4]}{\text{Var}(L)^2}. \quad (4.2.19)$$

Using the moment identities above, we first obtain

$$\text{Var}(L) = \mathbb{E}[L^2] - \mathbb{E}[L]^2 = \frac{1}{(2\alpha - 1)\Gamma(2\alpha)} - \frac{\beta^2}{\Gamma(\alpha + 1)^2}.$$

Moreover,

$$\begin{aligned} \mathbb{E}[(L - \mathbb{E}[L])^3] &= \mathbb{E}[L^3] - 3\mathbb{E}[L]\mathbb{E}[L^2] + 2\mathbb{E}[L]^3 \\ &= \frac{(\alpha + 1)\beta}{\alpha(2\alpha - 1)\Gamma(3\alpha)} - \frac{3\beta}{\Gamma(\alpha + 1)(2\alpha - 1)\Gamma(2\alpha)} + \frac{2\beta^3}{\Gamma(\alpha + 1)^3}, \end{aligned}$$

and

$$\begin{aligned} \mathbb{E}[(L - \mathbb{E}[L])^4] &= \mathbb{E}[L^4] - 4\mathbb{E}[L]\mathbb{E}[L^3] + 6\mathbb{E}[L]^2\mathbb{E}[L^2] - 3\mathbb{E}[L]^4 \\ &= \frac{6(2\alpha^2 + 2\alpha - 1)}{(4\alpha - 1)(2\alpha - 1)^2\Gamma(4\alpha)} - \frac{4(\alpha + 1)\beta^2}{\alpha(2\alpha - 1)\Gamma(\alpha + 1)\Gamma(3\alpha)} \\ &\quad + \frac{6\beta^2}{(2\alpha - 1)\Gamma(\alpha + 1)^2\Gamma(2\alpha)} - \frac{3\beta^4}{\Gamma(\alpha + 1)^4}. \end{aligned}$$

Substituting these expressions into (4.2.19), we find

$$\gamma(L) = \frac{\frac{(\alpha + 1)\beta}{\alpha(2\alpha - 1)\Gamma(3\alpha)} - \frac{3\beta}{\Gamma(\alpha + 1)(2\alpha - 1)\Gamma(2\alpha)} + \frac{2\beta^3}{\Gamma(\alpha + 1)^3}}{\left(\frac{1}{(2\alpha - 1)\Gamma(2\alpha)} - \frac{\beta^2}{\Gamma(\alpha + 1)^2}\right)^{3/2}},$$

and

$$\begin{aligned} \kappa(L) &= \frac{\frac{6(2\alpha^2 + 2\alpha - 1)}{(4\alpha - 1)(2\alpha - 1)^2\Gamma(4\alpha)} - \frac{4(\alpha + 1)\beta^2}{\alpha(2\alpha - 1)\Gamma(\alpha + 1)\Gamma(3\alpha)}}{\left(\frac{1}{(2\alpha - 1)\Gamma(2\alpha)} - \frac{\beta^2}{\Gamma(\alpha + 1)^2}\right)^2} \\ &\quad + \frac{\frac{6\beta^2}{(2\alpha - 1)\Gamma(\alpha + 1)^2\Gamma(2\alpha)} - \frac{3\beta^4}{\Gamma(\alpha + 1)^4}}{\left(\frac{1}{(2\alpha - 1)\Gamma(2\alpha)} - \frac{\beta^2}{\Gamma(\alpha + 1)^2}\right)^2}. \end{aligned}$$

A Gaussian random variable is characterized by the standardized moment identities

$$\gamma = 0, \quad \kappa = 3.$$

Thus, to rule out Gaussianity, it is enough to show that at least one of these two identities fails.

For the limiting variable L , the explicit formulas above show that this always happens in the superdiffusive regime. Indeed, two cases must be distinguished.

If $\beta \neq 0$, then the distribution of L is asymmetric, and the formula for $\gamma(L)$ does not vanish. Hence L cannot be Gaussian.

If $\beta = 0$, then the distribution is symmetric, so $\gamma(L) = 0$, but the formula for $\kappa(L)$ remains different from the Gaussian value 3. Hence L is again not Gaussian.

We conclude that, for every admissible value of β , the pair $(\gamma(L), \kappa(L))$ cannot equal the Gaussian pair $(0, 3)$. Therefore the limiting distribution of L is necessarily **non-Gaussian**.

Finally, since

$$M = \Gamma(\alpha + 1)L - \beta,$$

and skewness and kurtosis are invariant under shifts and positive rescalings, M and L have the same skewness and kurtosis. Hence M is non-Gaussian, which completes the proof. $\square$

The non-Gaussian character of the limiting random variable M is further illustrated by the numerical simulations in Figure 4.6, which display the empirical distribution of the rescaled walk for three representative values of p in the superdiffusive regime.

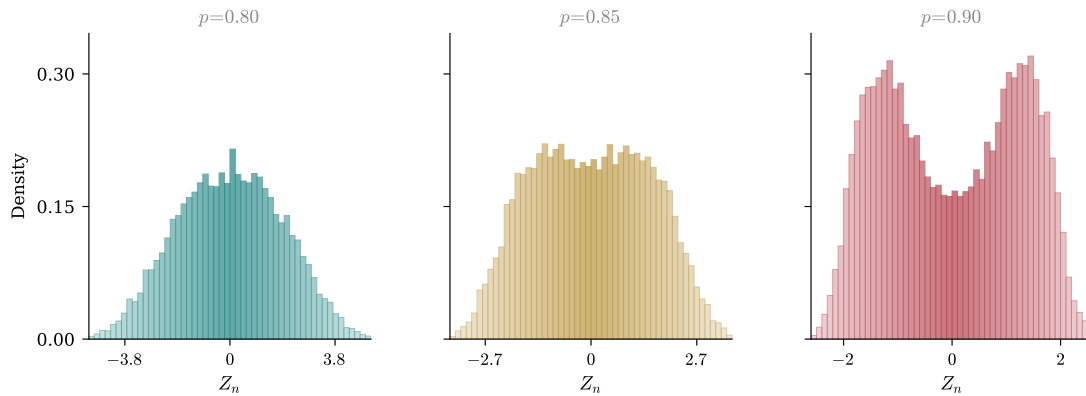


Figure 4.6: Histograms of $Z_n := S_n / (n^\alpha \Gamma(\alpha + 1)^{-1})$ over 15,000 paths at $n = 5,000$, $q = \frac{1}{2}$, for three values of p in the superdiffusive regime.

As p moves towards 1, the shape of the limiting density changes notably, and the role of q becomes increasingly pronounced. Both effects are explored further in the next section.

4.2.4 Exploration of the Superdiffusive Limit Law

In Theorem 4.10, we established that in the **superdiffusive regime**, the rescaled walk

$$Z_n := \Gamma(\alpha + 1) \frac{S_n}{n^\alpha} - \beta$$

converges almost surely to a non-degenerate, mean-zero random variable M . Moreover, M is not Gaussian. The proof of non-Gaussianity was based on explicit formulas for the skewness and kurtosis of the limiting random variable. However, these moment computations do not give a closed form for the full limiting distribution. To the best of our knowledge, no explicit expression for this distribution is currently known. This section therefore takes a **numerical** approach: we simulate the superdiffusive regime and explore how the shape of the limiting distribution changes across the parameter space.

We begin by examining the role of the initial bias parameter q . Recall from [Theorem 3.2](#) that q enters the dynamics of the ERW exclusively through the parameter $\beta = 2q - 1$, which controls the expected displacement. In particular, when $q = \frac{1}{2}$, one has $\beta = 0$ and the walk is unbiased in the sense that $\mathbb{E}[S_n] = 0$ for all $n \geq 0$. The centering by β in the definition of Z_n is therefore designed to remove this bias. Nevertheless, as the following figure reveals, the distribution of the rescaled walk is not simply shifted by β , the entire shape of the distribution depends on q in a more fundamental way.

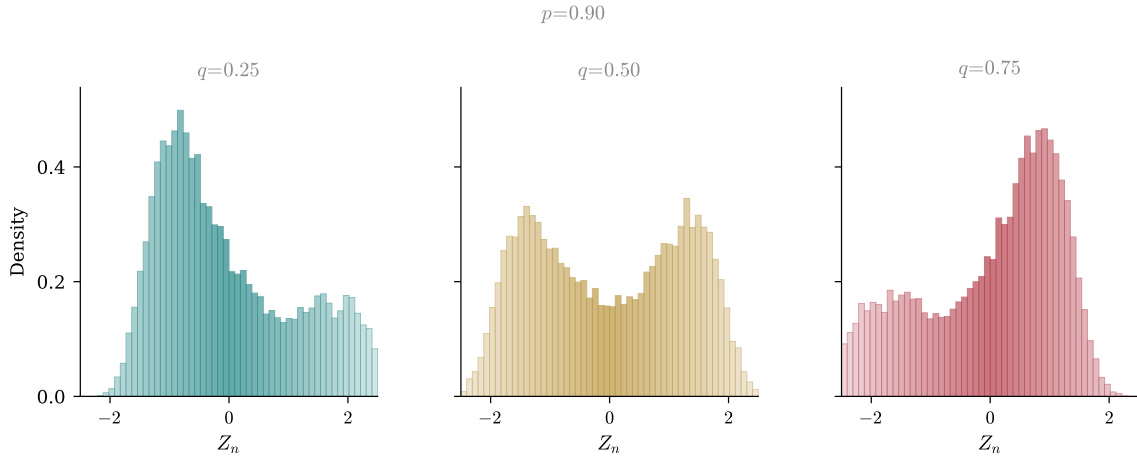


Figure 4.7: Histograms of $Z_n := S_n / (n^\alpha \Gamma(\alpha + 1)^{-1}) - \beta$ over 15,000 paths at $n = 5,000$, $p = 0.90$, for $q \in \{0.25, 0.50, 0.75\}$.

In the unbiased case $q = 0.50$, the histogram exhibits the near-symmetric shape already visible in [Figure 4.6](#). For $q = 0.25$ and $q = 0.75$, however, the distributions are markedly skewed: the histogram for $q = 0.25$ is skewed to the left, while that for $q = 0.75$ is skewed to the right. This agrees with the skewness computed in the proof of [Theorem 4.10](#). For $\beta \neq 0$, the formula for $\gamma(L)$ does not vanish, so the limiting distribution is genuinely asymmetric.

The intuition behind this behaviour is straightforward. In the superdiffusive regime, the memory is strong enough that the first increment ζ_1 has a lasting effect on the entire path. When $q > \frac{1}{2}$, the first step is more likely to be $+1$, and the strong memory keeps

reinforcing this, causing paths to drift in the positive direction. Although the centering by β removes this drift on average, it cannot fix the resulting asymmetry in the distribution. When $q < \frac{1}{2}$, the same effect occurs in the opposite direction, and the distribution is mirrored accordingly.

To obtain a more complete picture of how both parameters jointly shape the limiting distribution, [Figure 4.8](#) presents a systematic overview:

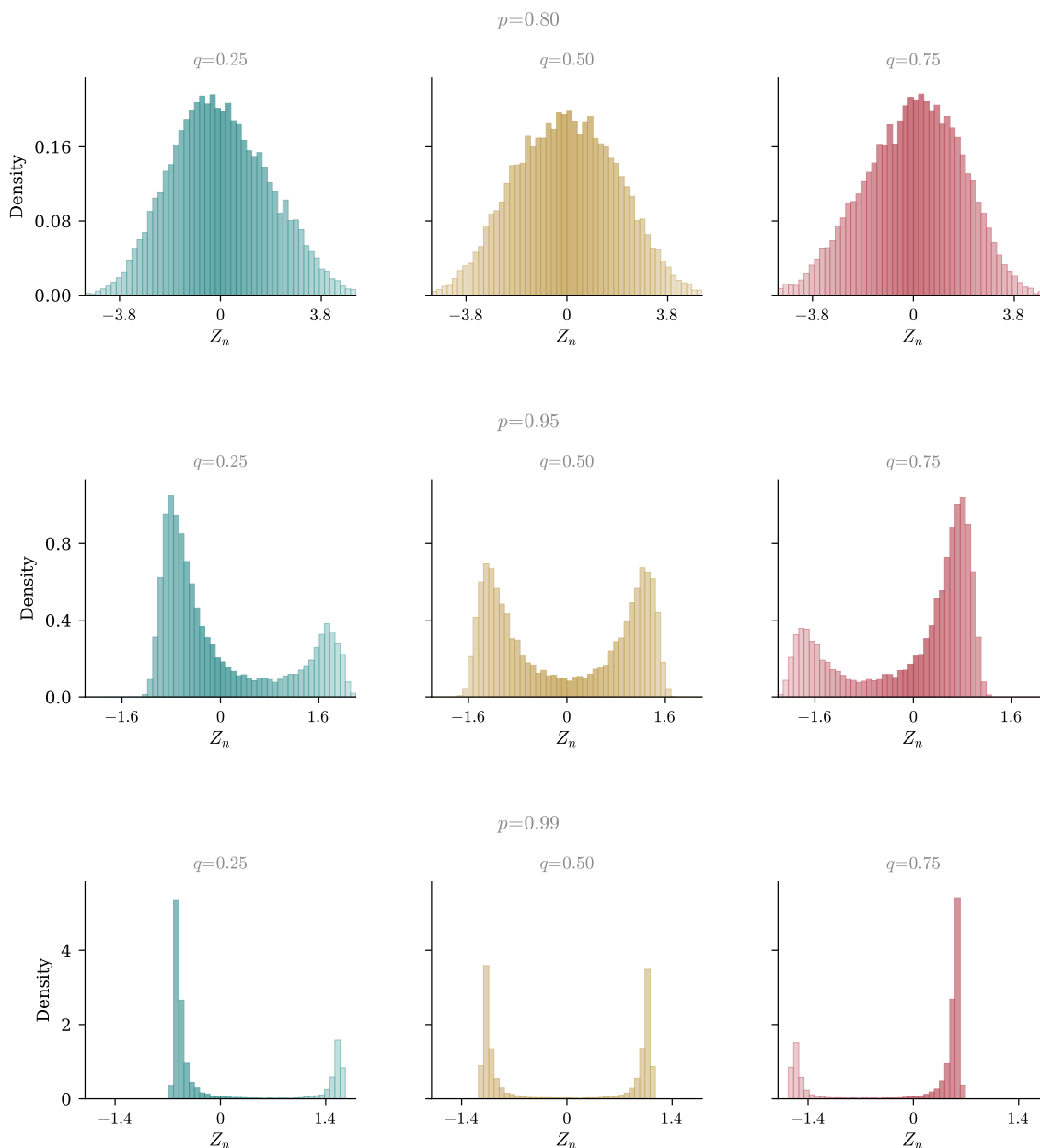


Figure 4.8: Histograms of $Z_n := S_n / (n^\alpha \Gamma(\alpha + 1)^{-1}) - \beta$ over 15,000 paths at $n = 5,000$, for $p \in \{0.80, 0.95, 0.99\}$ (top to bottom) and $q \in \{0.25, 0.50, 0.75\}$ (left to right, blue, yellow, red).

Two clear patterns emerge from the figure. Along each row, increasing q away from $\frac{1}{2}$ introduces skewness, with the direction determined by the sign of β . As p increases, the effect stands out

more clearly, because stronger memory gives the initial bias greater influence.

Along each column, increasing p causes the distribution to concentrate sharply around a single peak. This makes sense: as p grows towards 1, every increment tends to repeat the first one, so the path becomes nearly deterministic and Z_n collapses onto essentially two values.

Together, the figures show that p and q each shape the limit law in a distinct way: p controls how concentrated the distribution is, while q controls how symmetric it is.

To complement the histograms in [Figure 4.8](#), we present a numerical summary table for the same rescaled variable Z_n . We consider the parameter grid

$$p \in \{0.80, 0.85, 0.90, 0.95\}, \quad q \in \{0.25, 0.50, 0.75\},$$

and report empirical summary statistics for each pair (p, q) . In particular, the skewness measures the asymmetry of the distribution, while the kurtosis measures how peaked or heavy-tailed the distribution is, compared to the Gaussian value 3.

p	q	Emp. mean	Emp. variance	Emp. skewness	Emp. kurtosis
0.80	0.25	−0.001	3.668	0.153	2.725
0.80	0.50	0.019	3.958	−0.007	2.636
0.80	0.75	0.015	3.729	−0.145	2.708
0.85	0.25	0.003	2.066	0.369	2.474
0.85	0.50	0.016	2.307	−0.018	2.125
0.85	0.75	−0.018	2.061	−0.345	2.449
0.90	0.25	0.009	1.372	0.594	2.250
0.90	0.50	0.014	1.620	0.003	1.682
0.90	0.75	−0.006	1.376	−0.590	2.222
0.95	0.25	0.008	1.004	0.844	2.157
0.95	0.50	0.019	1.241	−0.029	1.308
0.95	0.75	0.005	0.977	−0.873	2.224

Table 4.1: Empirical statistics of $Z_n := S_n / (n^\alpha \Gamma(\alpha + 1)^{-1}) - \beta$ over 15,000 paths at $n = 10^5$, for $p \in \{0.80, 0.85, 0.90, 0.95\}$ and $q \in \{0.25, 0.50, 0.75\}$.

The plots above suggest that the superdiffusive limit law has a more detailed structure than the convergence theorem alone reveals. In fact, some results about this limiting distribution are now known in the literature. We briefly mention a few of them below, without going into their proofs, since those methods fall outside the scope of the present thesis.

In contrast to the regimes $p \leq \frac{3}{4}$, where the appropriately normalized Elephant Random Walk has a Gaussian limit law, the distribution of the superdiffusive limit remained for some time much less explicit. In the regime $p > \frac{3}{4}$, writing $\alpha := 2p - 1$, one has

$$\frac{S_n}{n^\alpha} \xrightarrow{a.s.} L_q,$$

where L_q denotes the limiting random variable corresponding to the initial bias parameter $q \in [0, 1]$. A major step in the understanding of this limit law was made in [GLR23], where the authors proved the existence of a density for the limiting distribution. Moreover, if L_1 denotes the special case corresponding to $q = 1$, then the general limit law is related to L_1 through

$$L_q \stackrel{\mathcal{D}}{=} (2\vartheta_q - 1)L_1,$$

where ϑ_q is a Bernoulli random variable with parameter q , independent of L_1 . This relation is also established in [GLR23]. Recently, [GLRS25] proved that L_1 is **unimodal**. This means that the density of L_1 has a unique local maximum, or informally, a single peak. Note however that this only holds for the $q = 1$ case. Since the proofs of these results rely on advanced techniques, we do not pursue them here.

ALMOST SURE PROPERTIES OF THE ERW

In this chapter we change our focus. Instead of studying averaged or distributional properties of the Elephant Random Walk, we now study its almost sure long-term behaviour. In particular, we ask whether, with probability one, the walk returns to its starting point infinitely often (recurrence), or whether it visits the starting point only finitely many times (transience). And how this almost sure behaviour depends on the memory parameter p .

A central limit theorem is not sufficient to answer this question. What is required instead is precise control over the fluctuations of the walk on events of probability one. This is where two powerful tools come in: the **Quadratic Strong Law** and the **Law of the Iterated Logarithm**. Essentially, these laws give us bounds on how much the walk can swing back and forth.

With those tools available to us, [Section 5.3](#) will use them to show that in the diffusive and critical regimes, the walk must cross the origin infinitely often, meaning the walk is recurrent.

5.1 THE QUADRATIC STRONG LAW

Before we can tackle recurrence and transience, we need to understand how fast the walk grows on events of probability one. The **Quadratic Strong Law** (QSL) provides this by giving the precise almost sure growth rate of $(S_n)_{n \geq 0}^2$. The approach we follow is built on the martingale framework developed by Bercu [\[Ber18\]](#), which reduces the analysis of the ERW's fluctuations to the study of a carefully constructed martingale transform. Specifically, we will show that the correct normalization of $(S_n)_{n \geq 0}^2$ converges almost surely to a finite limit, where the normalization depends on the regime of p .

More precisely, we aim to establish the results presented by [\[Ber18\]](#), that in the **diffusive regime** (i.e. $p < 3/4$), one has

$$\lim_{n \rightarrow \infty} \frac{1}{\log n} \sum_{k=1}^n \left(\frac{S_k}{k} \right)^2 \stackrel{a.s.}{=} \frac{1}{3 - 4p},$$

and that in the **critical regime** (i.e. $p = 3/4$), one has

$$\lim_{n \rightarrow \infty} \frac{1}{\log \log n} \sum_{k=2}^n \left(\frac{S_k}{k \log k} \right)^2 \stackrel{a.s.}{=} 1.$$

We begin by introducing the sequence $(\chi_n)_{n \geq 1}$ defined by

$$\chi_1 := 1, \quad \chi_n := \frac{\Gamma(n) \Gamma(\alpha + 1)}{\Gamma(n + \alpha)}, \quad (n \geq 2).$$

By [Property 1.4.2](#), the sequence $(\chi_n)_{n \geq 1}$ satisfies the recurrence relation

$$\chi_n = \left(1 + \frac{\alpha}{n}\right) \chi_{n+1}, \quad (n \geq 1). \quad (5.1.1)$$

We define $(\mathcal{M}_n)_{n \geq 0}$ by

$$\mathcal{M}_n = \chi_n S_n,$$

together with $\mathcal{M}_0 := 0$. Combining [\(5.1.1\)](#) with [\(3.2.2\)](#), we obtain, for every $n \geq 1$,

$$\begin{aligned} \mathbb{E}[\mathcal{M}_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[\chi_{n+1} S_{n+1} \mid \mathcal{F}_n] \\ &= \chi_{n+1} \mathbb{E}[S_{n+1} \mid \mathcal{F}_n] \\ &= \chi_{n+1} \left(1 + \frac{\alpha}{n}\right) S_n \\ &= \chi_n S_n \\ &= \mathcal{M}_n. \end{aligned}$$

Hence $(\mathcal{M}_n)_{n \geq 0}$ is a **martingale** with respect to the filtration $(\mathcal{F}_n)_{n \geq 0}$. Denoting the increments of $(\mathcal{M}_n)_{n \geq 0}$ by

$$\Delta \mathcal{M}_k := \mathcal{M}_k - \mathcal{M}_{k-1},$$

a short computation gives, for every $k \geq 2$,

$$\begin{aligned} \Delta \mathcal{M}_k &= \chi_k S_k - \chi_{k-1} S_{k-1} \\ &= \chi_k S_k - \left(1 + \frac{\alpha}{k-1}\right) \chi_k S_{k-1} \\ &= \chi_k \left(S_k - \left(1 + \frac{\alpha}{k-1}\right) S_{k-1} \right). \end{aligned}$$

Accordingly, we define

$$\varepsilon_1 := S_1, \quad \varepsilon_{n+1} := S_{n+1} - \left(1 + \frac{\alpha}{n}\right) S_n, \quad (n \geq 1),$$

so that

$$\Delta \mathcal{M}_k = \chi_k \varepsilon_k, \quad (k \geq 1),$$

and the martingale $(\mathcal{M}_n)_{n \geq 0}$ admits the representation

$$\mathcal{M}_n = \sum_{k=1}^n \Delta \mathcal{M}_k = \sum_{k=1}^n \chi_k \varepsilon_k.$$

We first examine $\mathbb{E}[\varepsilon_{n+1} \mid \mathcal{F}_n]$. Since

$$\varepsilon_{n+1} = S_{n+1} - \left(1 + \frac{\alpha}{n}\right) S_n = \tilde{\varepsilon}_{n+1} - \frac{\alpha}{n} S_n,$$

a direct computation yields

$$\begin{aligned}
 \mathbb{E}[\varepsilon_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\left[S_{n+1} - \left(1 + \frac{\alpha}{n}\right) S_n \mid \mathcal{F}_n\right] \\
 &= \mathbb{E}\left[\xi_{n+1} - \frac{\alpha}{n} S_n \mid \mathcal{F}_n\right] \\
 &= \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] - \frac{\alpha}{n} S_n \\
 &= \frac{\alpha}{n} S_n - \frac{\alpha}{n} S_n \\
 &= 0.
 \end{aligned}$$

Hence

$$\mathbb{E}[\varepsilon_{n+1} \mid \mathcal{F}_n] = 0. \quad (5.1.2)$$

For the second conditional moment, we again use $\varepsilon_{n+1} = \xi_{n+1} - \frac{\alpha}{n} S_n$. Applying the standard properties of conditional expectation,

$$\begin{aligned}
 \mathbb{E}[\varepsilon_{n+1}^2 \mid \mathcal{F}_n] &= \mathbb{E}\left[\left(\xi_{n+1} - \frac{\alpha}{n} S_n\right)^2 \mid \mathcal{F}_n\right] \\
 &= \mathbb{E}\left[\xi_{n+1}^2 - 2\frac{\alpha}{n} S_n \xi_{n+1} + \frac{\alpha^2}{n^2} S_n^2 \mid \mathcal{F}_n\right] \\
 &= \mathbb{E}[\xi_{n+1}^2 \mid \mathcal{F}_n] - 2\frac{\alpha}{n} S_n \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] + \frac{\alpha^2}{n^2} S_n^2 \\
 &= 1 - 2\frac{\alpha}{n} S_n \left(\frac{\alpha}{n} S_n\right) + \frac{\alpha^2}{n^2} S_n^2 \\
 &= 1 - \frac{\alpha^2}{n^2} S_n^2 \\
 &= 1 - \alpha^2 \left(\frac{S_n}{n}\right)^2.
 \end{aligned} \quad (5.1.3)$$

By [Corollary 4.5](#), it follows that

$$\mathbb{E}[\varepsilon_{n+1}^2 \mid \mathcal{F}_n] \xrightarrow{a.s.} 1. \quad (5.1.4)$$

We now turn to the examination of $\mathbb{E}[\varepsilon_{n+1}^4 \mid \mathcal{F}_n]$. Writing $\varepsilon_{n+1} = \xi_{n+1} - \left(\frac{\alpha}{n}\right) S_n$, and setting

$$\mu_n := \alpha \frac{S_n}{n},$$

we have $\varepsilon_{n+1} = \xi_{n+1} - \mu_n$. Expanding the fourth power gives

$$\begin{aligned}
 \varepsilon_{n+1}^4 &= (\xi_{n+1} - \mu_n)^4 \\
 &= \xi_{n+1}^4 - 4\mu_n \xi_{n+1}^3 + 6\mu_n^2 \xi_{n+1}^2 - 4\mu_n^3 \xi_{n+1} + \mu_n^4 \\
 &= 1 - 4\mu_n \xi_{n+1} + 6\mu_n^2 - 4\mu_n^3 \xi_{n+1} + \mu_n^4,
 \end{aligned}$$

where we have used that $\xi_{n+1}^2 = \xi_{n+1}^4 = 1$ and $\xi_{n+1}^3 = \xi_{n+1}$. Recalling that

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = \alpha \frac{S_n}{n} = \mu_n,$$

we obtain

$$\begin{aligned}\mathbb{E}[\varepsilon_{n+1}^4 \mid \mathcal{F}_n] &= 1 - 4\mu_n \mathbb{E}[\zeta_{n+1} \mid \mathcal{F}_n] + 6\mu_n^2 - 4\mu_n^3 \mathbb{E}[\zeta_{n+1} \mid \mathcal{F}_n] + \mu_n^4 \\ &= 1 - 4\mu_n^2 + 6\mu_n^2 - 4\mu_n^4 + \mu_n^4 \\ &= 1 + 2\mu_n^2 - 3\mu_n^4.\end{aligned}$$

Substituting $\mu_n = \alpha \frac{S_n}{n}$ back yields

$$\mathbb{E}[\varepsilon_{n+1}^4 \mid \mathcal{F}_n] = 1 + 2\alpha^2 \left(\frac{S_n}{n}\right)^2 - 3\alpha^4 \left(\frac{S_n}{n}\right)^4.$$

Setting $y := \mu_n^2$, this may be written as

$$\mathbb{E}[\varepsilon_{n+1}^4 \mid \mathcal{F}_n] = 1 + 2y - 3y^2.$$

Since $0 \leq y \leq 1$, the quadratic function $1 + 2y - 3y^2$ is maximised at $y = \frac{1}{3}$, and therefore

$$\mathbb{E}[\varepsilon_{n+1}^4 \mid \mathcal{F}_n] \leq \frac{4}{3} \quad \forall n \geq 1. \quad (5.1.5)$$

In the same manner, it follows from (5.1.3) that

$$\mathbb{E}[\varepsilon_{n+1}^2 \mid \mathcal{F}_n] \leq 1 \quad \forall n \geq 1.$$

We now have all the necessary ingredients to apply the following theorem due to Bernard Bercu [Ber04]. We state it here in the specific form needed for our purposes. The original result is more general.

Theorem 5.1. *Let $M_n := \sum_{k=1}^n \chi_k \varepsilon_k$, $\nu_n := \sum_{k=1}^n \chi_k^2$, and $f_n := \chi_n^2 / \nu_n$, where $(M_n)_{n \geq 0}$ is a real martingale transform with respect to $(\mathcal{F}_n)$ and (ε_n) is a martingale difference sequence. If*

$$\mathbb{E}[\varepsilon_{n+1}^2 \mid \mathcal{F}_n] \xrightarrow{a.s.} 1, \quad f_n \xrightarrow{a.s.} 0,$$

and for some $a > 2$

$$\sup_{n \geq 0} \mathbb{E}[|\varepsilon_{n+1}|^a \mid \mathcal{F}_n] < \infty \text{ a.s.,}$$

then

$$\frac{1}{\log \nu_n} \sum_{k=1}^n f_k \left(\frac{M_k^2}{\nu_k} \right) \xrightarrow{a.s.} 1.$$

In line with Theorem 5.1, we define

$$\nu_n := \sum_{k=1}^n \chi_k^2, \quad f_n := \frac{\chi_n^2}{\nu_n}, \quad (n \geq 1).$$

We first address the **diffusive regime** (i.e. $p < \frac{3}{4}$). From [Lemma 1.4](#),

$$\chi_n \sim \Gamma(2p)n^{1-2p}, \quad \chi_n^2 \sim \Gamma(2p)^2 n^{2-4p}. \quad (5.1.6)$$

Together with [Lemma 1.2](#), this amounts to

$$\nu_n = \sum_{k=1}^n \chi_k^2 \sim \Gamma(2p)^2 \sum_{k=1}^n k^{2-4p} \sim \Gamma(2p)^2 \frac{n^{3-4p}}{3-4p}. \quad (5.1.7)$$

Consequently,

$$f_n \sim \frac{\Gamma(2p)^2 n^{2-4p}}{\frac{\Gamma(2p)^2}{3-4p} n^{3-4p}} = (3-4p) \frac{1}{n}. \quad (5.1.8)$$

In particular, $f_n \rightarrow 0$ as $n \rightarrow \infty$. Combining this with [\(5.1.4\)](#) and [\(5.1.5\)](#), [Theorem 5.1](#) yields

$$\frac{1}{\log \nu_n} \sum_{k=1}^n f_k \left(\frac{\mathcal{M}_k^2}{\nu_k} \right) \xrightarrow{a.s.} 1.$$

By definition of f_k ,

$$\frac{1}{\log \nu_n} \sum_{k=1}^n \left(\frac{\chi_k^2 M_k^2}{\nu_k^2} \right) \xrightarrow{a.s.} 1. \quad (5.1.9)$$

Recall from [\(5.1.7\)](#) that we obtained the limit

$$\frac{\nu_n}{n^{3-4p}} \rightarrow \ell, \quad \text{where } \ell := \frac{\Gamma(2p)^2}{3-4p}. \quad (5.1.10)$$

This is equivalent to $\nu_n \sim n^{3-4p} \ell$. By definition, this means

$$\log(\nu_n) = \log \left(\ell n^{3-4p} (1 + o(1)) \right) = \log \ell + (3-4p) \log n + \log(1 + o(1)).$$

Note that $\log(1 + o(1)) \rightarrow 0$. Dividing both sides by $\log n$ yields

$$\frac{\log \nu_n}{\log n} = \frac{\log \ell}{\log n} + (3-4p) + \frac{\log(1 + o(1))}{\log n} \rightarrow 3-4p,$$

as $n \rightarrow \infty$. As a result

$$\frac{\log \nu_n}{\log n} \rightarrow 3-4p. \quad (5.1.11)$$

Setting $A_n = \sum_{k=1}^n \left(\frac{\chi_k^2 M_k^2}{\nu_k^2} \right)$, the left-hand side of [\(5.1.9\)](#) factorizes as

$$\frac{A_n}{\log n} = \left(\frac{A_n}{\log \nu_n} \right) \cdot \left(\frac{\log \nu_n}{\log n} \right).$$

Combining (5.1.9) and (5.1.11) therefore yields

$$\frac{1}{\log n} \sum_{k=1}^n \left(\frac{a_k^2 M_k^2}{v_k^2} \right) \xrightarrow{a.s.} 3 - 4p.$$

Since $M_n = \chi_n S_n$, this is equivalent to

$$\frac{1}{\log n} \sum_{k=1}^n \left(\frac{k^2 \chi_k^4}{v_k^2} \right) \left(\frac{S_k^2}{k^2} \right) \xrightarrow{a.s.} 3 - 4p. \quad (5.1.12)$$

We now set $c_k := \left(\frac{k^2 \chi_k^4}{v_k^2} \right)$, and show that

$$c_k \rightarrow (3 - 4p)^2.$$

Indeed, (5.1.6) gives

$$\chi_n^2 \sim \Gamma(2p)^2 n^{2-4p}, \quad \chi_n^4 \sim \Gamma(2p)^4 n^{4-8p},$$

so that

$$c_k = \left(\frac{k^2 \chi_k^4}{v_k^2} \right) \sim \frac{k^2 \Gamma(2p)^4 k^{4-8p}}{\Gamma(2p)^4 \frac{k^{6-8p}}{(3-4p)^2}} = (3-4p)^2.$$

Consequently,

$$k^2 \chi_k^4 \sim (3-4p)^2 v_k^2.$$

Substituting this result in (5.1.12) yields

$$\begin{aligned} \frac{1}{\log n} \sum_{k=1}^n \left(\frac{k^2 \chi_k^4}{v_k^2} \right) \left(\frac{S_k^2}{k^2} \right) &\sim \frac{1}{\log n} \sum_{k=1}^n \left(\frac{(3-4p)^2 v_k^2}{v_k^2} \right) \left(\frac{S_k^2}{k^2} \right) \\ &= \frac{(3-4p)^2}{\log n} \sum_{k=1}^n \left(\frac{S_k^2}{k^2} \right) \end{aligned}$$

Combining this with (5.1.12) we conclude

$$\frac{(3-4p)^2}{\log n} \sum_{k=1}^n \left(\frac{S_k^2}{k^2} \right) \xrightarrow{a.s.} 3 - 4p. \quad (5.1.13)$$

This establishes the **Quadratic Strong Law** (QSL) in the diffusive regime:

$$\lim_{n \rightarrow \infty} \frac{1}{\log n} \sum_{k=1}^n \left(\frac{S_k}{k} \right)^2 \stackrel{a.s.}{=} \frac{1}{3-4p}, \quad (p < \frac{3}{4}). \quad (5.1.14)$$

To visualize the convergence in Figure 5.1, we simulate the quantity $Q_n := (\log n)^{-1} \sum_{k=1}^n (S_k/k)^2$ for three values of p in the diffusive regime. As n grows, the quantity Q_n converges to the limit $1/(3-4p)$.

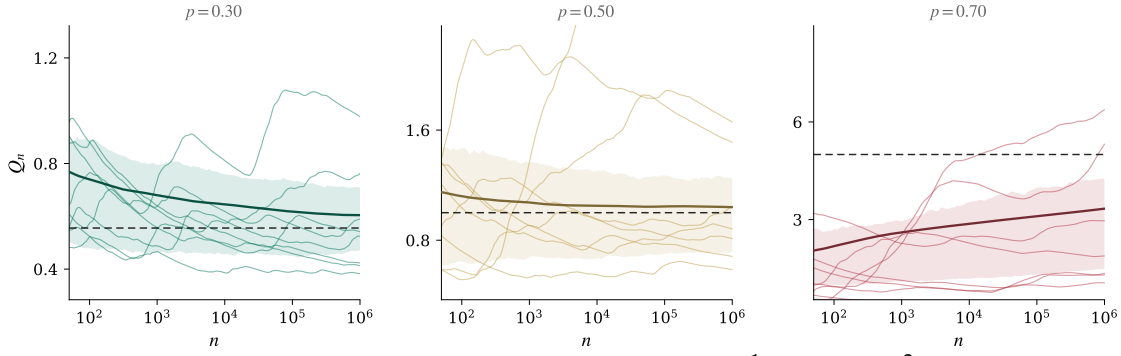


Figure 5.1: Running QSL statistic $Q_n = (\log n)^{-1} \sum_{k=1}^n (S_k/k)^2$ for $p \in \{0.30, 0.50, 0.70\}$: eight sample paths (thin), ensemble mean over 400 paths (bold), interquartile band (shaded), and theoretical limit $1/(3-4p)$ (dashed).

This only holds for the **diffusive regime**, since for $p = \frac{3}{4}$ the asymptotics differ, and a separate argument is required.

We now set $p = \frac{3}{4}$, so that $\alpha = \frac{1}{2}$, and

$$\chi_n = \frac{\Gamma(n)\Gamma(\alpha+1)}{\Gamma(n+\alpha)} = \frac{\Gamma(n)\Gamma(\frac{3}{2})}{\Gamma(n+\frac{1}{2})}.$$

But now **Stirling's approximation** (Lemma A.2) yields

$$\chi_n \sim \Gamma(\frac{3}{2})n^{-\frac{1}{2}} = \frac{\sqrt{\pi}}{2} \frac{1}{\sqrt{n}}.$$

And therefore

$$\chi_n^2 \sim \frac{\pi}{4n},$$

and therefore

$$v_n = \sum_{k=1}^n \chi_k^2 \sim \frac{\pi}{4} \sum_{k=1}^n \frac{1}{k} = \frac{\pi}{4} H_n \sim \frac{\pi}{4} \log n.$$

Defining $f_n := \frac{\chi_n^2}{v_n}$ as before,

$$\frac{\chi_n^2}{v_n} \sim \frac{\frac{\pi}{4n}}{\frac{\pi}{4} \log n} = \frac{1}{n \log n}. \quad (5.1.15)$$

Hence, $f_n \rightarrow 0$ as $n \rightarrow \infty$. The remaining hypotheses of **Theorem 5.1** were already verified above and continue to hold, so we retain

$$\frac{1}{\log v_n} \sum_{k=1}^n \left(\frac{\chi_k^4 S_k^2}{v_k^2} \right) \xrightarrow{a.s.} 1. \quad (5.1.16)$$

From (5.1.15) it follows that

$$\frac{\chi_n^4}{v_n^2} \sim \frac{\frac{\pi^2}{16n^2}}{\frac{\pi^2}{16} (\log n)^2} = \frac{1}{n^2 (\log n)^2},$$

so that (5.1.16) becomes

$$\frac{1}{\log v_n} \sum_{k=2}^n \left(\frac{S_k^2}{k^2 (\log k)^2} \right) \xrightarrow{a.s.} 1. \quad (5.1.17)$$

It remains to replace $\log v_n$ by $\log \log n$. Since $v_n \sim \frac{\pi}{4} \log n$,

$$\begin{aligned} \log v_n &= \log \left(\frac{\pi}{4} \log n (1 + o(1)) \right) \\ &= \log \left(\frac{\pi}{4} \right) + \log \log n + \log(1 + o(1)). \end{aligned}$$

Because $\log(\frac{\pi}{4})$ is constant, and $\log(1 + o(1)) \rightarrow 0$, the only asymptotically relevant term is $\log \log n$ since this term diverges to infinity. More precisely,

$$\frac{\log v_n}{\log \log n} = 1 + \frac{\log \frac{\pi}{4} + \log(1 + o(1))}{\log \log n} \rightarrow 1,$$

as $n \rightarrow \infty$, so that

$$\log v_n \sim \log \log n. \quad (5.1.18)$$

Inserting this into (5.1.17), we arrive at the **Quadratic Strong Law (QSL)** for the critical regime:

$$\lim_{n \rightarrow \infty} \frac{1}{\log \log n} \sum_{k=2}^n \left(\frac{S_k}{k \log k} \right)^2 \stackrel{a.s.}{=} 1, \quad (p = \frac{3}{4}). \quad (5.1.19)$$

This completes the proof of the Quadratic Strong Law in both the diffusive and critical regimes.

The critical regime convergence is illustrated in Figure 5.2, where the Q_n quantity now carries an additional $\log k$ in the denominator.

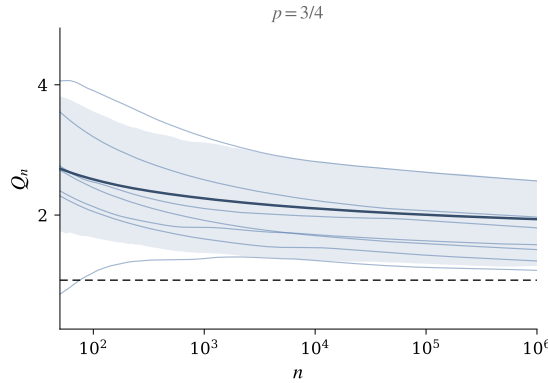


Figure 5.2: Running critical-regime QSL statistic $Q_n = (\log \log n)^{-1} \sum_{k=2}^n (S_k / (k \log k))^2$ for the ERW at $p = 3/4$: eight sample paths (thin), ensemble mean over 400 paths (bold), interquartile band (shaded), and theoretical limit 1 (dashed).

This completes the first half of the tools we need for the main statement of Section 5.3. The Law of the Iterated Logarithm, treated in the next section, completes the other half.

5.2 THE LAW OF THE ITERATED LOGARITHM

The Quadratic Strong Law controls the growth of $(S_n)_{n \geq 0}^2$ in an average sense: it gives the almost sure limit of the running average $\frac{1}{\log n} \sum_{k=1}^n (S_k/k)^2$, which tells us something about the typical size of the walk, but not how large or small S_n itself can get on a single path. The Law of the Iterated Logarithm fills this gap. As established by [Ber18], it gives the almost sure upper and lower bounds on the fluctuations of $(S_n)_{n \geq 0}$, for $p < 3/4$,

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} = \frac{1}{\sqrt{3-4p}}, \quad \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} = -\frac{1}{\sqrt{3-4p}},$$

while for $p = 3/4$,

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log n}} = 1, \quad \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log n}} = -1.$$

These bounds will be the key ingredient in proving recurrence in [Section 5.3](#).

We now follow the proof of [Ber18] to establish these bounds. First we define the **Predictable Quadratic Variation** $\langle \mathcal{M}_n \rangle$ associated with $(\mathcal{M}_n)_{n \geq 0}$ as $\langle \mathcal{M}_0 \rangle = 0$ and

$$\langle \mathcal{M}_n \rangle = \sum_{k=1}^n \mathbb{E}[\Delta \mathcal{M}_k^2 \mid \mathcal{F}_{k-1}].$$

Since $\Delta \mathcal{M}_k = \chi_k \epsilon_k$, this may be rewritten as

$$\begin{aligned} \langle \mathcal{M}_n \rangle &= \sum_{k=1}^n \mathbb{E}[(\chi_k \epsilon_k)^2 \mid \mathcal{F}_{k-1}] \\ &= \sum_{k=1}^n \chi_k^2 \mathbb{E}[\epsilon_k^2 \mid \mathcal{F}_{k-1}] \end{aligned}$$

We begin with the **diffusive regime** (i.e. $p < 3/4$). Recall from (5.1.4) that $\mathbb{E}[\epsilon_k^2 \mid \mathcal{F}_{k-1}] \xrightarrow{a.s.} 1$, so **Toeplitz lemma** ([Lemma A.5](#)) gives

$$\frac{\langle \mathcal{M}_n \rangle}{v_n} = \frac{\sum_{k=1}^n \chi_k^2 \mathbb{E}[\epsilon_k^2 \mid \mathcal{F}_{k-1}]}{\sum_{k=1}^n \chi_k^2} \xrightarrow{a.s.} 1. \quad (5.2.1)$$

To obtain the iterated logarithm bounds, we apply a martingale version of **Stout's law of the iterated logarithm**, which we state below in the form given by Hall and Heyde [[HH80](#), Theorem 4.8].

Theorem 5.2 (Martingale law of the iterated logarithm). *Let $(\mathcal{M}_n, \mathcal{F}_n)_{n \geq 0}$ be the martingale introduced above. Assume that there exists a deterministic sequence $(v_n)_{n \geq 1}$ such that*

1. $v_n \uparrow \infty$,
2. $\frac{\langle \mathcal{M} \rangle_n}{v_n} \xrightarrow{a.s.} 1$,
3. $\sum_{n=1}^{\infty} \frac{\mathbb{E}[(\Delta \mathcal{M}_n)^4 | \mathcal{F}_{n-1}]}{v_n^2} < \infty \quad a.s..$

Then

$$\limsup_{n \rightarrow \infty} \frac{\mathcal{M}_n}{\sqrt{2v_n \log \log v_n}} \stackrel{a.s.}{=} 1,$$

and

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{M}_n}{\sqrt{2v_n \log \log v_n}} \stackrel{a.s.}{=} -1.$$

We now verify the three conditions of [Theorem 5.2](#). Condition (1) was established in [\(5.1.7\)](#), where we showed that $v_n \rightarrow \infty$, and condition (2) was precisely [\(5.2.1\)](#). It remains to verify condition (3). Recall that

$$\Delta \mathcal{M}_n = \chi_n \epsilon_n, \quad \text{with} \quad \epsilon_n = \xi_n - \alpha \frac{S_{n-1}}{n-1}.$$

Since $|\xi_n| = 1$,

$$|\epsilon_n| \leq |\xi_n| + \left| \alpha \frac{S_{n-1}}{n-1} \right| \leq 1 + |\alpha| \leq 2,$$

and consequently

$$\mathbb{E}[(\Delta \mathcal{M})^4 | \mathcal{F}_{n-1}] = \chi_n^4 \mathbb{E}[\epsilon_n^4 | \mathcal{F}_{n-1}] \leq 16 \chi_n^4.$$

From the asymptotics established earlier,

$$\chi_n^4 \sim \Gamma(2p)^4 n^{4-8p}, \quad v_k^2 \sim \Gamma(2p)^4 \left(\frac{n^{3-4p}}{3-4p} \right)^2,$$

so that

$$\frac{\chi_n^4}{v_n^2} \sim (3-4p)^2 n^{-2}.$$

Using the classical identity $\sum_{k=1}^{\infty} k^{-2} = \frac{\pi^2}{6}$, we obtain

$$\sum_{k=1}^{\infty} \frac{\chi_k^4}{v_k^2} \sim (3-4p)^2 \sum_{k=1}^{\infty} k^{-2} < \infty. \quad (5.2.2)$$

It follows that

$$\sum_{n=1}^{\infty} \frac{\mathbb{E}[(\Delta \mathcal{M}_n)^4 | \mathcal{F}_{n-1}]}{v_n^2} \leq 16 \sum_{n=1}^{\infty} \frac{\chi_n^4}{v_n^2} < \infty,$$

which verifies condition (3). [Theorem 5.2](#) therefore applies

$$\limsup_{n \rightarrow \infty} \frac{M_n}{\sqrt{2\nu_n \log \log \nu_n}} \stackrel{a.s.}{=} 1, \quad \liminf_{n \rightarrow \infty} \frac{M_n}{\sqrt{2\nu_n \log \log \nu_n}} \stackrel{a.s.}{=} -1. \quad (5.2.3)$$

Writing $\mathcal{M}_n = \chi_n S_n$, this becomes

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\chi_n^{-1} \sqrt{2\nu_n \log \log \nu_n}} \stackrel{a.s.}{=} 1, \quad \liminf_{n \rightarrow \infty} \frac{S_n}{\chi_n^{-1} \sqrt{2\nu_n \log \log \nu_n}} \stackrel{a.s.}{=} -1.$$

It remains to simplify the normalising sequence. Combining [\(5.1.6\)](#) and

$$\nu_n = \Gamma(\alpha + 1)^2 \sum_{k=1}^n k^{-2\alpha} \sim \frac{\Gamma(\alpha + 1)^2}{1 - 2\alpha} n^{1-2\alpha},$$

together with

$$\log \log \nu_n \sim \log \log n,$$

we find

$$\chi_n^{-1} \sqrt{2\nu_n \log \log \nu_n} \sim \frac{n^\alpha}{\Gamma(\alpha + 1)} \sqrt{\frac{\Gamma(\alpha + 1)^2}{1 - 2\alpha} n^{1-2\alpha} \log \log n},$$

which simplifies to

$$\chi_n^{-1} \sqrt{2\nu_n \log \log \nu_n} \sim \sqrt{\frac{2n \log \log n}{1 - 2\alpha}}.$$

Expressing this in terms of p ,

$$\chi_n^{-1} \sqrt{2\nu_n \log \log \nu_n} \sim \frac{1}{\sqrt{3 - 4p}} \sqrt{2n \log \log n}.$$

Substituting back into the earlier result, we obtain the **Law of the Iterated Logarithm** (LIL) for the diffusive regime:

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} &\stackrel{a.s.}{=} \frac{1}{\sqrt{3 - 4p}} \\ \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} &\stackrel{a.s.}{=} -\frac{1}{\sqrt{3 - 4p}}. \end{aligned} \quad (5.2.4)$$

[Figure 5.3](#) provides numerical confirmation of [\(5.2.4\)](#): the scaled paths remain within the envelope $\pm 1/\sqrt{3 - 4p}$ and touch its boundary infinitely often, but never exceed it permanently. As p increases toward $3/4$, the envelope widens, reflecting that stronger memory causes S_n to reach larger almost sure values.

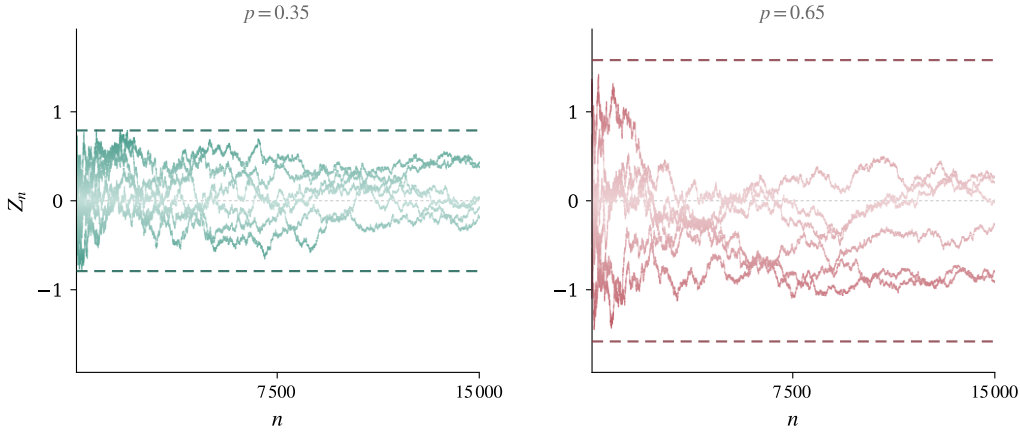


Figure 5.3: Scaled paths $Z_n = S_n / \sqrt{2n \log \log n}$ for the ERW with $q = \frac{1}{2}$, $n = 15,000$. Dashed lines show the LIL envelope $\pm 1 / \sqrt{3 - 4p}$.

We now turn to the **critical regime** $p = \frac{3}{4}$, for which $\alpha = \frac{1}{2}$. **Identity 1.4.3** gives

$$\chi_n \sim \Gamma\left(\frac{3}{2}\right) n^{-\frac{1}{2}} = \frac{\sqrt{\pi}}{2\sqrt{n}},$$

so that $\chi_n^2 \sim \frac{\pi}{4n}$ and

$$v_n \sim \frac{\pi}{4} \sum_{k=1}^n \frac{1}{k} \sim \frac{\pi}{4} \log n.$$

In particular $v_n \rightarrow \infty$. Note that all conditions of **Stout's martingale LIL** continue to hold, so (5.2.3) remains valid. In the critical case, however, the normalizing sequence behaves differently:

$$\chi_n^{-1} \sqrt{2v_n \log \log v_n} \sim \frac{2\sqrt{n}}{\sqrt{\pi}} \sqrt{\frac{\pi}{2} \log n \log \log v_n}.$$

From (5.1.18) we have

$$\log \log v_n \sim \log \log \log n,$$

and therefore

$$\chi_n^{-1} \sqrt{2v_n \log \log v_n} \sim \sqrt{2n \log n \log \log \log n}.$$

Substituting this back into (5.2.3) yields the **Law of the Iterated Logarithm** (LIL) for the critical regime:

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log \log n}} &\stackrel{a.s.}{=} 1. \\ \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log \log n}} &\stackrel{a.s.}{=} -1. \end{aligned} \tag{5.2.5}$$

Just like before, **Figure 5.4** provides an illustration of (5.2.5).

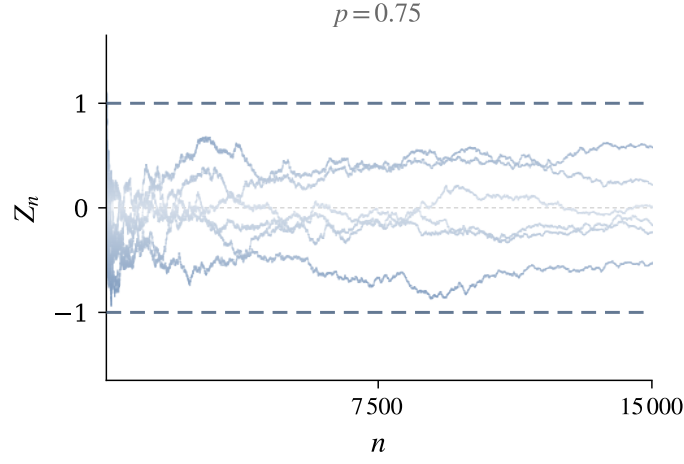


Figure 5.4: Sample paths of $Z_n := S_n / \sqrt{2n \log n \log \log \log n}$ for the critical ERW with $q = \frac{1}{2}$, $n = 15,000$. The dashed lines show the LIL envelope ± 1 .

The purpose of this section is to derive the fluctuation estimates needed for the recurrence result, and this only concerns the regimes $p \leq \frac{3}{4}$. For this reason, we do not discuss the superdiffusive case $p > \frac{3}{4}$ here. That regime will be treated separately in [Section 5.4](#), where transience is obtained from the corresponding almost sure limit theorem.

5.3 RECURRENCE OF THE ELEPHANT RANDOM WALK

A natural question to ask about the ERW is whether it returns to the origin infinitely often, or eventually drifts away never to return. These two behaviours are known as recurrence and transience, respectively, and we now make this distinction precise. Recall that we have assumed throughout that $S_0 = 0$.

Recall that, for each site $x \in \mathbb{Z}$, we define the occupation time $\mathcal{L}_n(x)$ as the number of visits to x up to time n , that is,

$$\mathcal{L}_n(x) := \sum_{k=0}^n \mathbb{1}_{\{S_k=x\}}.$$

As $n \rightarrow \infty$, the total number of visits to x is given by the limit

$$\mathcal{L}_\infty(x) := \lim_{n \rightarrow \infty} \mathcal{L}_n(x) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}.$$

The ERW is said to be **recurrent** if it returns to the origin infinitely often almost surely, that is, if

$$\mathbb{P}(\mathcal{L}_\infty(0) = \infty) = 1.$$

Conversely, the ERW is said to be **transient** if it visits the origin only finitely many times almost surely, meaning

$$\mathbb{P}(\mathcal{L}_\infty(0) < \infty) = 1.$$

In the diffusive and critical regimes, the Elephant Random Walk is recurrent. We follow the argument of [CGS17a].

Theorem 5.3. *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with $q \in [0, 1]$ and $p \in (0, \frac{3}{4}]$. Then $(S_n)_{n \geq 0}$ is **recurrent**, i.e.*

$$\mathbb{P}(\mathcal{L}_\infty(0) = \infty) = 1.$$

Proof. We split the argument according to the diffusive regime $p < \frac{3}{4}$ and the critical regime $p = \frac{3}{4}$.

We first consider the diffusive regime, i.e. $p < \frac{3}{4}$. Recall from Equation 5.2.4,

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} \stackrel{a.s.}{=} \frac{1}{\sqrt{3-4p}}.$$

Set

$$c := \frac{1}{\sqrt{3-4p}} > 0.$$

By the definition of the lim sup, on an event of probability one, there exists infinitely many n such that

$$\frac{S_n}{\sqrt{2n \log \log n}} \geq \frac{c}{2},$$

or equivalently,

$$S_n \geq \frac{c}{2} \sqrt{2n \log \log n}.$$

Since $\sqrt{2n \log \log n} \rightarrow \infty$ as $n \rightarrow \infty$, it follows that for every $N > 0$ there exist infinitely many n such that $S_n \geq N$, and hence

$$\limsup_{n \rightarrow \infty} S_n \stackrel{a.s.}{=} \infty.$$

The symmetric argument, starting from

$$\liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} \stackrel{a.s.}{=} -\frac{1}{\sqrt{3-4p}},$$

gives infinitely many n for which

$$\frac{S_n}{\sqrt{2n \log \log n}} \leq -\frac{c}{2},$$

and therefore

$$S_n \leq -\frac{c}{2} \sqrt{2n \log \log n}.$$

The same reasoning now yields, for every $N > 0$, infinitely many n with $S_n \leq -N$, so that

$$\liminf_{n \rightarrow \infty} S_n \stackrel{a.s.}{=} -\infty.$$

Combining the two conclusions,

$$\mathbb{P} \left(\left\{ \liminf_{n \rightarrow \infty} S_n = -\infty \text{ and } \limsup_{n \rightarrow \infty} S_n = \infty \right\} \right) = 1.$$

Now recall that the ERW is a **nearest-neighbor walk** on $\mathbb{Z}$, since each increment satisfies $\zeta_n \in \{-1, +1\}$. Therefore, on every sample path for which

$$\liminf_{n \rightarrow \infty} S_n = -\infty \quad \text{and} \quad \limsup_{n \rightarrow \infty} S_n = \infty,$$

the walk takes arbitrarily large positive values and arbitrarily large negative values. Since the walk moves in steps of size one, every passage from a positive value to a negative value, or conversely, must include a visit to the origin. Consequently,

$$S_n = 0 \quad \text{for infinitely many } n \quad \text{a.s.},$$

which shows that the walk is recurrent for $p < \frac{3}{4}$.

Suppose now $p = \frac{3}{4}$. By (5.2.5),

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log \log n}} \stackrel{\text{a.s.}}{=} 1, \quad \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log \log n}} \stackrel{\text{a.s.}}{=} -1.$$

Since

$$\sqrt{2n \log n \log \log \log n} \rightarrow \infty \quad \text{as } n \rightarrow \infty,$$

the same argument as in the diffusive case yields

$$\limsup_{n \rightarrow \infty} S_n = \infty \quad \text{and} \quad \liminf_{n \rightarrow \infty} S_n = -\infty \quad \text{a.s.}$$

Again using that the walk is nearest-neighbour on $\mathbb{Z}$, we conclude that it must visit the origin infinitely often almost surely. Hence the ERW is recurrent at $p = \frac{3}{4}$ as well and the proof is complete for all $p \leq \frac{3}{4}$. $\square$

To complement the previous statement, we conclude this section with a numerical illustration (Figure 5.5) of the counting zero process in the $p \leq \frac{3}{4}$ regime.

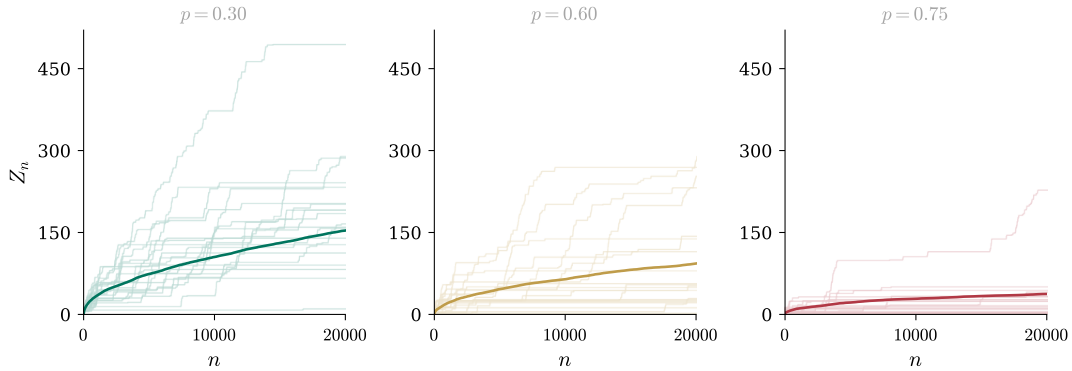


Figure 5.5: Simulated counting-zero process $Z_n = \sum_{k=0}^n \mathbb{1}_{\{S_k=0\}}$ for the ERW with $q = \frac{1}{2}$, $n = 20,000$. The faint curves represent sample paths, and the solid curve the empirical mean.

This identifies $p = \frac{3}{4}$ as the threshold between recurrence and transience for the ERW. Beyond this critical value, the walk enters the superdiffusive regime $p > \frac{3}{4}$, where its long-term behaviour changes fundamentally and recurrence no longer holds. This transient case will be studied in the next section.

A stronger notion than recurrence is that of **positive recurrence**. Whereas recurrence only requires that the walk returns to the origin almost surely, positive recurrence also asks whether the first return occurs in finite expected time. More precisely, if

$$\tau_0 := \inf\{n \geq 1 : S_n = 0\},$$

then positive recurrence of the origin corresponds to the condition

$$\mathbb{E}[\tau_0] < \infty.$$

For the Elephant Random Walk, Coletti and Papageorgiou [CP20] proved that this condition holds whenever

$$p < \frac{1}{6}.$$

Thus, for sufficiently small memory parameter p , the walk not only returns to the origin almost surely, but does so quickly enough that the expected first return time is finite. Intuitively, this is consistent with the behaviour of the ERW for small p : the walk then has a stronger tendency to reverse previously chosen directions, which keeps it close to the origin.

This result was later sharpened by Bertoin [Ber22], who proved the full characterization

$$\mathbb{E}[\tau_0] < \infty \quad \Longleftrightarrow \quad p < \frac{1}{4}.$$

Hence, inside the recurrent regime $p \leq \frac{3}{4}$, there is a second threshold at $p = \frac{1}{4}$. For $p < \frac{1}{4}$, the ERW is positive recurrent. For $\frac{1}{4} \leq p \leq \frac{3}{4}$, the walk is still recurrent, but the expected first return time to the origin is infinite. The proof of this stronger result falls outside the scope of the thesis, and we therefore restrict ourselves here to stating it as an additional fact.

5.4 TRANSIENCE OF THE ELEPHANT RANDOM WALK

In the superdiffusive regime, the **Elephant Random Walk** is no longer guaranteed to return to its starting point infinitely often, that is, the ERW becomes **transient**. Intuitively, the memory is now so persistent that the walk is consistently pulled in one direction, making indefinite return to the origin impossible. This section makes that intuition precise.

Recall from the proof of [Theorem 4.10](#) that in the superdiffusive regime,

$$\frac{S_n}{n^\alpha} \xrightarrow{a.s.} L,$$

where L is a non-degenerate, non-Gaussian random variable. In particular, for almost every $\omega \in \Omega$, the convergence holds pointwise:

$$\frac{S_n(\omega)}{n^\alpha} \rightarrow L(\omega). \quad (5.4.1)$$

Moreover, Qin establishes in [[Qin23](#), Proposition 1.5] that

$$\mathbb{P}(L = 0) = 0. \quad (5.4.2)$$

A proof of this fact falls outside the scope of the thesis and is therefore omitted. Together, however, (5.4.1) and (5.4.2) provide exactly what is needed to establish transience in the superdiffusive regime.

Theorem 5.4. *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with $q \in [0, 1]$ and $p \in (\frac{3}{4}, 1)$. Then $(S_n)_{n \geq 0}$ is **transient**, i.e.*

$$\mathbb{P}(\mathcal{L}_\infty(0) < \infty) = 1.$$

Proof. Define

$$\mathcal{A} := \{\omega \in \Omega : L(\omega) \neq 0\},$$

so that $\mathbb{P}(\mathcal{A}) = 1$ by (5.4.2), and set

$$\Omega_0 := \left\{ \omega \in \Omega : \frac{S_n(\omega)}{n^\alpha} \rightarrow L(\omega) \right\}.$$

Fix $\omega \in \Omega_0 \cap \mathcal{A}$, and note that

$$\mathbb{P}(\Omega_0 \cap \mathcal{A}) = 1.$$

By construction,

$$\frac{S_n(\omega)}{n^\alpha} \rightarrow L(\omega).$$

Choosing $\epsilon_\omega := \frac{|L(\omega)|}{2}$ in the definition of convergence, there exists $N_1(\omega) \in \mathbb{N}$ such that for all $n \geq N_1(\omega)$,

$$\left| \frac{S_n(\omega)}{n^\alpha} - L(\omega) \right| < \frac{|L(\omega)|}{2}.$$

By the **reverse triangle inequality**, this gives, for all $n \geq N_1(\omega)$,

$$\left| \frac{S_n(\omega)}{n^\alpha} \right| \geq |L(\omega)| - \left| \frac{S_n(\omega)}{n^\alpha} - L(\omega) \right| > \frac{|L(\omega)|}{2},$$

and hence

$$|S_n(\omega)| > \frac{|L(\omega)|}{2} n^\alpha, \quad (n \geq N_1(\omega)).$$

Since $\alpha > 0$,

$$\frac{|L(\omega)|}{2} n^\alpha \rightarrow \infty,$$

and consequently $|S_n(\omega)| \rightarrow \infty$. Now fix any site $x \in \mathbb{Z}$. There exists $N_2(\omega, x) \in \mathbb{N}$ such that for all $n \geq N_2(\omega, x)$,

$$\frac{|L(\omega)|}{2} n^\alpha > |x|.$$

Setting $N(\omega, x) := \max(N_1(\omega), N_2(\omega, x))$, it follows that

$$|S_n(\omega)| > |x| \quad \forall n \geq N(\omega, x),$$

and therefore $S_n(\omega) \neq x$ for all sufficiently large n . Hence every site $x \in \mathbb{Z}$ is visited only finitely often, and in particular

$$\mathcal{L}_\infty(0) < \infty.$$

Since this holds for every $\omega \in \Omega_0 \cap \mathcal{A}$ and $\mathbb{P}(\Omega_0 \cap \mathcal{A}) = 1$, we conclude

$$\mathbb{P}(\mathcal{L}_\infty(0) < \infty) = 1.$$

Thus the ERW is transient for $p > \frac{3}{4}$. □

INFERRING MEMORY FROM A PATH

Throughout the preceding chapters, the memory parameter p has been treated as a fixed and known quantity, and the analysis has focused on understanding how its value shapes the large scale behaviour of the walk. In practice, however, p is rarely known in advance. A natural question therefore arises: given an observed path $S_0, S_1, \dots, S_n$ of the ERW, how well can one recover the parameter p that generated it?

This chapter takes up that question, following the approach of Bercu and Laulin in [BL24]. We construct a quasi-maximum likelihood estimator $\hat{p}_n$ for the memory parameter from a single observed path and prove that it converges almost surely to the true parameter p . We also prove asymptotic normality of the estimator in the diffusive and superdiffusive regimes. Finally, we use these estimation results to construct confidence intervals for p , and study their finite-sample performance through simulations.

6.1 A QUASI-MAXIMUM LIKELIHOOD ESTIMATOR

In this section we construct an estimator for the unknown memory parameter p from a single observed trajectory of the ERW. The approach is as follows. We first compute the conditional probability of the next increment ξ_{n+1} given the history $\mathcal{F}_n$. This shows that ξ_{n+1} , conditional on the past, has a Rademacher distribution with a parameter depending on p and on the current position S_n . We then use these conditional probabilities to write down the likelihood of the full observed path $(\xi_1, \dots, \xi_n)$.

In principle, one could estimate p by maximizing this likelihood directly. However, the resulting likelihood equation does not lead to an explicit solution. We therefore replace the exact log-likelihood by its second order Taylor approximation. This approximation keeps the relevant dependence on p , while turning the maximization problem into one that can be solved in closed form. The maximizer of this method is called the **quasi-maximum likelihood estimator**. This procedure leads to an explicit estimator $\hat{p}_n$, whose strong consistency will be proved in [Theorem 6.1](#).

Recall that $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ contains all information about the walk up to time n . Conditional on $\mathcal{F}_n$, the event $\{\xi_{n+1} = 1\}$ can occur

in two ways: either a positive past increment is selected and repeated, or a negative past increment is selected and reversed. Hence

$$\mathbb{P}(\xi_{n+1} = 1 \mid \mathcal{F}_n) = p \frac{\sum_{k=1}^n \mathbb{1}_{\{\xi_k=1\}}}{n} + (1-p) \frac{\sum_{k=1}^n \mathbb{1}_{\{\xi_k=-1\}}}{n}. \quad (6.1.1)$$

The numbers of positive and negative increments satisfy

$$\sum_{k=1}^n \mathbb{1}_{\{\xi_k=1\}} + \sum_{k=1}^n \mathbb{1}_{\{\xi_k=-1\}} = n, \quad \sum_{k=1}^n \mathbb{1}_{\{\xi_k=1\}} - \sum_{k=1}^n \mathbb{1}_{\{\xi_k=-1\}} = S_n.$$

Solving these two equations gives

$$\sum_{k=1}^n \mathbb{1}_{\{\xi_k=1\}} = \frac{n + S_n}{2}, \quad \sum_{k=1}^n \mathbb{1}_{\{\xi_k=-1\}} = \frac{n - S_n}{2}.$$

Substitution into (6.1.1) yields

$$\mathbb{P}(\xi_{n+1} = 1 \mid \mathcal{F}_n) = \frac{p}{n} \frac{n + S_n}{2} + \frac{1-p}{n} \frac{n - S_n}{2} = \frac{1}{2} \left(1 + \alpha \frac{S_n}{n} \right),$$

where $\alpha = 2p - 1$. We denote this conditional probability by

$$p_n := \frac{1}{2} \left(1 + \alpha \frac{S_n}{n} \right). \quad (6.1.2)$$

Since $\xi_{n+1} \in \{-1, 1\}$, we also have

$$\mathbb{P}(\xi_{n+1} = -1 \mid \mathcal{F}_n) = 1 - p_n.$$

Thus, conditional on the past, ξ_{n+1} has a **Rademacher distribution** with parameter p_n , and for $\theta \in \{-1, 1\}$,

$$\mathbb{P}(\xi_{n+1} = \theta \mid \mathcal{F}_n) = p_n^{(1+\theta)/2} (1 - p_n)^{(1-\theta)/2}. \quad (6.1.3)$$

We now use this conditional distribution to construct the likelihood. For a fixed observed path $(\xi_1, \dots, \xi_n)$, the likelihood is the joint probability of this path, viewed as a function of p . By the chain rule,

$$\mathcal{L}_n(p) = \mathbb{P}(\xi_1) \prod_{k=1}^{n-1} \mathbb{P}(\xi_{k+1} \mid \mathcal{F}_k).$$

Using (6.1.3) and the initial distribution

$$\mathbb{P}(\xi_1 = 1) = q, \quad \mathbb{P}(\xi_1 = -1) = 1 - q,$$

we obtain

$$\mathcal{L}_n(p) = q^{(1+\xi_1)/2} (1 - q)^{(1-\xi_1)/2} \prod_{k=1}^{n-1} p_k^{(1+\xi_{k+1})/2} (1 - p_k)^{(1-\xi_{k+1})/2}, \quad (6.1.4)$$

where $p_k = \frac{1}{2} \left(1 + \alpha \frac{S_k}{k}\right)$. Maximizing $\mathcal{L}_n(p)$ is equivalent to maximizing its logarithm, since the logarithm is strictly increasing. We therefore define the log-likelihood $\ell_n(p) := \log \mathcal{L}_n(p)$. From (6.1.4),

$$\begin{aligned} \ell_n(p) = & \sum_{k=1}^{n-1} \left[\frac{1 + \xi_{k+1}}{2} \log p_k + \frac{1 - \xi_{k+1}}{2} \log(1 - p_k) \right] \\ & + \frac{1 + \xi_1}{2} \log q + \frac{1 - \xi_1}{2} \log(1 - q). \end{aligned} \quad (6.1.5)$$

The last two terms do not depend on p . Moreover, since

$$p_k = \frac{1}{2} \left(1 + (2p - 1) \frac{S_k}{k}\right),$$

we have

$$\frac{dp_k}{dp} = \frac{S_k}{k}.$$

Differentiating (6.1.5) then gives

$$\ell'_n(p) = \sum_{k=1}^{n-1} \frac{S_k}{k} \left(\frac{1 + \xi_{k+1}}{2p_k} - \frac{1 - \xi_{k+1}}{2(1 - p_k)} \right).$$

Substituting the expressions for p_k and $1 - p_k$, this becomes

$$\ell'_n(p) = \sum_{k=1}^{n-1} \frac{S_k}{k} \left(\frac{1 + \xi_{k+1}}{1 + \alpha S_k/k} - \frac{1 - \xi_{k+1}}{1 - \alpha S_k/k} \right).$$

Combining the two fractions yields

$$\ell'_n(p) = \sum_{k=1}^{n-1} \frac{2(S_k/k) (\xi_{k+1} - \alpha S_k/k)}{1 - \alpha^2 (S_k/k)^2}. \quad (6.1.6)$$

Using $\xi_{k+1}^2 = 1$, this can also be written as

$$\ell'_n(p) = \sum_{k=1}^{n-1} \frac{2\xi_{k+1}(S_k/k)}{1 + \alpha \xi_{k+1}(S_k/k)}.$$

Although this gives the likelihood equation explicitly, solving $\ell'_n(p) = 0$ does not lead to a closed form solution. We therefore replace $\ell_n(p)$ by its second order **Taylor approximation**.

The approximation is taken around the point 0 in the variables S_k/k . This is asymptotically justified because **Corollary 4.5** gives

$$\frac{S_n}{n} \xrightarrow{a.s.} 0,$$

as $n \rightarrow \infty$, so the logarithmic terms in (6.1.5) are asymptotically evaluated near their expansion point. Recall that, for $x \rightarrow 0$,

$$\log(1 + x) = x - \frac{x^2}{2} + o(x^2).$$

Applying this expansion to

$$\log p_k = -\log 2 + \log \left(1 + \alpha \frac{S_k}{k} \right)$$

and

$$\log(1 - p_k) = -\log 2 + \log \left(1 - \alpha \frac{S_k}{k} \right),$$

and keeping terms up to second order, gives the approximation

$$\begin{aligned} \lambda_n(p) &= \sum_{k=1}^{n-1} \alpha \frac{S_k}{k} \left(\xi_{k+1} - \frac{\alpha}{2} \frac{S_k}{k} \right) - (n-1) \log 2 \\ &\quad + \frac{1 + \xi_1}{2} \log q + \frac{1 - \xi_1}{2} \log(1 - q). \end{aligned} \quad (6.1.7)$$

This is why the resulting estimator is called a **quasi-maximum likelihood estimator**: it maximizes an approximation of the exact log-likelihood rather than the exact log-likelihood itself.

Since $\alpha = 2p - 1$, differentiating (6.1.7) with respect to p gives

$$\lambda'_n(p) = \sum_{k=1}^{n-1} 2\xi_{k+1} \frac{S_k}{k} \left(1 - \alpha \xi_{k+1} \frac{S_k}{k} \right), \quad \lambda''_n(p) = -4 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2.$$

For $n \geq 2$, the sum in the second derivative is positive. Hence $\lambda''_n(p) < 0$, and λ_n is strictly concave. Therefore its unique global maximum is obtained by solving $\lambda'_n(p) = 0$ [Dir25, Theorem 2.17]. The equation $\lambda'_n(p) = 0$ gives

$$\sum_{k=1}^{n-1} 2\xi_{k+1} \frac{S_k}{k} \left(1 - \alpha \xi_{k+1} \frac{S_k}{k} \right) = 0.$$

Dividing by 2 and expanding gives

$$\sum_{k=1}^{n-1} \xi_{k+1} \frac{S_k}{k} = \alpha \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2,$$

where we used $\xi_{k+1}^2 = 1$. Since $\alpha = 2p - 1$, this gives

$$2p - 1 = \frac{\sum_{k=1}^{n-1} \xi_{k+1} S_k / k}{\sum_{k=1}^{n-1} (S_k / k)^2}.$$

Solving for p , we obtain the estimator

$$\hat{p}_n = \frac{\sum_{k=1}^{n-1} \frac{S_k}{k} \left(\xi_{k+1} + \frac{S_k}{k} \right)}{2 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2}. \quad (6.1.8)$$

This expression depends only on the observed path $(\xi_1, \dots, \xi_n)$, and therefore provides an explicit estimator for the memory parameter p .

Because $\hat{p}_n$ is obtained from a Taylor approximation rather than from the exact likelihood, its reliability is not certain. However, the simulations in Figure 6.1 paint a promising picture. In the diffusive, critical, and superdiffusive regimes, the running estimator settles near the true value of p as n grows.

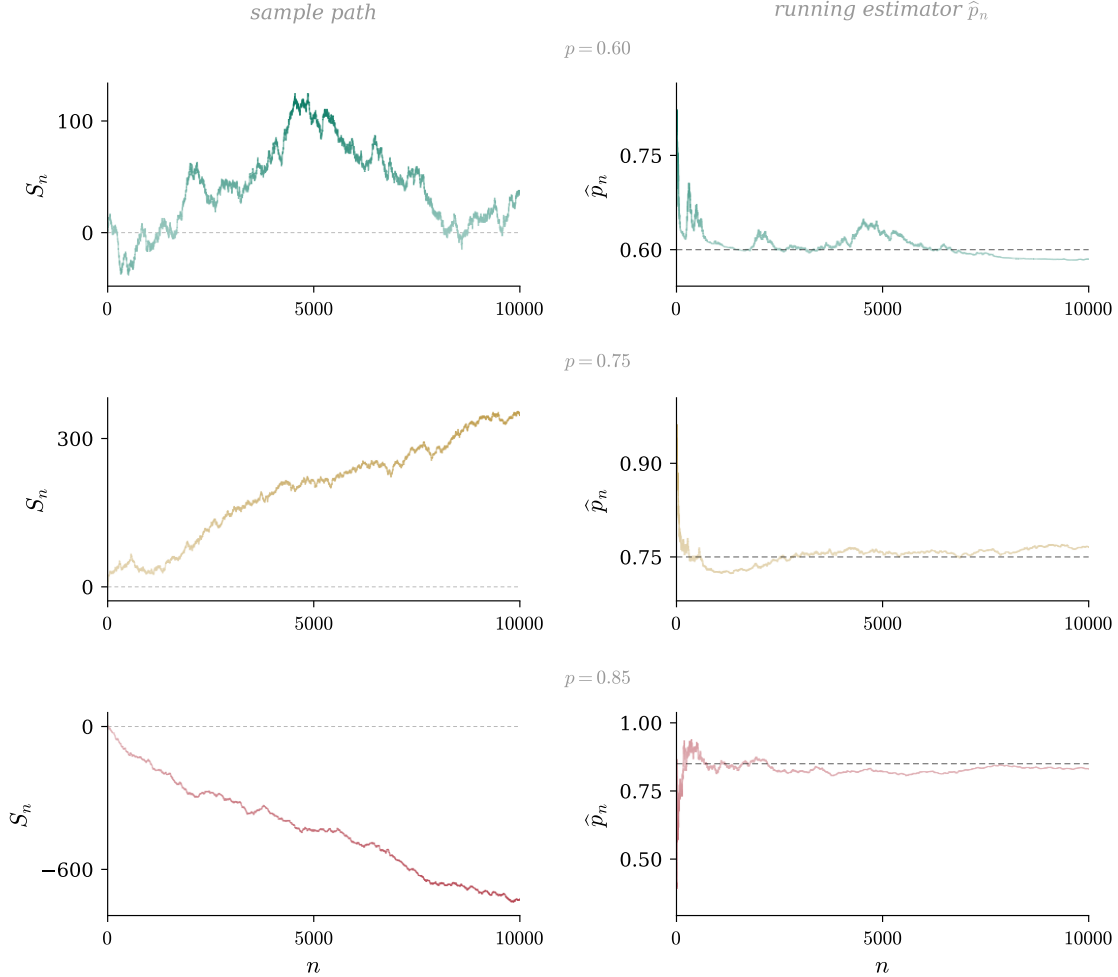


Figure 6.1: Sample paths of the ERW (left) and the running estimator $\hat{p}_n$ (right) for $p \in \{0.60, 0.75, 0.85\}$, with $q = \frac{1}{2}$ and $n = 10,000$. Dashed lines indicate the true value of p .

6.2 CONSISTENCY AND ASYMPTOTIC NORMALITY

In Section 6.1 we constructed an explicit estimator $\hat{p}_n$ for the memory parameter based on a single observed path of the ERW. We now show that this estimator is **strongly consistent**, in the sense that $\hat{p}_n \rightarrow p$ almost surely as $n \rightarrow \infty$. Thus, as the length of the observed trajectory increases, the estimator converges with probability one to the true underlying memory parameter. The following theorem from Bercu and Lailin in [BL24, Theorem 3.1] proves this.

Theorem 6.1. Let $(S_n)_{n \geq 0}$ be an Elephant Random Walk with parameters $(p, q) \in (0, 1) \times [0, 1]$, and let $\hat{p}_n$ be the estimator defined in (6.1.8). Then $\hat{p}_n$ is a strongly consistent estimator of p , i.e.,

$$\hat{p}_n \xrightarrow{a.s.} p \quad \text{as } n \rightarrow \infty.$$

Proof. We introduce two auxiliary processes. For $n \geq 2$, set

$$U_n := \sum_{k=1}^{n-1} \frac{S_k}{k} \left(\xi_{k+1} - \alpha \frac{S_k}{k} \right), \quad V_n := 4 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2,$$

and set $U_1 := 0$. The increments of U_n are given by

$$\Delta U_{k+1} := U_{k+1} - U_k = \frac{S_k}{k} \left(\xi_{k+1} - \alpha \frac{S_k}{k} \right), \quad k \geq 1.$$

Note that $|\Delta U_{k+1}| \leq 2$, since $|S_k/k| \leq 1$, $|\xi_{k+1}| = 1$, and $|\alpha| < 1$. Hence U_n is integrable, and even square-integrable. Moreover, using $\mathbb{E}[\xi_{k+1} | \mathcal{F}_k] = \alpha S_k/k$, we obtain

$$\mathbb{E}[\Delta U_{k+1} | \mathcal{F}_k] = \frac{S_k}{k} \left(\mathbb{E}[\xi_{k+1} | \mathcal{F}_k] - \alpha \frac{S_k}{k} \right) = 0.$$

Therefore $(U_n)_{n \geq 1}$ is a square-integrable **martingale** with respect to $(\mathcal{F}_n)_{n \geq 0}$. Recall the explicit expression (6.1.8),

$$\hat{p}_n = \frac{\sum_{k=1}^{n-1} \frac{S_k}{k} \left(\xi_{k+1} + \frac{S_k}{k} \right)}{2 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2}.$$

Adding and subtracting the term $\alpha \frac{S_k}{k}$ in the numerator gives

$$\hat{p}_n = \frac{\sum_{k=1}^{n-1} \frac{S_k}{k} \left(\xi_{k+1} - \alpha \frac{S_k}{k} \right) + (\alpha + 1) \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2}{2 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2}.$$

Since $\alpha = 2p - 1$, we have $\alpha + 1 = 2p$, so that

$$\hat{p}_n = \frac{U_n + 2p \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2}{2 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2} = p + \frac{U_n}{2 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2} = p + \frac{2U_n}{V_n}.$$

Equivalently,

$$\hat{p}_n - p = \frac{2U_n}{V_n}. \quad (6.2.1)$$

We next compute the predictable quadratic variation $\langle U \rangle_n$. By (1.2.1),

$$\begin{aligned} \langle U \rangle_n &= \sum_{k=1}^{n-1} \mathbb{E} \left[(\Delta U_{k+1})^2 \mid \mathcal{F}_k \right] \\ &= \sum_{k=1}^{n-1} \mathbb{E} \left[\left(\frac{S_k}{k} \left(\xi_{k+1} - \alpha \frac{S_k}{k} \right) \right)^2 \mid \mathcal{F}_k \right] \\ &= \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2 \mathbb{E} \left[\left(\xi_{k+1} - \alpha \frac{S_k}{k} \right)^2 \mid \mathcal{F}_k \right] \\ &= \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2 \left[1 - \alpha^2 \left(\frac{S_k}{k} \right)^2 \right]. \end{aligned}$$

So that

$$\langle U \rangle_n = \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2 - \alpha^2 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^4. \quad (6.2.2)$$

Dividing V_n by $\langle U \rangle_n$ yields

$$\frac{V_n}{\langle U \rangle_n} = \frac{4}{1 - \alpha^2 \left(\frac{\sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^4}{\sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2} \right)}. \quad (6.2.3)$$

We now aim to show that

$$\frac{\sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^4}{\sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2} \xrightarrow{a.s.} 0,$$

as $n \rightarrow \infty$. Once $V_n \rightarrow \infty$ almost surely is established, this convergence is an immediate consequence of **Toeplitz's Lemma** (Lemma A.5). We verify $V_n \rightarrow \infty$ separately by regime.

Diffusive case: By the **Quadratic Strong Law** from (5.1.13),

$$\frac{(3-4p)}{\log n} \sum_{k=1}^n \left(\frac{S_k}{k} \right)^2 \xrightarrow{a.s.} 1.$$

Equivalently

$$\frac{\frac{1}{4}(3-4p)V_n}{\log n} \xrightarrow{a.s.} 1,$$

and since $\log n \rightarrow \infty$, we must have that $V_n \rightarrow \infty$ almost surely.

Critical case: Similarly, by (5.1.19),

$$\frac{1}{\log \log n} \sum_{k=2}^n \left(\frac{S_k}{k \log k} \right)^2 \xrightarrow{a.s.} 1.$$

For every $k \geq 3$,

$$\left(\frac{S_k}{k}\right)^2 = (\log k)^2 \left(\frac{S_k}{k \log k}\right)^2 \geq \left(\frac{S_k}{k \log k}\right)^2.$$

Hence,

$$V_n = 4 \sum_{k=1}^{n-1} \left(\frac{S_k}{k}\right)^2 \geq 4 \sum_{k=2}^{n-1} \left(\frac{S_k}{k \log k}\right)^2.$$

After division by $\log \log n$, the right-hand side converges almost surely to 4. Consequently, $V_n \rightarrow \infty$ almost surely.

Superdiffusive case: Note that in this regime $2\alpha - 1 > 0$. We saw in [Chapter 4](#) that,

$$\frac{S_n}{n^\alpha} \xrightarrow{a.s.} L,$$

where L is a random variable. In addition, Qin establishes in [[Qin23](#), Proposition 1.5] that

$$\mathbb{P}(L = 0) = 0.$$

Hence,

$$\left(\frac{S_k}{k^\alpha}\right)^2 \xrightarrow{a.s.} L^2,$$

with $L^2 > 0$. Now observe that

$$\left(\frac{S_k}{k}\right)^2 = k^{2\alpha-2} \left(\frac{S_k}{k^\alpha}\right)^2.$$

Since $2\alpha - 2 > -1$, we have by [Lemma 1.2](#),

$$\sum_{k=1}^n k^{2\alpha-2} \sim \frac{n^{2\alpha-1}}{2\alpha-1}.$$

[Toeplitz's lemma](#) ([Lemma A.5](#)) applied to

$$\left(\frac{S_k}{k^\alpha}\right)^2 \xrightarrow{a.s.} L^2$$

with weights $k^{2\alpha-2}$, then yields

$$\frac{1}{n^{2\alpha-1}} \sum_{k=1}^n \left(\frac{S_k}{k}\right)^2 = \frac{1}{n^{2\alpha-1}} \sum_{k=1}^n k^{2\alpha-2} \left(\frac{S_k}{k^\alpha}\right)^2 \xrightarrow{a.s.} \frac{L^2}{2\alpha-1}.$$

Substituting $2\alpha - 1 = 4p - 3$,

$$\frac{V_n}{n^{4p-3}} = \frac{4}{n^{4p-3}} \sum_{k=1}^{n-1} \left(\frac{S_k}{k}\right)^2 \xrightarrow{a.s.} \frac{4L^2}{4p-3}. \quad (6.2.4)$$

Because $L^2 > 0$ almost surely, the limit on the right-hand side is almost surely strictly positive. Therefore, $V_n \rightarrow \infty$ almost surely. Therefore, we have now shown that

$$V_n \xrightarrow{a.s.} \infty,$$

for all $p \in (0, 1)$. **Toeplitz Lemma** (Lemma A.5) combined with the fact that $\left(\frac{S_n}{n}\right)^2 \rightarrow 0$ almost surely, gives

$$\frac{\sum_{k=1}^{n-1} \left(\frac{S_k}{k}\right)^4}{\sum_{k=1}^{n-1} \left(\frac{S_k}{k}\right)^2} \xrightarrow{a.s.} 0.$$

Combining this with (6.2.3) yields

$$\frac{V_n}{\langle U \rangle_n} \xrightarrow{a.s.} 4. \quad (6.2.5)$$

In particular

$$\langle U \rangle_n \xrightarrow{a.s.} \infty.$$

We now invoke the **Law of Large Numbers for Martingales** (see, e.g., [Duf97, Theorem 1.3.15]).

Theorem 6.2 (Law of Large Numbers for Martingales).
Let $(M_n)_{n \geq 1}$ be a square-integrable martingale with predictable quadratic variation $\langle M \rangle_n$. If

$$\langle M \rangle_n \xrightarrow{a.s.} \infty,$$

then

$$\frac{M_n}{\langle M \rangle_n} \xrightarrow{a.s.} 0.$$

Applying Theorem 6.2 yields

$$\frac{U_n}{\langle U \rangle_n} \xrightarrow{a.s.} 0.$$

Combined with the inverse of (6.2.5),

$$\frac{\langle U \rangle_n}{V_n} \xrightarrow{a.s.} \frac{1}{4},$$

we deduce that

$$\frac{U_n}{V_n} = \frac{U_n}{\langle U \rangle_n} \cdot \frac{\langle U \rangle_n}{V_n} \xrightarrow{a.s.} 0.$$

In view of (6.2.1),

$$\hat{p}_n - p \xrightarrow{a.s.} 0,$$

and hence

$$\hat{p}_n \xrightarrow{a.s.} p,$$

for all $p \in (0, 1)$.

□

By [Theorem 6.1](#), $\hat{p}_n$ is strongly consistent. Thus, asymptotically, the estimator does find the true memory parameter. That alone is not enough for confidence intervals. For those, we also need to know how large the error $\hat{p}_n - p$ is and how it is distributed.

The next theorem gives us this in the diffusive regime: once rescaled by $\sqrt{\log n}$, the estimator is **asymptotically normal**. The result, and the proof we present below, are due to Bercu and Laulin [[BL24](#), Theorem 3.2].

Theorem 6.3. *Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with parameters $p \in (0, \frac{3}{4})$ and $q \in [0, 1]$, and let $\hat{p}_n$ be the estimator defined in [\(6.1.8\)](#). Then*

$$\frac{2\sqrt{\log n} (\hat{p}_n - p)}{\sqrt{3 - 4p}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

In particular,

$$\frac{2\sqrt{\log n} (\hat{p}_n - p)}{\sqrt{3 - 4\hat{p}_n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1). \quad (6.2.6)$$

Proof. From the computations in the proof of [Theorem 6.1](#), we know that

$$\frac{V_n}{\langle U \rangle_n} \xrightarrow{\text{a.s.}} 4. \quad (6.2.7)$$

Moreover, in the diffusive regime $p < 3/4$, the **Quadratic Strong Law** from [\(5.1.13\)](#) gives

$$\frac{V_n}{\log n} \xrightarrow{\text{a.s.}} \frac{4}{3 - 4p}. \quad (6.2.8)$$

Combining [\(6.2.7\)](#) and [\(6.2.8\)](#), we also obtain

$$\frac{\langle U \rangle_n}{\log n} \xrightarrow{\text{a.s.}} \frac{1}{3 - 4p}. \quad (6.2.9)$$

Consider the following **Martingale Central Limit Theorem** [HH80, Corollary 3.1]

Theorem 6.4. *Let $(D_{n,k})_{1 \leq k \leq r_n}$ be a square-integrable martingale difference array with respect to filtrations $(\mathcal{G}_{n,k})_{0 \leq k \leq r_n}$, meaning that*

$$\mathbb{E}[D_{n,k} \mid \mathcal{G}_{n,k-1}] = 0, \quad D_{n,k} \in \mathcal{G}_{n,k}.$$

If the conditional variance condition

$$\sum_{k=1}^{r_n} \mathbb{E}[D_{n,k}^2 \mid \mathcal{G}_{n,k-1}] \xrightarrow{\text{a.s.}} \sigma^2$$

holds for some constant $\sigma^2 \geq 0$, and if the conditional Lindeberg condition

$$\sum_{k=1}^{r_n} \mathbb{E}\left[D_{n,k}^2 \mathbb{1}_{\{|D_{n,k}| > \varepsilon\}} \mid \mathcal{G}_{n,k-1}\right] \xrightarrow{\text{a.s.}} 0$$

holds for every $\varepsilon > 0$, then

$$\sum_{k=1}^{r_n} D_{n,k} \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2).$$

We now apply **Theorem 6.4** to

$$D_{n,k} := \frac{\Delta U_{k+1}}{\sqrt{\log n}}, \quad 1 \leq k \leq n-1.$$

Since $\mathbb{E}[\Delta U_{k+1} \mid \mathcal{F}_k] = 0$, this is a martingale difference array. Moreover, from (6.2.9),

$$\frac{1}{\log n} \sum_{k=1}^{n-1} \mathbb{E}[(\Delta U_{k+1})^2 \mid \mathcal{F}_k] = \frac{\langle U \rangle_n}{\log n} \xrightarrow{\text{a.s.}} \frac{1}{3-4p}.$$

Thus the **conditional variance condition** holds with $\sigma^2 = \frac{1}{3-4p}$. It remains to check the **conditional Lindeberg condition**. Since

$$|\Delta U_{k+1}| = \left| \frac{S_k}{k} \left(\xi_{k+1} - \alpha \frac{S_k}{k} \right) \right| \leq 2,$$

we have, for every $\varepsilon > 0$,

$$\{|D_{n,k}| > \varepsilon\} = \{|\Delta U_{k+1}| > \varepsilon \sqrt{\log n}\} = \emptyset$$

for all sufficiently large n . Hence

$$\sum_{k=1}^{n-1} \mathbb{E}\left[D_{n,k}^2 \mathbb{1}_{\{|D_{n,k}| > \varepsilon\}} \mid \mathcal{F}_k\right] = 0$$

for all sufficiently large n . Therefore the conditional Lindeberg condition holds. [Theorem 6.4](#) thus yields

$$\sum_{k=1}^{n-1} \frac{\Delta U_{k+1}}{\sqrt{\log n}} = \frac{U_n}{\sqrt{\log n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{3-4p}\right).$$

Thus, by [Slutsky's Theorem](#),

$$\frac{U_n}{\sqrt{\langle U \rangle_n}} = \frac{U_n}{\sqrt{\log n}} \sqrt{\frac{\log n}{\langle U \rangle_n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

We now return to the estimator. Since

$$\hat{p}_n - p = \frac{2U_n}{V_n},$$

we can write

$$\sqrt{\log n}(\hat{p}_n - p) = \frac{2U_n \sqrt{\log n}}{V_n} = \frac{2 \frac{U_n}{\sqrt{\log n}}}{\frac{V_n}{\log n}}.$$

Using

$$\frac{U_n}{\sqrt{\log n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{3-4p}\right) \quad \text{and} \quad \frac{V_n}{\log n} \xrightarrow{\text{a.s.}} \frac{4}{3-4p},$$

[Slutsky's Theorem](#) yields

$$\sqrt{\log n}(\hat{p}_n - p) \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{3-4p}{4}\right).$$

Equivalently,

$$\frac{2\sqrt{\log n}(\hat{p}_n - p)}{\sqrt{3-4p}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

In addition, Bercu and Laulin [\[BL24\]](#) proved the asymptotic normality result

$$\frac{2\sqrt{\log n}(\hat{p}_n - p)}{\sqrt{3-4\hat{p}_n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1),$$

for which we shall not provide a proof. $\square$

Bercu and Laulin show in [\[BL24, Remark 3.6\]](#) that the critical regime does not admit the same type of asymptotic normality. For the superdiffusive regime, however, it can be proved.

The next theorem establishes asymptotic normality in the superdiffusive regime, following Bercu and Laulin [\[BL24, Theorem 3.8\]](#). In their paper, the result is stated conditionally on the event $\{L^2 > 0\}$; however, Qin [\[Qin23\]](#) showed this to be an event of probability 1. Hence, no additional assumption is imposed.

Theorem 6.5. Let $(S_n)_{n \geq 0}$ be the Elephant Random Walk with parameters $p \in (\frac{3}{4}, 1)$ and $q \in [0, 1]$, and let $\hat{p}_n$ be the estimator defined in (6.1.8). Then

$$\sqrt{V_n}(\hat{p}_n - p) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Proof. We already showed that

$$\hat{p}_n - p = \frac{2U_n}{V_n},$$

and that

$$\frac{\langle U \rangle_n}{V_n} \xrightarrow{\text{a.s.}} \frac{1}{4}.$$

Since $p > \frac{3}{4}$, we have $\alpha > \frac{1}{2}$. By the superdiffusive limit theorem (4.2.18),

$$\frac{S_n}{n^\alpha} \xrightarrow{\text{a.s.}} L,$$

as $n \rightarrow \infty$, with $\mathbb{P}(\{L^2 > 0\}) = 1$. As in the proof of strong consistency (6.2.4), **Toeplitz's lemma** (Lemma A.5) gives

$$\frac{V_n}{n^{4p-3}} = \frac{4}{n^{2\alpha-1}} \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2 \xrightarrow{\text{a.s.}} \frac{4L^2}{4p-3}.$$

Since $\mathbb{P}(L^2 > 0) = 1$, the preceding convergence implies that $V_n \rightarrow \infty$ almost surely. Together with $\langle U \rangle_n / V_n \rightarrow 1/4$ almost surely, this also gives $\langle U \rangle_n \rightarrow \infty$ almost surely.

We now apply the **Martingale Central Limit Theorem** to $(U_n)_{n \geq 1}$ [HH80, Corollary 3.2]. Its increments are

$$|\Delta U_{k+1}| \leq \frac{|S_k|}{k} \left(|\xi_{k+1}| + |\alpha| \frac{|S_k|}{k} \right) \leq 1 + |\alpha| \leq 2.$$

Since $|S_k/k| \leq 1$, $|\xi_{k+1}| \leq 1$, and $|\alpha| \leq 1$, we have

$$|\Delta U_{k+1}| \leq 2.$$

Thus the **conditional Lindeberg condition** is satisfied. On the almost sure event $\{L^2 > 0\}$, we have

$$\frac{U_n}{\sqrt{\langle U \rangle_n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Finally, using $\hat{p}_n - p = 2U_n/V_n$, we write

$$\sqrt{V_n}(\hat{p}_n - p) = \frac{2U_n}{\sqrt{V_n}} = \frac{U_n}{\sqrt{\langle U \rangle_n}} \cdot 2\sqrt{\frac{\langle U \rangle_n}{V_n}}.$$

The first factor converges in distribution to $\mathcal{N}(0, 1)$, while the second factor converges almost surely to 1. By **Slutsky's Theorem** ([Theorem A.7](#)),

$$\sqrt{V_n}(\hat{p}_n - p) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

□

In [Section 6.3](#) we apply what we have developed here to construct confidence intervals for the memory parameter p .

6.3 EXAMPLE: CONFIDENCE INTERVALS

Building on the strong consistency ([Theorem 6.1](#)) and asymptotic normality ([Theorem 6.3](#) & [Theorem 6.5](#)) of the estimator $\hat{p}_n$, this section constructs **confidence intervals** for p and examines their behaviour for finite n via a **Monte Carlo study**. Since in practice one observes only a single trajectory $(\xi_1, \dots, \xi_n)$, it is useful to have a measure of how reliable the estimate is; the Monte Carlo study provides this.

Confidence Intervals for the Diffusive Regime

We begin with the **diffusive regime**, i.e. $p < 3/4$, and construct a confidence interval for the memory parameter p . Given an observed trajectory up to time n and a prescribed confidence level $1 - \delta$, where $\delta \in (0, 1)$, the goal is to find a random interval $\mathcal{I}_n$, built from the path $\xi_1, \dots, \xi_n$, such that

$$\mathbb{P}(p \in \mathcal{I}_n) \approx 1 - \delta$$

for large n . The construction rests on the **asymptotic normality** established in [Theorem 6.3](#), which gives

$$\frac{2\sqrt{\log n}(\hat{p}_n - p)}{\sqrt{3 - 4\hat{p}_n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1). \quad (6.3.1)$$

Writing Φ for the **distribution function** of the standard normal and setting $z_{1-\delta/2} := \Phi^{-1}(1 - \delta/2)$, we have

$$\mathbb{P}(-z_{1-\delta/2} \leq Z \leq z_{1-\delta/2}) = \Phi(z_{1-\delta/2}) - \Phi(-z_{1-\delta/2}) = 1 - \delta$$

for $Z \sim \mathcal{N}(0, 1)$. Applying this to (6.3.1) yields, for large n ,

$$\mathbb{P}\left(-z_{1-\delta/2} \leq \frac{2\sqrt{\log n}(\hat{p}_n - p)}{\sqrt{3 - 4\hat{p}_n}} \leq z_{1-\delta/2}\right) \approx 1 - \delta.$$

Rearranging for p gives

$$\mathbb{P}\left(\hat{p}_n - \frac{\sqrt{3 - 4\hat{p}_n}}{2\sqrt{\log n}} z_{1-\delta/2} \leq p \leq \hat{p}_n + \frac{\sqrt{3 - 4\hat{p}_n}}{2\sqrt{\log n}} z_{1-\delta/2}\right) \approx 1 - \delta.$$

A level- $(1 - \delta)$ confidence interval for p is therefore

$$\mathcal{I}_n^{\text{diff}} = \left[\hat{p}_n - \frac{\sqrt{3 - 4\hat{p}_n}}{2\sqrt{\log n}} z_{1-\delta/2}, \hat{p}_n + \frac{\sqrt{3 - 4\hat{p}_n}}{2\sqrt{\log n}} z_{1-\delta/2} \right],$$

provided $\hat{p}_n < p_c$, otherwise the diffusive interval is undefined. The lower and upper endpoints of $\mathcal{I}_n^{\text{diff}}$ are

$$p_n^- := \hat{p}_n - z_{1-\delta/2} \frac{\sqrt{3 - 4\hat{p}_n}}{2\sqrt{\log n}}, \quad p_n^+ := \hat{p}_n + z_{1-\delta/2} \frac{\sqrt{3 - 4\hat{p}_n}}{2\sqrt{\log n}},$$

and the construction is illustrated in [Figure 6.2](#).

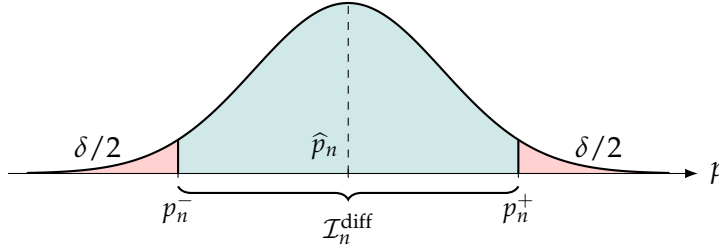


Figure 6.2: Schematic construction of the interval $\mathcal{I}_n^{\text{diff}} = [p_n^-, p_n^+]$ around the estimator $\hat{p}_n$.

Confidence Intervals for the Superdiffusive Regime

We now turn to the **superdiffusive regime**, i.e., $p > 3/4$. The construction of a confidence interval for p follows the same principle as in the diffusive case, but rests on a different normalisation. By [Theorem 6.5](#), the estimator $\hat{p}_n$ satisfies

$$\sqrt{V_n} (\hat{p}_n - p) \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1).$$

Applying the same symmetry argument as before yields, for large n ,

$$\mathbb{P}\left(-z_{1-\delta/2} \leq \sqrt{V_n} (\hat{p}_n - p) \leq z_{1-\delta/2}\right) \approx 1 - \delta.$$

Rearranging for p , a level- $(1 - \delta)$ confidence interval for the superdiffusive regime is

$$\mathcal{I}_n^{\text{sup}} = \left[\hat{p}_n - \frac{1}{\sqrt{V_n}} z_{1-\delta/2}, \hat{p}_n + \frac{1}{\sqrt{V_n}} z_{1-\delta/2} \right].$$

A General Confidence Interval

A more general result is due to Bercu and Lailin [[BL24](#), Theorem 3.11], who construct a confidence interval for p that requires

no prior knowledge of the regime. Their argument relies on the [Azuma–Hoeffding inequality](#) [Azu67]. As the proof falls outside the scope of this thesis, we state only the conclusion and refer the reader to [BL24, Theorem 3.11] for further details. For any $\delta \in (0, 1)$, the interval

$$\mathcal{I}_n^{\text{gen}} = \left[\hat{p}_n - \frac{2\sqrt{3n \log(2/\delta)}}{V_n}, \hat{p}_n + \frac{2\sqrt{3n \log(2/\delta)}}{V_n} \right],$$

satisfies $\mathbb{P}(p \in \mathcal{I}_n^{\text{gen}}) \geq 1 - \delta$ for all large n and every $p \in (0, 1)$.

Monte Carlo Study

The goal of the [Monte Carlo study](#) is to examine how the confidence intervals derived above perform for finite values of n . The theoretical results describe the asymptotic behaviour as $n \rightarrow \infty$, whereas in practice only a finite path is observed. To assess the finite-sample performance, we simulate independent ERW paths and compare the resulting confidence intervals with the true value of the memory parameter p .

We fix $\delta = 0.05$ throughout, corresponding to a confidence level of 95%. For each fixed value of p , we simulate N_{sim} independent paths up to the maximal time $n = 250,000$, with initial bias $q = 0.75$. At each recorded time n , every simulated path produces a confidence interval for p . To keep the notation general, let $\mathcal{I}_n^{(i)}$ denote any such confidence interval computed from the i -th simulated path. We define its [empirical coverage](#) by

$$\widehat{\text{Cov}}(\mathcal{I}_n) := \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} \mathbb{1}\{p \in \mathcal{I}_n^{(i)}\}.$$

Thus, $\widehat{\text{Cov}}_n(\mathcal{I})$ is the proportion of simulated confidence intervals that contain the true value of p .

This procedure is applied to the confidence intervals considered above. Regime-specific intervals are reported only in the regimes in which their derivation is justified: the diffusive interval is reported for $p < p_c$, while the superdiffusive interval is reported for $p > p_c$. At the critical value $p = p_c$, no regime-specific interval is reported. The general confidence interval is evaluated across all regimes. All intervals are computed using the same simulated paths.

The intervals are evaluated over the grid

$$(p, n) \in \{0.25, 0.50, 0.60, 0.65, 0.75, 0.80, 0.90\} \times \{500, 1000, 10000, 50000, 250000\}.$$

The results are presented in [Table 6.1](#). The table was produced with $N_{\text{sim}} = 5,000$ simulated paths and $q = 0.75$.

p	n	Mean $\hat{p}_n$	Diff. cov.	Mean diff. length	Sup. cov.	Mean sup. length	Gen. cov.	Mean gen. length
0.25	500	0.1620	0.9834	1.1782	–	–	1.0000	22.0449
0.25	1000	0.1646	0.9848	1.1166	–	–	1.0000	28.5248
0.25	10000	0.1780	0.9794	0.9588	–	–	1.0000	69.8282
0.25	50000	0.1847	0.9780	0.8812	–	–	1.0000	134.5237
0.25	250000	0.1915	0.9740	0.8187	–	–	1.0000	263.5251
0.50	500	0.3897	0.9504	0.8973	–	–	1.0000	14.6995
0.50	1000	0.3962	0.9524	0.8451	–	–	1.0000	18.7340
0.50	10000	0.4143	0.9546	0.7185	–	–	1.0000	43.8958
0.50	50000	0.4251	0.9584	0.6555	–	–	1.0000	82.5097
0.50	250000	0.4303	0.9568	0.6088	–	–	1.0000	159.3457
0.60	500	0.4916	0.9100	0.7543	–	–	1.0000	11.4017
0.60	1000	0.4960	0.9170	0.7076	–	–	1.0000	14.4189
0.60	10000	0.5170	0.9248	0.5885	–	–	1.0000	32.5483
0.60	50000	0.5274	0.9370	0.5327	–	–	1.0000	59.9444
0.60	250000	0.5347	0.9386	0.4906	–	–	1.0000	113.4589
0.65	500	0.5442	0.8654	0.6845	–	–	1.0000	9.7673
0.65	1000	0.5518	0.8766	0.6332	–	–	1.0000	12.1447
0.65	10000	0.5691	0.8978	0.5160	–	–	1.0000	26.7929
0.65	50000	0.5777	0.9074	0.4649	–	–	1.0000	48.8233
0.65	250000	0.5861	0.9110	0.4238	–	–	1.0000	91.3268
0.75	500	0.6530	–	–	–	–	1.0000	6.4701
0.75	1000	0.6599	–	–	–	–	1.0000	7.8067
0.75	10000	0.6733	–	–	–	–	1.0000	15.9271
0.75	50000	0.6828	–	–	–	–	1.0000	27.2414
0.75	250000	0.6900	–	–	–	–	1.0000	47.7767
0.80	500	0.7123	–	–	0.9502	0.4441	1.0000	4.8348
0.80	1000	0.7188	–	–	0.9474	0.4020	1.0000	5.6977
0.80	10000	0.7345	–	–	0.9412	0.2991	1.0000	10.4435
0.80	50000	0.7429	–	–	0.9452	0.2498	1.0000	16.8474
0.80	250000	0.7490	–	–	0.9380	0.2120	1.0000	28.1651
0.90	500	0.8520	–	–	0.9486	0.2602	1.0000	1.9705
0.90	1000	0.8582	–	–	0.9334	0.2173	1.0000	2.0573
0.90	10000	0.8738	–	–	0.9226	0.1223	1.0000	2.6048
0.90	50000	0.8823	–	–	0.9270	0.0819	1.0000	3.1649
0.90	250000	0.8884	–	–	0.9256	0.0547	1.0000	3.9315

Table 6.1: Monte Carlo coverage study for the ERW with $\delta = 0.05$, $q = 0.75$ and $N_{\text{sim}} = 5000$ simulated paths.

Several results of [Table 6.1](#) are worth highlighting.

- First, the sample mean of $\hat{p}_n$ in [Table 6.1](#) is often still noticeably below the true value of p , especially for smaller values of p . This does not contradict the strong consistency result of [Theorem 6.1](#), since that theorem is asymptotic and does not describe the speed of convergence. A way to understand this slow finite-sample behaviour is through the quantity

$$V_n = 4 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2,$$

which appears in the identity

$$\hat{p}_n - p = \frac{2U_n}{V_n}.$$

In this sense, V_n measures the amount of information about the memory parameter contained in a single observed path. By the Law of Large Numbers, $S_k/k \rightarrow 0$ almost surely, so the summands in V_n become small. In the diffusive regime, V_n grows only logarithmically, which means that the effective amount of information increases very slowly. Consequently, even for large but finite n , the estimator may still be far from its limiting value. For larger values of p , the walk is more persistent and the values of $|S_k|$ are typically larger, causing V_n to grow faster. This explains why the estimator improves more quickly in the superdiffusive part of Table 6.1.

- Second, the empirical coverage of the diffusive interval $\mathcal{I}_n^{\text{diff}}$ decreases as p grows towards the critical value $p_c = 3/4$. The reason for this is, for finite n , the estimator $\hat{p}_n$ fluctuates around p . If $\hat{p}_n > p_c$, then $\sqrt{3 - 4\hat{p}_n}$ is not real and $\mathcal{I}_n^{\text{diff}}$ is undefined; we count such realizations as non-coverage. As p approaches p_c , the probability that $\hat{p}_n$ exceeds p_c increases, so this source of non-coverage becomes more frequent. As n grows, $\hat{p}_n$ gets closer to p , which explains why the coverage for $p = 0.60$ and $p = 0.65$ improves a lot over time.

To visualize the coverage results of $\mathcal{I}_n^{\text{diff}}$ more clearly, they are plotted in Figure 6.3.

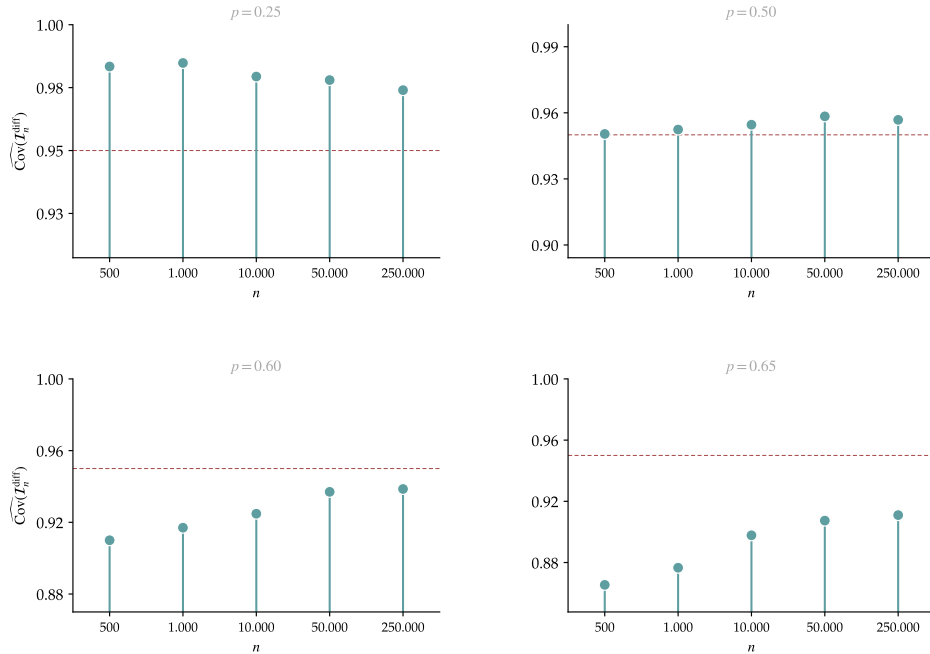


Figure 6.3: Empirical coverage of $\mathcal{I}_n^{\text{diff}}$ for $p \in \{0.25, 0.50, 0.60, 0.65\}$ across $n \in \{500, 1000, 10000, 50000, 250000\}$, based on $N_{\text{sim}} = 5000$ simulated paths with $q = 0.75$. The dashed line marks the confidence level $1 - \delta = 0.95$.

- In the superdiffusive regime, the empirical coverage of $\mathcal{I}_n^{\text{sup}}$ is slightly below the nominal level of $1 - \delta = 0.95$. This can be explained by the finite-sample behaviour of the estimator. As seen in Table 6.1, the sample mean of $\hat{p}_n$ still tends to lie below the true value of p . Since $\mathcal{I}_n^{\text{sup}}$ is centred around $\hat{p}_n$, this shifts the interval slightly to the left. As a consequence, the true value of p falls above the upper endpoint more often than expected under the limiting normal approximation.

The empirical coverage of $\mathcal{I}_n^{\text{sup}}$ is plotted in Figure 6.4.

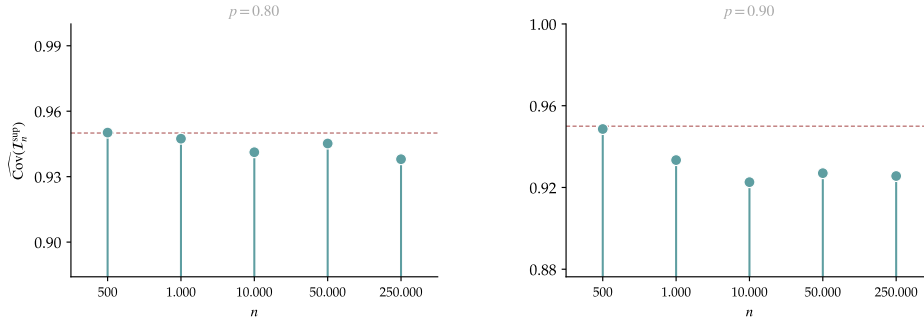


Figure 6.4: Empirical coverage of $\mathcal{I}_n^{\text{sup}}$ for $p \in \{0.80, 0.90\}$ across $n \in \{500, 1000, 10000, 50000, 250000\}$, based on $N_{\text{sim}} = 5000$ simulated paths with $q = 0.75$. The dashed line marks the confidence level $1 - \delta = 0.95$.

- The general confidence interval $\mathcal{I}_n^{\text{gen}}$ has empirical coverage equal to 1 across the whole grid in Table 6.1. However, this high coverage mainly comes from the fact that the interval is extremely wide. Since $\mathcal{I}_n^{\text{gen}}$ is constructed to work in all regimes at once, it does not use the sharper information available in the diffusive or superdiffusive case. Its width depends on $\sqrt{n}/V_n$, and in the simulations V_n is not large enough to make the interval informative. Thus, the interval is theoretically safe, but has limited practical value.

The Monte Carlo study suggests a practical recipe, but also shows its limitations near the critical value. Given a single dataset that is assumed to be generated by an Elephant Random Walk, the estimator $\hat{p}_n$ provides a point estimate of the underlying memory parameter p . The value of $\hat{p}_n$ then indicates which regime the walk most plausibly belongs to, and the corresponding regime-specific confidence interval can be computed from the same observed path.

The results of Table 6.1 show that this procedure is reasonably reliable whenever $\hat{p}_n$ lies safely within a regime, that is, away from the critical value $p_c = 3/4$. For such values the empirical coverage of the regime-specific intervals stays close to the nominal level $1 - \delta = 0.95$, and this already holds at moderate sample sizes rather than only in the asymptotic limit.

Near the critical value, however, the situation is different. If $\hat{p}_n$ lies in a prescribed neighborhood of p_c , the observed path contains too little information about the regime to make a safe choice between the diffusive and superdiffusive intervals. In that case one should not simply choose the interval that appears most favorable, since the true parameter is unknown in applications. The general interval $\mathcal{I}_n^{\text{gen}}$ is theoretically valid across regimes, but the Monte Carlo study shows that it is often so wide that it has little practical value. Thus, close to p_c , the most careful interpretation is that the data do not provide enough evidence to justify a regime specific choice.

OVERVIEW & OUTLOOK

This final chapter has two parts. We first revisit the main results of this thesis. We then look at a few topics that lie beyond the scope of this work but are a next step for further reading on the Elephant Random Walk.

7.1 SUMMARY OF RESULTS

Building on the foundations established in [Chapter 1](#), we began by analyzing the **Simple Random Walk** on $\mathbb{Z}^d$ in [Chapter 2](#). We showed that $\mathbb{E}[X_n] = 0$ and $\text{Var}(X_n) = n$, proved the Law of Large Numbers $X_n/n \xrightarrow{a.s.} 0$, and established the Central Limit Theorem

$$\frac{X_n}{\sqrt{n}} \xrightarrow{\mathcal{D}} \mathcal{N}\left(0, \frac{1}{d} I_d\right).$$

The chapter concluded with **Pólya's theorem**: the SRW is recurrent for $d = 1, 2$ and transient for $d \geq 3$. These results served both as an introduction and as a baseline for comparison, since the Simple Symmetric Random Walk is a special case of the ERW by setting $p = q = \frac{1}{2}$.

In [Chapter 3](#) we introduced the Elephant Random Walk and derived exact formulas for the expected value and second moment. For the expectation, we obtained

$$\mathbb{E}[S_n] = \beta \frac{\Gamma(n + \alpha)}{\Gamma(\alpha + 1) \Gamma(n)}, \quad \text{where } \alpha = 2p - 1, \beta = 2q - 1,$$

which grows as $\mathbb{E}[S_n] \sim \beta n^\alpha / \Gamma(\alpha + 1)$. The second moment revealed a phase transition at the critical value $p_c = \frac{3}{4}$, dividing the parameter space into three regimes:

$$\mathbb{E}[S_n^2] \sim \begin{cases} \frac{n}{3 - 4p}, & p < \frac{3}{4}, \\ n \log n, & p = \frac{3}{4}, \\ \frac{n^{4p-2}}{(4p-3) \Gamma(4p-2)}, & p > \frac{3}{4}. \end{cases}$$

This division into a diffusive, critical, and superdiffusive regime set the stage for the rest of the thesis. The chapter closed with an explicit computation of the increment correlations.

The limit theorems of [Chapter 4](#) made the consequences of this phase transition precise. The Law of Large Numbers showed that

$(S_n - \mathbb{E}[S_n])/n \xrightarrow{a.s.} 0$ for all parameters in the parameter space. The **Central Limit Theorem**, however, revealed fundamentally different behaviour in each regime. In the diffusive case, we proved

$$\frac{S_n - \frac{\beta}{\Gamma(\alpha+1)} n^\alpha}{\sqrt{\frac{n}{3-4p}}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1), \quad p < \frac{3}{4}.$$

At the critical point, a Gaussian limit still holds but with an adjusted normalization:

$$\frac{S_n - \frac{2\beta}{\sqrt{\pi}} \sqrt{n}}{\sqrt{n \log n}} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1), \quad p = \frac{3}{4}.$$

In the superdiffusive regime, Gaussian behaviour breaks down and the rescaled walk converges almost surely to a non-degenerate, non-Gaussian random variable:

$$\frac{S_n}{n^\alpha \Gamma(\alpha+1)^{-1}} - \beta \xrightarrow{a.s.} M, \quad p > \frac{3}{4}.$$

The non-Gaussianity of M was established through explicit moment computations, and a numerical exploration illustrated how both p and q jointly shape the limiting distribution when $p > \frac{3}{4}$.

Chapter 5 turned to the pathwise behaviour of the ERW. In the diffusive regime, we established

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} \stackrel{a.s.}{=} \frac{1}{\sqrt{3-4p}}, \quad \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log \log n}} \stackrel{a.s.}{=} -\frac{1}{\sqrt{3-4p}},$$

while at the critical point the envelope takes the form

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log \log n}} \stackrel{a.s.}{=} 1, \quad \liminf_{n \rightarrow \infty} \frac{S_n}{\sqrt{2n \log n \log \log \log n}} \stackrel{a.s.}{=} -1.$$

These bounds were then used to settle the question of recurrence and transience: for $p \leq \frac{3}{4}$, the walk oscillates between arbitrarily large positive and negative values and must therefore cross the origin infinitely often, while for $p > \frac{3}{4}$, the strong memory drives the walk to infinity eventually, making the walk transient. The critical value $p_c = \frac{3}{4}$ thus emerges once more as a threshold, this time between recurrence and transience.

In **Chapter 6** we introduced the estimator

$$\hat{p}_n = \frac{\sum_{k=1}^{n-1} \frac{S_k}{k} \left(\tilde{\zeta}_{k+1} + \frac{S_k}{k} \right)}{2 \sum_{k=1}^{n-1} \left(\frac{S_k}{k} \right)^2}$$

for the memory parameter, constructed from a single observed path. We established strong consistency $\hat{p}_n \xrightarrow{a.s.} p$ and asymptotic normality, building on the regime-specific growth results of the preceding chapters. In [Section 6.3](#), these properties were used to construct confidence intervals for p , whose finite-sample quality was then studied via Monte Carlo simulation.

Taken together, all these results paint a picture of how memory shapes the behaviour of a random walk. The critical value $p_c = \frac{3}{4}$ appears as a boundary throughout: it separates diffusive from superdiffusive growth, Gaussian from non-Gaussian limit laws, and recurrent from transient behaviour.

7.2 DIRECTIONS FOR FURTHER STUDY

The Elephant Random Walk raises many further questions beyond those addressed in this thesis, and the literature devoted to them has grown rapidly in recent years. In the remainder of this section we briefly state a number of these further directions, accompanied by references to the papers in which they are studied in detail.

Multi-Dimensional Elephant Random Walk

In this thesis we restricted the Elephant Random Walk to a single dimension. It is, however, possible to extend the model to a walk on $\mathbb{Z}^d$, in the same way as for the SRW. This extension is called the **Multi-Dimensional Elephant Random Walk** (MERW), and is studied by Bercu and Laulin in [\[BL17\]](#).

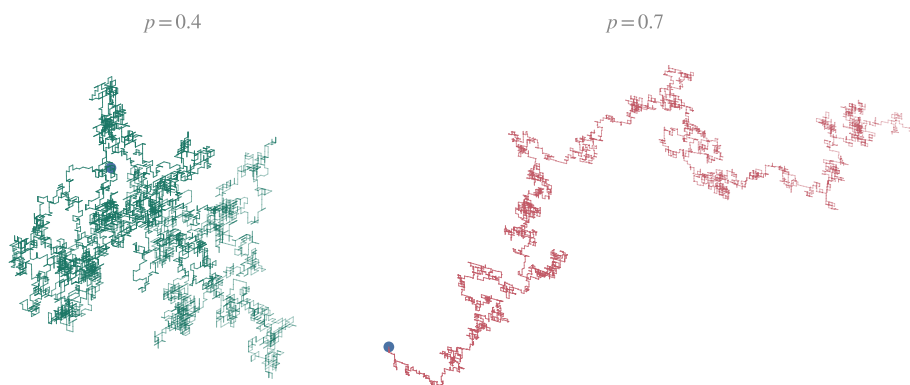


Figure 7.1: Sample paths of the MERW on $\mathbb{Z}^3$ for $n = 6,000$. Left: $p = 0.4$. Right: $p = 0.7$.

They introduce the appropriate d -dimensional dynamics, identify the critical value of the memory parameter as $p_d = \frac{2d+1}{4d}$, and extend the main asymptotic results of the one-dimensional ERW to arbitrary

dimension in all three regimes. The recurrence and transience of the MERW was settled recently by Qin [Qin23]: for $d \geq 3$ the walk is transient for every $p \in [0, 1]$, resolving a conjecture of Bertoin, while for $d = 2$ a phase transition occurs at $p_2 = 5/8$, with the MERW being recurrent for $p < 5/8$ and transient for $p \geq 5/8$. The interested reader is referred to [BL17, Qin23] for further details.

Elephant Random Walk with Delays

Throughout this thesis the elephant has been required to move at every step, with $\xi_n \in \{-1, +1\}$. A variation, suggested by Bercu, Chabanol, and Ruch and developed in detail by Gut and Stadtmüller in [GS19], is to allow the walker to remain in place at each time with some positive probability. In this extension, called the **Elephant Random Walk with delays** (ERWD), the increment ξ_{n+1} copies a uniformly chosen past step with probability p , reverses it with probability q , and equals zero with probability r , where $p + q + r = 1$. The role of the memory parameter is now played by $\alpha = p - q$, and the critical value separating the three regimes is $\alpha = 1/2$. Setting $r = 0$ recovers the familiar threshold $p = 3/4$ from the standard ERW. Gut and Stadtmüller establish a strong law of large numbers and a central limit theorem in each of the three regimes, with the limit laws now containing an additional Dirac mass at the origin to account for the steps spent in place. We refer to [GS19] for the full picture.

What Remains Open

THE SUPERDIFFUSIVE LIMIT LAW. A first natural direction for further research is the limiting random variable in the superdiffusive regime. In this thesis we have seen that, for $p > 3/4$, the properly rescaled ERW converges almost surely to a non-degenerate and non-Gaussian random variable. Compared with the Gaussian limits in the diffusive and critical regimes, this limit law is much less explicit. Its precise density, tail behaviour, and dependence on the parameters p and q remain open questions. Recent work has made progress by relating the limit law to fixed-point equations and Pólya-type urn models [GLR23], but a complete description is still an active topic of study.

GENERALIZED ELEPHANT RANDOM WALKS. A second direction is to study generalizations of the ERW. In the classical model, the probability of the next step is a linear function of the past proportion of positive and negative increments. This linearity makes the model tractable, but it is also a restrictive assumption. In more realistic

models, reinforcement may be nonlinear: it may appear only after a threshold, grow at a different rate, or saturate over time. This leads to Generalized Elephant Random Walks, where the memory mechanism is governed by a more general function of the history. The main question is then which features of the classical ERW are robust. Does one still obtain a sharp phase transition? Are the limiting laws Gaussian in the diffusive regime? What replaces the superdiffusive limit? In [MRS24] the authors defined the Generalized Elephant Random Walk and proposed a few open questions which are still open for study.

AUXILIARY LEMMAS

This appendix collects a number of classical results that are used at various points in the main text. They are stated here for ease of reference and without proof.

Lemma A.1 (*Stolz–Cesàro theorem* [Fur13]). Let $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ be real sequences. Suppose that $(b_n)_{n \geq 1}$ is strictly increasing and satisfies $b_n \rightarrow \infty$ as $n \rightarrow \infty$. If the limit

$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L$$

exists in $\mathbb{R}$, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L.$$

Lemma A.2 (*Stirling’s approximation* [Jam15]). As $x \rightarrow \infty$, the Gamma function satisfies

$$\Gamma(x) \sim \sqrt{2\pi} x^{x-\frac{1}{2}} e^{-x}.$$

In particular, for integers $n \rightarrow \infty$,

$$n! = \Gamma(n+1) \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

Lemma A.3 (*Borel–Cantelli lemma* [Sch17]). Let $(A_n)_{n \geq 1}$ be a sequence of events.

1. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then

$$\mathbb{P}(A_n \text{ infinitely often}) = 0.$$

2. If the events $A_1, A_2, \dots$ are independent and

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

then

$$\mathbb{P}(A_n \text{ infinitely often}) = 1.$$

Lemma A.4 (*Lévy's continuity theorem [Kle20]*). Let $(X_n)_{n \geq 1}$ be a sequence of random variables in $\mathbb{R}^d$, with characteristic functions

$$\varphi_n(t) := \mathbb{E} \left[e^{i \langle t, X_n \rangle} \right], \quad t \in \mathbb{R}^d.$$

Suppose that there exists a function $\varphi : \mathbb{R}^d \rightarrow \mathbb{C}$ such that

$$\varphi_n(t) \rightarrow \varphi(t) \quad \text{for all } t \in \mathbb{R}^d,$$

and suppose that φ is continuous at 0. Then φ is the characteristic function of some random variable X in $\mathbb{R}^d$, and

$$X_n \xrightarrow{\mathcal{D}} X.$$

Lemma A.5 (*Toeplitz lemma [LH15]*). Let $(x_n)_{n \geq 1}$ be a sequence of real numbers such that

$$x_n \rightarrow x \quad \text{as } n \rightarrow \infty,$$

and let $(a_n)_{n \geq 1}$ be a sequence of nonnegative real numbers such that

$$A_n := \sum_{k=1}^n a_k \rightarrow \infty.$$

Then

$$\frac{1}{A_n} \sum_{k=1}^n a_k x_k \rightarrow x \quad \text{as } n \rightarrow \infty.$$

Lemma A.6 (*Raabe's criterion [Ham20]*). Let $(a_n)_{n \geq 1}$ be a sequence of strictly positive real numbers, and suppose that the limit

$$R := \lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right)$$

exists.

Then:

1. if $R > 1$, the series $\sum_{n=1}^{\infty} a_n$ converges;
2. if $R < 1$, the series $\sum_{n=1}^{\infty} a_n$ diverges.

In the case $R = 1$, the criterion is inconclusive.

Theorem A.7 (*Slutsky's Theorem* [vdV98]). Let $(X_n)_{n \geq 1}$ and $(Y_n)_{n \geq 1}$ be sequences of random variables such that

$$X_n \xrightarrow{d} X \quad \text{and} \quad Y_n \xrightarrow{\mathbb{P}} c$$

for some random variable X and some constant $c \in \mathbb{R}$. Then

$$X_n + Y_n \xrightarrow{d} X + c, \quad X_n Y_n \xrightarrow{d} cX.$$

If moreover $c \neq 0$, then

$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}.$$

SIMULATION CODE AND REPRODUCIBILITY

The Python code for the simulations, figures, and Monte Carlo studies in this thesis is available at

<https://github.com/Antonie2004/Thesis-Simulation-Code>

Reproduction requires Python 3.10 or later together with `numpy`, `scipy`, `matplotlib`, and `pandas`.

Most figures draw a fresh random sample on each run. The Monte Carlo studies instead fix the seed of the random number generator, so that their tables can be reproduced. The study reported in [Table 6.1](#) uses the seed 20260525, and the table in [Table 4.1](#) uses the seed 20260612.

COLOPHON

This document was typeset using the typographical look-and-feel `classicthesis`, developed by André Miede and Ivo Pletikosić. The style was inspired by Robert Bringhurst's book on typography, *The Elements of Typographic Style*. `classicthesis` is available for L^AT_EX at:

<https://bitbucket.org/amiede/classicthesis/>

During the writing of this thesis, I used generative AI tools (ChatGPT and Claude) in several ways. First, I used them to improve the language of text that I had written myself, such as grammar and word choice. Second, I used them to help find and fix errors in the code I wrote for the simulations and figures. Third, I used them to help resolve L^AT_EX problems while formatting this document.

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